

The Complexity of Pooling: Algebraic Barriers and Structural Algorithms

Abstract

In pooling, every stream leaving a pool carries the pool’s mixture, so blending-network flows are coupled by bilinear quality equations. Rational threshold decision is complete for the existential theory of the reals ($\exists\mathbb{R}$) with one two-sided attribute and no bypasses, even with bounded degrees and fixed finite data apart from the threshold, and also with one pool, bypasses, and a growing number of attributes. Hence pooling lies in **NP** only if $\mathbf{NP} = \exists\mathbb{R}$; the reductions also give rational instances whose unique optimal flow has arbitrarily high algebraic degree. If the number of pools times the affine quality rank, or times the number of products, is fixed, LP basis indices give **NP** certificates. One-attribute pooling is strongly **NP**-complete, and its profit has no PTAS unless $\mathbf{P} = \mathbf{NP}$, when all four layer degrees equal two. With bypasses, one pool with one scalar quality, two feed arcs, two outlet arcs, and fixed finite physical data is strongly **NP**-complete. For this interface with input degree at most two and positive flow lower bounds allowed, feasibility is polynomial at product degree at most two and strongly **NP**-complete at degree three. Polynomial exact algorithms follow from independent bounded blocks, planar composition along degree-two bypass paths, and exact supply and demand contracts. For the rank-one block used in pooling relaxations, we prove the general-cost hardness conjectured by Dey, Kocuk and Santana, polynomial optimization at fixed cost interaction rank, and exponential or superpolynomial lower bounds on exact conic lifts of its unit-margin hull and on outer approximations within inverse-quadratic ℓ_1 error.

Introduction

Pooling models blending networks in refining and other process industries [27]. Inputs of known quality are mixed in pools, every stream leaving a pool carries the pool’s mixture, and products must meet quality specifications. Direct input-to-product arcs, called *bypasses*, compete with pool feeds for source capacity. Once pool compositions are fixed, flow conservation, capacities, and product specifications form a linear system; choosing the compositions makes the problem a nonconvex bilinear program. The classical p-, q-, and pq-formulations [8, 27, 43] use pool qualities or source proportions, and later work strengthens their relaxations and discretizations [9, 15, 16, 23]. This paper asks which restrictions on the network, data, and contracts make exact pooling tractable, and what an exact answer must contain.

Known complexity. Alfaki and Haugland [4] proved that pooling is strongly **NP**-hard, even with one pool and no bypasses; their reduction uses a growing number of attributes. Haugland [24] showed that **NP**-hardness persists with one attribute, two sources, and two products; that it becomes strong with one attribute and either two sources or two products; and that strong **NP**-hardness also holds when all in-degrees, or all out-degrees, are at most two. One pool without bypasses is polynomial with a fixed number of attributes [4, Proposition 2], products [24, Theorem 2], or inputs [10, 26]. Baltean-Lugojan and Misener [6] give strongly polynomial algorithms for one-quality subclasses

with fixed positive demands and without feed-availability and pool-capacity bounds. Boland et al. [10, author manuscript, Section 4] ask whether scalar-quality pooling becomes polynomial when all in-degrees, or all out-degrees, are at most two. They and Haugland [24, Section 6] ask which one-pool results extend to more pools, Haugland specifically for a fixed number of pools with bounded numbers of inputs, products, or qualities.

These results classify combinatorial hardness. A more basic question remains: does exact pooling decision have short discrete certificates at all? A rational instance need not have a rational optimal flow (Example 2.9). The quadratic formulation places the threshold problem in $\exists\mathbb{R}$, the class of problems polynomially reducible to deciding whether a system of rational polynomial equations and inequalities has a real solution [39]; here $\text{NP} \subseteq \exists\mathbb{R} \subseteq \text{PSPACE}$ [1, Section 1.1].

Contributions. Table 1 summarizes the main physical boundaries.

- (1) *Existential-real completeness* (Section 2). Threshold decision is $\exists\mathbb{R}$ -complete with one attribute having lower and upper product specifications, binary source qualities, and no bypasses (Theorem 2.2), also with in-degrees at most two, out-degrees at most three, and all numbers except the threshold in fixed finite sets (Theorem 2.5), and with one pool, bypasses, and a growing attribute dimension (Theorem 2.7). Hence threshold decision lies in NP only if $\text{NP} = \exists\mathbb{R}$. With exact positive node contracts, feasibility is $\exists\mathbb{R}$ -complete in both classes (Corollary 2.6 and the paragraph after Theorem 2.7); whether one upper-bounded attribute suffices remains open. The reductions preserve solution sets up to rational bijections, so for every irrational real algebraic number α some rational instance has a unique optimal flow whose coordinates generate $\mathbb{Q}(\alpha)$ (Theorem 2.8). Irrationality alone would not exclude NP membership, because a certificate need not list a flow. The earlier reductions [4, 24] encode combinatorial problems; ours realize addition and inversion through exact product quality specifications. To the best of our knowledge, the pooling literature compared in Section 2 contains no previous $\exists\mathbb{R}$ -completeness result for these two classes. Without bypasses, one-pool pooling lies in NP for any number of attributes (Proposition 2.12), so the one-pool hardness needs bypasses unless $\text{NP} = \exists\mathbb{R}$. If the number of pools times the affine input-quality rank, or times the number of products, is fixed, a guessed LP basis is a polynomial-size certificate (Corollary 2.11); this gives NP membership, not a polynomial algorithm or rational optimal flows.
- (2) *Hardness near the algorithmic regimes* (Section 3). One-attribute pooling is strongly NP -complete when all four layer degrees equal two, with unit capacities and only upper specifications, and its profit has no PTAS unless $\text{P} = \text{NP}$ (Theorem 3.3). If its zero upper quality bounds are replaced by a suitable fixed positive bound, exact optimization remains strongly NP -hard and the profit still has no PTAS unless $\text{P} = \text{NP}$ (Proposition 3.5). This answers both degree questions of Boland et al. [10] negatively. With two pools and two products on an undirected tree, and growing input and attribute counts, threshold decision is NP -complete in the ordinary sense (Theorem 3.9), which answers the product-bounded case of the two-pool question negatively; the subfamily used in the reduction has an FPTAS in its given representation (Proposition 3.10). With bypasses, one pool with one upper-bounded scalar quality, two actual feed arcs, two actual outlet arcs, and fixed finite physical data is strongly NP -complete (Theorem 3.19); the many bypass sources are not mixing feeds. Broad one-pool hardness with capacities was already stated by Baltean-Lugojan and Misener [6, Remark 4.6]; our contribution is the explicit reduction under these simultaneous restrictions. Feasibility remains strongly NP -complete when only five external nodes lack exact contracts and ordinary products may have three bypass inlets (Theorem 3.20).

- (3) *Structural exact algorithms* (Section 4). These apply when the linear network can be eliminated without losing pool balances or the objective. A fixed nonlinear core coupled to independent polyhedral blocks of fixed dimension admits polynomial global optimization with an exact algebraic optimizer (Theorem 4.1). Consequences include fixed input and pool counts with arbitrary bypasses, fixed pool and attribute counts without bypasses (Corollary 4.3 and Corollary 4.4), which answer the input- and quality-bounded cases of Haugland’s question, and fixed pool count, affine quality rank, and bypass vertex integrity (Theorem 4.5). On degree-two bypass graphs, composition of planar relations along paths (Theorem 4.8) gives polynomial feasibility with a fixed number of pool attachments and arbitrary attributes, and optimization of objectives on a fixed set of retained arcs (Theorem 4.9 and Theorem 4.11). For one pool with two feed and two outlet arcs, input out-degree at most two, and positive flow lower bounds allowed, feasibility is polynomial at product in-degree at most two and strongly NP-complete at three (Corollary 4.13). This boundary concerns feasibility; the degree-two algorithm does not cover dense objectives such as the economic threshold of Theorem 3.19.
- (4) *Contract algorithms* (Section 5). Exact contracts turn pool mass and attribute mass into conserved boundary quantities, or into signed node divergences. For one pool with bypass degree at most two this gives polynomial algorithms for exact supplies with a fixed number of receiving products (Theorem 5.3); for feasibility under full contracts with scalar or affine-rank-one qualities, with quadratic-field witnesses when the common pool bound has lower bound zero and a redundant upper bound (Theorem 5.6), and with restrictive common bounds and throughput objectives (Theorem 5.13 and Corollary 5.14); and for a fixed number of exceptions to otherwise exact contracts, with arbitrary quality rank and redundant common bounds (Theorem 5.10), in contrast to the five-exception hardness of Theorem 3.20. Feasibility with two distinct full input-quality vectors is polynomial on arbitrary bypass graphs (Theorem 5.15). Table 3 gives the precise scopes; restrictions from different theorems do not combine without an additional argument.
- (5) *Response geometry and convexification* (Section 6 and the appendices). As one product price varies, a pool-free blending path has 2^n distinct unique optimal flows (Theorem A.3). A one-pool extension has 2^n isolated global optima and, after a price perturbation, at least 2^n strict local optima, yet it has a polynomial-size exact LP hull (Theorem A.6, Corollary A.7, and Theorem A.9). For the isolated rank-one margin block used in pooling relaxations [16], we prove the general-cost hardness conjectured by Dey, Kocuk, and Santana (Theorem B.4 and Theorem B.7) and polynomial exact optimization for fixed cost interaction rank (Theorem B.1). The unit-margin hull K_{2m} has a correlation-polytope face (Theorem C.1). With known correlation-polytope bounds [12, 17, 33], this gives exponential or superpolynomial size lower bounds for its exact conic lifts (Corollary C.2) and for outer approximations within total entrywise ℓ_1 error $1/A_m$ (LP) or $1/(2A_m)$ (SDP), where $A_m = m(184m + 6)$ (Corollary C.5 and Theorem C.8). These uniform bounds do not concern ordinary additive operating costs or a single objective, and are consistent with the exponential second-order-cone representation of Jalilian and Kocuk [31].
- (6) *Physical foundations* (Section 1). We give physical-flow proofs of established tools: destination decomposition and the zero-optimum test of Xiang et al. [44] (Theorem 1.3), an exact uncapacitated conic hull and its cyclic extension (Theorem 1.5 and Theorem 1.7), and a characterization of the product specifications that guarantee integral optimal flows (Theorem 1.8).

Organization. Section 1 fixes the model conventions (exact contracts, standard economics) and exact arithmetic; fixed parameters may enter polynomial exponents, and no fixed-parameter

Physical restriction	Decision or objective scope	Result
One attribute, no bypasses; or one pool and growing attribute dimension	Rational thresholds, two-sided qualities, zero flow lower bounds	$\exists\mathbb{R}$ -complete (Theorem 2.2 and Theorem 2.7)
Fixed pool count times quality rank or product count	Feasibility and linear arc-cost thresholds; unrestricted bypasses	In NP (Corollary 2.11)
All four layer degrees two; growing pool count	Upper-only scalar qualities, zero flow lower bounds; profit	Strongly NP-complete; no PTAS (Theorem 3.3)
One pool, two feeds and two outlets, input degree two	Feasibility with contracts allowed: arbitrary attributes at product degree two; fixed scalar data suffice at degree three	Degree two: polynomial; degree three: strongly NP-complete (Corollary 4.13)
Fixed pools, quality rank and bypass vertex integrity	Feasibility and arbitrary linear arc-cost optimization	Polynomial (Theorem 4.5)
Degree-two bypasses; fixed total pool attachments	Feasibility and objectives on a fixed retained arc set	Polynomial (Theorem 4.9 and Theorem 4.11)

Table 1: Selected physical boundaries, with finite rational capacities. The cited statements give the full hypotheses. The no-PTAS conclusion assumes $P \neq NP$.

tractability claim is implied. Sections 2 and 3 prove algebraic completeness and restricted hardness; Sections 4 and 5 develop structural and contract algorithms. Section 6 discusses representation consequences and the remaining degree-two bypass question. Appendix A proves the response-geometry results and limits of elimination, including dense-slab hardness. Appendices B and C treat costs and convexification of the isolated margin block (115) and can be read independently of Sections 2–5.

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1 Models, certificates, and elementary structure

The requirement that all streams leaving a pool have the same composition is local, but it couples decisions at different products. Which complexity results apply depends on the precise conventions for flow contracts, quality specifications, and direct arcs, so we fix them first. The basic formulations and the rank-one interpretation are established background. The destination decomposition shows which parts of profitable-flow detection follow from existing single-product and multi-commodity arguments; the conic-hull, cyclic, and facial-specification results give the further structural consequences used later. None of the sign tests solves the general objective-threshold problem.

1.1 Physical models and decision problems

Let I, L, J be disjoint finite sets of inputs, pools, and products. In the *standard pooling model*, the arc set satisfies

$$A \subseteq (I \times L) \cup (L \times J) \cup (I \times J).$$

An arc in $I \times J$ is a *bypass*. The bypass graph is the undirected bipartite graph on $I \cup J$ with these arcs as edges. Its degree, the *bypass degree*, differs from the total degree of an external node, which also counts incident pool arcs. Write $y_{i\ell}, v_{\ell j}, z_{ij}$ for input–pool, pool–product, and bypass flows, respectively. A missing arc has flow zero. Input i has a rational quality vector $\lambda_i \in \mathbb{Q}^K$. The quality of pool ℓ is $q_\ell \in \mathbb{R}^K$. Define input withdrawals, pool throughputs, and product deliveries by

$$s_i = \sum_{\ell} y_{i\ell} + \sum_j z_{ij}, \quad T_\ell = \sum_i y_{i\ell} = \sum_j v_{\ell j}, \quad (1)$$

$$t_j = \sum_{\ell} v_{\ell j} + \sum_i z_{ij}. \quad (2)$$

The perfect-mixing equations and product quality masses are

$$\sum_i \lambda_{ik} y_{i\ell} = q_{\ell k} T_\ell \quad (\ell \in L, k = 1, \dots, K), \quad (3)$$

$$m_{jk} = \sum_{\ell} q_{\ell k} v_{\ell j} + \sum_i \lambda_{ik} z_{ij}. \quad (4)$$

Ordinary lower and upper specifications mean

$$\underline{\mu}_{jk} t_j \leq m_{jk} \leq \bar{\mu}_{jk} t_j. \quad (5)$$

For the geometric results below we also allow a rational polyhedral product region $R_j = \{u : A_j u \leq b_j\}$, represented in flow variables by $A_j m_j \leq b_j t_j$. This broader specification format is used only when explicitly stated. A scalar quality with both lower and upper bounds counts as one attribute. Replacing a lower bound by an upper bound on the negative of that attribute creates another coordinate in an upper-bound-only formulation; these are different restrictions.

All flows are nonnegative. Individual arcs and the totals s_i, T_ℓ, t_j may have finite rational lower and upper bounds. Initially intersect each such flow-bound interval with $[0, \infty)$ and reject the

instance if any intersection is empty. In every capacitated model, these bounds give a finite rational upper bound on every arc flow, either directly or through finite node capacities. The explicitly uncapacitated sets introduced below are exempt from this convention. A bound with equal endpoints is an *exact flow contract*. An *exact product contract* fixes both $t_j = d_j$ and $m_j = d_j B_j$ for given demand d_j and quality vector B_j . A *common pool bound* constrains T_ℓ ; it is additional to individual feed and outlet bounds. A theorem that assumes this bound redundant does not allow an arbitrary restrictive capacity in its place.

Product quality constraints use their homogeneous mass form even at zero flow. Since incoming flows are nonnegative, $t_j = 0$ forces every incoming flow to zero, so $m_j = 0$ and the constraints impose no composition requirement. In particular, an exact quality assigned to a zero-demand product is immaterial. Likewise, $T_\ell = 0$ makes every incident pool flow zero. Such a pool may receive any finite quality without changing feasibility. Removing an unusable pool requires checking that zero belongs to its throughput interval and every incident-arc interval; external-node contracts must remain, since bypasses or other pools can satisfy them.

Costs are rational and linear in actual arc flows. We minimize $c^T f$, where f collects those flows, and use profit $-c^T f$ when convenient. *Standard economic costs* have the more restrictive form $\sum_i a_i s_i - \sum_j r_j t_j$. Arbitrary arc costs need not have this form. Under exact source and product flow contracts, standard economic cost is constant; arbitrary routing costs need not be constant.

The *threshold problem* asks whether a feasible flow satisfies $c^T f \leq \tau$ for a rational threshold τ . *Feasibility* omits this last inequality. With zero lower flow bounds and no positive flow contracts, the zero flow is feasible. Hardness of a nonzero profit threshold therefore does not imply hardness of feasibility for that same class. The *profitable-flow problem* asks whether some feasible flow has $c^T f < 0$; equivalently, when zero is feasible, whether the optimum is strictly negative.

We also consider a *generalized pooling network*, with $A \subseteq (I \cup L) \times (L \cup J)$, which permits pool–pool arcs. For a flow f_{uv} , put $\xi_u = \lambda_u$ at an input and $\xi_u = q_u$ at a pool. The corresponding equations are

$$F_v := \sum_u f_{uv} = \sum_w f_{vw}, \quad F_v q_v = \sum_u f_{uv} \xi_u \quad (v \in L), \quad m_j = \sum_u f_{uj} \xi_u. \quad (6)$$

An acyclic generalized network has no directed cycles. The cyclic extension in Section 1.6 uses the algebraic steady-state interpretation of (6), which permits an isolated positive circulation without source withdrawal. No theorem for that interpretation assumes an additional source-tracing or initialization condition. None of the models here includes losses, fixed charges, nonlinear blending laws, or intermediate quality restrictions unless explicitly added.

1.2 Encoding and exact certificates

Input size is the binary encoding length of the graph and all rational data. Polynomial time means bit complexity in this size. When dimensions are declared fixed, constants and polynomial exponents may depend on them. Such statements do not imply fixed-parameter tractability with a uniform exponent, or a practical running-time bound.

An exact real-algebraic answer specifies real roots of rational polynomials with isolating data and exact expressions for the coordinates. Whenever we claim a polynomial-time exact algorithm, its output representation, including degrees, coefficient lengths, and isolating data, must also have polynomial size. Rational data do not by themselves imply a rational optimal flow. Conversely, membership in NP does not require the certificate to be the flow: a short combinatorial certificate may be checked by a fixed-dimensional algebraic computation.

For standard and acyclic generalized pooling, active pool qualities are convex combinations of input qualities. This follows directly by weighted averaging in standard pooling and by induction in a topological order in an acyclic network. Inactive pool qualities can therefore be chosen in the coordinatewise input-quality box. If I is empty, acyclicity and conservation force all flows to zero, which is checked directly. When I is nonempty, the finite flow bounds together with this box make the lifted feasible set closed and bounded. Thus a feasible instance has an attained linear optimum, and its arc-flow projection is compact. This argument does not assign source-derived bounds to isolated cyclic circulations.

The polynomial formulation (1)–(5) immediately places the threshold problem in $\exists\mathbb{R}$, the class of decision problems polynomially reducible to existence of a real solution of rational polynomial equalities and inequalities. This observation is an upper bound; it supplies no general **NP** certificate bound.

A useful exact reduction replaces the attribute count by affine rank. Suppose $I \neq \emptyset$, and compute rational vectors λ_0 and independent columns of $B \in \mathbb{Q}^{K \times r}$ such that $\lambda_i = \lambda_0 + Ba_i$, $a_i \in \mathbb{Q}^r$. Rational Gaussian elimination computes these data with polynomial encoding length. For standard pooling, write $q_\ell = \lambda_0 + B\eta_\ell$. Equations (1) and (3) are then equivalent to

$$\sum_i a_i y_{i\ell} = \eta_\ell T_\ell. \quad (7)$$

Indeed, substitution cancels $\lambda_0 T_\ell$, and independence of the columns of B proves the converse. Every active pool is represented; an inactive pool can be assigned arbitrary coordinates in the box spanned coordinatewise by the a_i . Substitution into every original product inequality preserves all attributes, including when their number grows. If $r = 0$, all input qualities coincide and quality feasibility reduces to deletion of incompatible positive product flows and linear constraints. Affine quality rank and the number of distinct quality vectors are separate parameters.

Readers interested mainly in algebraic completeness can continue with Section 2; the remaining subsections of this section are independent of those constructions.

1.3 The rank-one flow block

For one standard pool, let y_i and v_j be its incoming and outgoing flows, with common total T . When $T > 0$, the amount of input i allocated to product j through that pool is

$$W_{ij} = \frac{y_i v_j}{T}. \quad (8)$$

Thus W is nonnegative, has rank one, and has row and column sums y, v . When $T = 0$, nonnegativity gives $y = v = 0$, and we set $W = 0$. Conversely, every nonzero nonnegative rank-one matrix has (8) with its own margins. To see this, write $W = ab^T$ with $a, b \geq 0$; such factors follow by taking a positive column and the proportional nonnegative column multipliers. Its margins are $y = a \sum_j b_j$, $v = b \sum_i a_i$, and $T = (\sum_i a_i)(\sum_j b_j)$, proving the identity. The mass of attribute k delivered along the pool's outlet to j is $\sum_i \lambda_{ik} W_{ij}$. This explains the role of rank-one matrix sets with row and column bounds in pooling formulations [16].

This correspondence requires care with objectives. Physical costs on feed and outlet arcs induce the matrix coefficients $C_{ij} = a_i + b_j$. An arbitrary cost matrix on W is a broader optimization model. In particular, hardness of arbitrary linear optimization over a rank-one block alone does not prove hardness of a single physical pool with only feed and outlet costs.

1.4 Single-product decomposition and profitable flow

Throughout this subsection the generalized network is acyclic, flow lower bounds are zero, and all other flow restrictions are nonnegative upper capacities. Only final products have quality restrictions. Let P be the set of feasible arc flows and $z^* = \min_{f \in P} c^T f \leq 0$. The destination-disaggregation construction is established pooling machinery [9]. The standard-pooling one-product approximation principle is due to Dey and Gupte [15]; the sign question appears explicitly in Gupte et al. [23, Remark 2.2]. Xiang et al. [44, Proposition 4 and Theorem 2] resolve that question by single-terminal LPs in a generalized terminal-proportion formulation, including pool-pool arcs, so polynomial zero-optimum detection is established. We give the full argument because it works directly with physical flows and supplies the decomposition used for the shortest-path and conic-hull statements.

Lemma 1.1 (Destination decomposition). *Every feasible physical flow has a decomposition $f = \sum_{j \in J} f^j$ into feasible flows, each serving only product j , with $0 \leq f^j \leq f$ coordinatewise.*

Proof. Fix feasible pool qualities. At products set $h_j(j') = 1$ if $j = j'$ and zero otherwise. In reverse topological order, set, for each positive-throughput pool v ,

$$h_j(v) = F_v^{-1} \sum_w f_{vw} h_j(w).$$

Set all $h_j(v)$ to zero at zero-throughput pools. Conservation and acyclicity imply that every positive-flow path can be continued to a product; hence reverse induction gives $0 \leq h_j(v) \leq 1$ and $\sum_j h_j(v) = 1$ at each active pool. Define $f_{uv}^j = f_{uv} h_j(v)$, using the head of the arc. All incoming flows of a pool v receive the same multiplier $h_j(v)$. Their sum is $F_v h_j(v)$; the sum of outgoing component flows has this same value by the recursion. Their incoming quality mass is the original quality mass multiplied by $h_j(v)$, so the original q_v still satisfies (6). At product j the original inflow is unchanged; other products receive zero. Coordinatewise domination preserves all upper capacities, including node throughputs. Finally, the head of any positive-flow arc is either an active pool or a product, where the fractions sum to one. Summing the components therefore gives f . $\square$

For each j , form a rational LP by retaining all ordinary flow balances and upper capacities, forcing delivery to other products to zero, and replacing all mixing equations by

$$\underline{\mu}_{jk} t_j \leq \sum_i \lambda_{ik} s_i \leq \bar{\mu}_{jk} t_j \quad (k = 1, \dots, K). \quad (9)$$

For polyhedral product regions the analogous row is $A_j \sum_i \lambda_i s_i \leq b_j t_j$.

Lemma 1.2 (Single-product projection). *This LP describes exactly the arc flows of the physical single-product problem.*

Proof. Summing all physical pool quality balances cancels internal quality flows, so the quality mass reaching the only served product is $\sum_i \lambda_i s_i$. This proves necessity. Conversely, take an LP flow and process pools in topological order. Give every active pool the weighted average of its incoming qualities and assign arbitrary finite qualities at inactive pools. All pool equations hold. Summing them again identifies product quality mass with the expression in (9), proving its specification. Zero delivery forces zero flow throughout an acyclic conserved network, so that case also satisfies all constraints. $\square$

Theorem 1.3 (Polynomial sign test; physical-flow version). *Let z_j be the minimum cost in the single-product LP. Then $z^* = 0$ if and only if every $z_j = 0$. These LPs also construct a profitable physical flow whenever one exists. If $m = |J| \geq 1$ and $z_{\text{best}} = \min_j z_j$, then*

$$z^* \leq z_{\text{best}} \leq z^*/m \leq 0. \quad (10)$$

Proof. Every single-product LP flow is physical by Lemma 1.2. Conversely, Lemma 1.1 decomposes an optimal physical flow into m feasible components whose costs sum to z^* . At least one component has cost at most z^*/m , and its LP optimum is no larger. This proves (10) and the sign equivalence. Rational LP optimization and exact comparison with zero take polynomial bit time. Reconstructing qualities as in Lemma 1.2 is exact. For rational flow data this reconstruction solves a rational triangular linear system on active pools, so determinant bounds give polynomial encoding length. If J is empty, acyclicity forces zero flow directly. $\square$

For $m = |J| \geq 1$, there is also a single destination-flow relaxation. Give every product its own nonnegative conserved single-product flow f^j satisfying (9), and impose all capacities on $\sum_j f^j$. Let z_{rel} be its minimum cost. Destination decomposition makes this a relaxation. Every component of a feasible relaxed solution is physical and respects the capacities by nonnegativity. Selecting the cheapest component of an optimal relaxed solution proves

$$z_{\text{rel}} \leq z^* \leq z_{\text{best}} \leq z_{\text{rel}}/m \leq 0.$$

In particular, this relaxation has the same zero-optimality status as pooling. These inequalities concern cost signs and product-count profit approximation; they are not exact optimization formulas for pooling.

Theorem 1.4 (Shortest-path sign criterion). *Delete zero-capacity arcs and nodes with their incident arcs. Let δ_{ij} be the minimum path cost from input i to product j on the remaining graph, and let I_j contain the inputs that reach j . Define β_j as the optimum of*

$$\begin{aligned} & \min \sum_{i \in I_j} \delta_{ij} \gamma_i \\ & \text{subject to } \gamma \geq 0, \quad \sum_{i \in I_j} \gamma_i = 1, \quad \underline{\mu}_{jk} \leq \sum_{i \in I_j} \lambda_{ik} \gamma_i \leq \bar{\mu}_{jk} \quad (k = 1, \dots, K), \end{aligned}$$

with value $+\infty$ for an infeasible LP. Then $z^ < 0$ if and only if some $\beta_j < 0$. A profitable instance has a rational polynomial-size witness obtained by routing at most $K + 1$ shortest input–product paths and scaling all flows by one positive factor.*

Proof. A negative-cost physical flow has a negative-cost single-product component. Decompose its ordinary conserved flow into input–product paths and divide source withdrawals by its positive total delivery t . The resulting weights $\gamma_i = s_i/t$ meet the quality bounds by aggregate conservation. Actual cost per unit is at least $\sum_i \delta_{ij} \gamma_i$, so $\beta_j < 0$. Conversely, route a negative-cost feasible mixture on one shortest path per supported input. Every retained capacity is strictly positive, so a sufficiently small positive common scale satisfies all capacities and preserves cost sign and quality ratios. The scaled conserved single-product flow is physical by Lemma 1.2.

The mixture LP is compact. Choose an optimal vertex and let s be its support size. Its active quality rows have rank at most K , and normalization adds one row. If $s > K + 1$, a nonzero direction on its support preserves every active row. Small perturbations in both signs preserve inactive inequalities and nonnegativity, contradicting extremality. Thus $s \leq K + 1$. Shortest paths in an acyclic graph have polynomial length and rational polynomial-size costs. The mixture vertex has polynomial rational encoding. For each capacity row r , let U_r be its upper bound and a_r its load in the unscaled flow. Choose the scale as $\min(\{1\} \cup \{U_r/a_r : a_r > 0\})$; zero-load rows impose no restriction. This positive rational scale has polynomial encoding. Quality reconstruction has the bit bound established in Theorem 1.3. $\square$

The support bound uses K coordinate quality bounds. For general polyhedral specifications it becomes one plus the rank of the active quality rows restricted to the input simplex. Positive capacity magnitudes affect attainable profit but affect its sign only through the support remaining after zero-capacity deletion.

1.5 An exact uncapacitated conic hull

Continue on the positive-capacity support. Let S be the physical flow set after removing all upper capacities, and let C_j be its single-product LP cone. We use $\text{cone}(X)$ for finite nonnegative linear combinations of points of X ; it is not assumed closed by definition.

Theorem 1.5. *For the acyclic model of Section 1.4,*

$$\text{conv}(S) = \text{cone}(S) = \sum_{j \in J} C_j = \text{cone}(P). \quad (11)$$

The sum is a Minkowski sum and has a polynomial-size linear extended formulation. In particular, this cone is polyhedral and closed.

Proof. Destination decomposition proves $S \subseteq \sum_j C_j$. Every $C_j \subseteq S$ by Lemma 1.2, so $\sum_j C_j \subseteq \text{cone}(S)$. Because S contains zero and is closed under nonnegative scaling, any finite conic combination can be written as a convex combination of scaled elements of S ; hence $\text{cone}(S) = \text{conv}(S)$. The Minkowski sum is a convex cone, proving the first three equalities. Since $P \subseteq S$, one inclusion for the last equality is immediate. Conversely every finite $f \in S$ can be scaled by a positive scalar into P , since all retained capacities are positive. Thus $S \subseteq \text{cone}(P)$. The formulation keeps a copy of the $O(|A|)$ flow variables for each product and the equation $f = \sum_j f^j$. It is a projection of a polyhedral cone and hence is polyhedral. With no products every cone in question is $\{0\}$. $\square$

Consequently all homogeneous linear inequalities valid for P are captured by (11). The profitable-flow test asks whether c lies outside its dual cone $\{c : c^T f \geq 0 \text{ for every } f \in S\}$; membership is equivalent to the absence of profitable flow.

Capacities cannot generally be reimposed after convexification to obtain $\text{conv}(P)$. For example, take input qualities zero and one, a single pool of capacity two, and two products requiring qualities exactly zero and exactly one. Allow only feed and outlet arcs and give each input, product, and arc capacity one. A physical flow serves at most one product, so total delivery is at most one throughout $\text{conv}(P)$. The sum of the two unit pure single-product flows belongs to $\text{conv}(S)$ and satisfies every stated capacity, but delivers two units.

1.6 Algebraic recirculation

The sign and conic-hull conclusions extend to cyclic generalized networks under the steady-state semantics of Section 1.1. All assumptions on zero lower bounds and absence of intermediate quality restrictions remain in force. The generalized sign test of Xiang et al. [44] already allows pool–pool arcs. Here we specify how the physical equations treat a singular case that a terminal-indexed formulation must handle separately: an isolated positive circulation, including one with no path to any product. This also identifies its contribution to the uncapacitated cone.

Lemma 1.6 (Cyclic single-product reconstruction). *Every ordinary nonnegative conserved flow serving at most one product admits pool qualities satisfying the mixing equations. Its delivered quality mass is $\sum_i \lambda_i s_i$. With rational flow data, qualities of polynomial rational encoding can be constructed.*

Proof. Consider the strongly connected components of the positive pool-flow support, ignoring inactive pools. Summing mass balances over any component shows that its total external inflow equals its total external outflow. A component with no external inflow therefore has no external outflow and is an isolated circulation. Assign a constant quality to all its pools. Process the other components in topological order. For one coordinate, the equations on a component D are

$$F_v q_v - \sum_{u \in D} f_{uv} q_u = b_v, \quad b_v = \sum_{u \notin D} f_{uv} \xi_u.$$

After division by $F_v > 0$, the internal matrix Q is nonnegative and has row sums at most one, with at least one deficient row. Strong connectivity implies that every state of the corresponding backward chain has a positive-probability path to a deficient row and then out of the component. Finiteness gives an integer h and $\varepsilon > 0$ such that every state exits within h steps with probability at least ε . Thus $\|Q^{rh}\|_\infty \leq (1 - \varepsilon)^r$, proving that $I - Q$ is invertible with inverse $\sum_{a \geq 0} Q^a$. The solution is a weighted average of the already known boundary qualities. This constructs all active pool qualities; inactive ones are arbitrary. Summing the quality balances cancels all internal mass, including cycles, and proves the aggregate identity. For rational flows, the equations on all nonisolated active pools form one nonsingular rational linear system. Cramer's rule gives polynomial encoding length; isolated components can use quality zero. $\square$

Theorem 1.7 (Cyclic sign and hull). *Let C_0 be the nonnegative circulation cone on the pools and C_j the uncapacitated single-product cone, now allowing cycles. Then*

$$\text{conv}(S) = \text{cone}(S) = C_0 + \sum_{j \in J} C_j = \text{cone}(P).$$

Profitable flow can be detected in polynomial time. On the positive-capacity support it exists precisely when there is a negative-cost directed cycle, or, in the absence of such a cycle, one of the shortest-path mixture LPs in Theorem 1.4 has negative optimum.

Proof. For any feasible flow, a terminal strongly connected component of its positive pool support with no positive arc to a product has no positive external inflow by conservation. It is therefore an isolated circulation. Remove all such components and call their flow f^0 . Every remaining active pool reaches a product along positive arcs. The forward routing chain with transition probabilities f_{vw}/F_v is absorbed at products with probability one: as in Lemma 1.6, there is a uniform positive chance of absorption within a fixed number of steps. Its absorption probabilities satisfy

$$F_v h_j(v) = \sum_w f_{vw} h_j(w), \quad \sum_j h_j(v) = 1.$$

Set $h_j(v) = 0$ on the removed circulation components and inactive pools, and set $h_j(j') = 1$ if $j = j'$ and zero otherwise at products. Using $f_{uv}^j = f_{uv} h_j(v)$ gives, by the same incoming-mass calculation as Lemma 1.1, physical dominated components and $f = f^0 + \sum_j f^j$. This proves one containment for the hull. Every nonnegative circulation is physical, by choosing a common constant quality on all its pools, and each C_j is physical by Lemma 1.6. Scaling and convexification then prove all hull equalities exactly as before.

A negative directed cycle is a feasible profitable circulation after small positive scaling. If there are no negative cycles, f^0 has nonnegative cost by cycle decomposition. A profitable flow therefore has a negative single-product component. Its ordinary path-and-cycle decomposition, followed by removal of nonnegative-cost cycles, yields a negative path mixture with unchanged source withdrawals

and delivery. The aggregate quality identity then proves a negative shortest-path mixture value. Conversely, route any such mixture on shortest paths and scale to capacities. Lemma 1.6 supplies feasible qualities even when the union of the paths contains cycles. Negative-cycle detection and shortest-path algorithms with signed arc costs, followed by rational LPs, take polynomial time. $\square$

If $J \neq \emptyset$, C_0 is contained in every C_j , so its extra summand is redundant. The capacitated single-product LP family also detects negative circulations, since each LP admits them. If J is empty, the sign problem is an ordinary circulation problem.

For $m = |J| \geq 1$, the approximation and relaxation inequalities of Section 1.4 hold in the cyclic model as well. In the decomposition used above, merge f^0 into one designated product component. Its circulation support is disjoint from the other components' positive support, so the merged component is physical and is still coordinatewise dominated by f . There are now exactly m feasible single-product components summing to f . Applying the averaging argument to an optimal physical flow proves (10). The same decomposition makes the destination-flow LP a relaxation, and each of its components is physical by Lemma 1.6. Selecting its cheapest component therefore gives $z_{\text{rel}} \leq z^* \leq z_{\text{best}} \leq z_{\text{rel}}/m \leq 0$. An optimum exists here as well: nonisolated pool qualities lie in the input-quality hull, and isolated circulations can all use quality zero. These choices put every feasible flow in the projection of one bounded closed lift, using the box containing zero and all input qualities (the singleton zero box when there are no inputs).

The distinction between active support and the original graph is essential. On the arcs $i \rightarrow a$, $a \rightarrow b$, $b \rightarrow a$, $b \rightarrow j$, let $f_{ab} = f_{ba} = 1$ and the other flows be zero. Both pool qualities may equal any common constant, although every vertex lies on an input-product path in the original graph. The mixing equations have a singular cycle block. More quantitatively, take $f_{ia} = \varepsilon$, $f_{ab} = 1$, $f_{ba} = 1 - \varepsilon$, $f_{bj} = \varepsilon$, with $0 < \varepsilon < 1$ and input quality λ . The pool equations are

$$\begin{pmatrix} 1 & -(1 - \varepsilon) \\ -1 & 1 \end{pmatrix} \begin{pmatrix} q_a \\ q_b \end{pmatrix} = \begin{pmatrix} \varepsilon\lambda \\ 0 \end{pmatrix}.$$

Their determinant is ε and the solution is $q_a = q_b = \lambda$. Assigning both qualities $\lambda + 1$ produces residual $(\varepsilon, 0)^T$ despite unit concentration error. Thus an absolute mixing residual alone does not control concentration error near a closed circulation. The treatment of cyclic formulation variables is part of the existing generalized-pooling literature [9, Section 4.3]; the arguments here explicitly separate the singular circulation case.

1.7 Facial product specifications and integral flow

Return to standard pooling, permitting arbitrary bypasses, rational polyhedral product regions, and lower as well as upper flow bounds. Assume a nonempty input set and put $C = \text{conv}\{\lambda_i : i \in I\}$ and $F_j = C \cap R_j$. We count $\emptyset$ and C as faces. A face F of C has the property that a strict convex combination of points of C belongs to F only if every positively weighted point belongs to F . For example, an upper quality bound zero with nonnegative input qualities selects a face: a positive-quality feed cannot contribute to such a product. This exclusion already appears in pooling hardness reductions, including Haugland [24, Theorem 4]. The result below identifies exactly when the same reasoning guarantees an integral optimum for every network and every linear objective on a given set of input qualities.

Theorem 1.8 (Universal flow integrality). *If every F_j is a face of C , the projected feasible flow set is a finite union of network flow polytopes. With integer flow bounds, these polytopes are integral; consequently every feasible instance has an integral optimal flow for every rational linear objective.*

Conversely, for any nonempty $F = C \cap R$ that is not a face, there is a one-pool, one-product network on the same input qualities with all arc and vertex upper bounds one, zero lower bounds, and costs in $\{0, -1\}$ for which no integral flow is optimal.

Proof. If product j receives positive flow, its blend lies in F_j . The face property forces every contributing pool quality and every contributing direct input quality into F_j . Applying it again to an active contributing pool forces $\lambda_i \in F_j$ whenever both $y_{il} > 0$ and $v_{lj} > 0$. Conversely, if this compatibility holds for every such pair and every positive direct arc, weighted averaging puts every delivered blend in F_j . If F_j is empty, these rules permit no positive delivery to j .

Fix an allowed arc set containing no incompatible direct arc or feed-outlet pair. All ordinary network flows restricted to that set are quality feasible by the preceding argument. Taking the finite union over compatible allowed sets gives exactly the pooling projection. Each branch has only conservation, arc bounds, and node-throughput bounds. Split nodes to represent the latter by bounded arcs and add a return arc from a common terminal to a common source to express the whole problem as a bounded circulation. A finite upper bound for the return arc is the sum of source upper bounds when these are supplied; otherwise the sum of the effective arc upper bounds suffices.

For completeness, a directed incidence matrix is totally unimodular. In any square submatrix, a column with at most one nonzero entry permits induction by determinant expansion; if no such column exists, every column has one $+1$ and one -1 , so the rows are dependent. Thus every square subdeterminant is 0 or ± 1 . Adding unit rows for bounds preserves this property by the same expansion. A vertex basis with integer right-hand side therefore has integer solution by Cramer's rule. This proves branch integrality, also after projecting auxiliary arcs. Boundedness and finiteness of the union give an attained integral optimum whenever at least one branch is nonempty.

For necessity, nonfaciality gives $x, y \in C$, $0 < \theta < 1$, with $q = \theta x + (1 - \theta)y \in F$ and, say, $x \notin F$. Expand x and y into input mixtures. Some input outside F has positive weight in the representation of x , since otherwise convexity would put x in F . Hence the representation of q uses positive total mass from inputs outside F . Connect all inputs to a single unit-capacity pool and that pool to a unit-capacity product with region R . Charge -1 on feeds from inputs outside F and zero on other arcs. The displayed unit mixture is feasible with negative cost. An integral feasible flow has throughput zero or one; in the latter case it comes entirely from one input whose quality must lie in F . Every integral feasible flow therefore has cost zero. The underlying single-product flow constraints are rational linear constraints, so a rational negative-cost mixture also exists, for example at an optimal LP vertex. $\square$

The converse is a necessity statement for this universal guarantee, quantified over networks and objectives, not a hardness theorem. It does not imply that every given network with a nonfacial product has a fractional optimum, or that a nonfacial class is computationally hard. The facial class itself will contain hard support-selection problems. Integer flow also does not imply pure input selection when a pool can process several integer units.

Proposition 1.9 (Recognition and bounded-pool algorithm). *Faciality of all product specifications is recognizable in polynomial bit time. For fixed pool count p and fixed affine input-quality dimension d , standard pooling with facial specifications admits exact polynomial-time optimization, including arbitrary bypasses and rational flow bounds.*

Proof. For product j , call input i forbidden if $\lambda_i \notin R_j$. Maximize the total weight of forbidden inputs over

$$w \geq 0, \quad \sum_i w_i = 1, \quad \sum_i \lambda_i w_i \in R_j.$$

Infeasibility means $F_j = \emptyset$. If feasible, the maximum is zero for a face by its defining property. Conversely, a zero maximum implies that every feasible mixture uses only allowed inputs. If an interior point of a segment in C belongs to F_j , expand both endpoints as input mixtures. A forbidden input in either expansion would yield positive objective in the LP. Both endpoints therefore are mixtures of allowed inputs and belong to F_j , proving faciality. All these tests are rational LPs of polynomial size.

Enumerate the nonempty faces H of C . For each tuple $(H_\ell)_{\ell \in L}$, retain feed (i, ℓ) only if $\lambda_i \in H_\ell$, outlet (ℓ, j) only if $H_\ell \subseteq F_j$, and bypass (i, j) only if $\lambda_i \in F_j$. Solve the resulting bounded min-cost network problem. Every resulting flow is physical by convexity. Conversely, given a feasible physical flow, choose for each active pool the smallest face of C containing its quality. The face property retains all its positive feeds; every receiving product face contains the chosen face, so all its positive outlets are retained. An inactive pool can choose any nonempty face. If an omitted arc has a positive lower bound, that branch is infeasible; it must not be removed without this check.

To bound enumeration, work in the rational affine hull of C . If $d > 0$, every facet contains d affinely independent input points. Enumerate all such subsets and test their supporting hyperplanes, giving at most $|I|^d$ facet candidates. A proper nonempty face is obtained by intersecting at most d facet hyperplanes with C : take a basis of its active facet normals. The other active hyperplanes follow because their normals are linear combinations and all equalities agree at a point of the face. Enumerating at most d facets gives at most $1 + |I|^{d^2}$ face descriptions, including duplicates and C itself. For $d = 0$, C is the only nonempty face. Membership and containment can be tested on the input points generating each face. Thus a polynomial number of tuples is inspected for fixed p, d , with polynomial-size rational data and polynomial-time network solves. $\square$

A particularly compact special case has all input attributes in $[0, 1]$ and product quality restrictions consisting only of coordinate upper bounds in $\{0, 1\}$. No additional nonredundant quality restrictions are imposed; lower flow bounds remain permitted. Put

$$D_{\ell k} = \sum_{i: \lambda_{ik} > 0} y_{i\ell}, \quad S_{\ell k} = \sum_{j: \bar{\mu}_{jk} = 0} v_{\ell j}.$$

Quality feasibility is equivalent to deletion of bypasses with $\lambda_{ik} > 0$ and $\bar{\mu}_{jk} = 0$ for some k , together with

$$D_{\ell k} = 0 \quad \text{or} \quad S_{\ell k} = 0 \quad (\ell \in L, k = 1, \dots, K). \quad (12)$$

Indeed, an upper bound zero is a sum of nonnegative quality masses bounded by zero. Every positive contributing pool must therefore have quality zero in that coordinate, which excludes every positive-quality feed. Conversely, the disjunction ensures this exclusion for every pool serving a strict product; upper bounds one accept every mixture. This proves both directions, including zero flow.

Choosing one side of each disjunction gives at most 2^{pK} ordinary network problems. An exact MILP uses one binary $b_{\ell k}$ per pair and, for a finite throughput upper bound U_ℓ , the rows

$$D_{\ell k} \leq U_\ell(1 - b_{\ell k}), \quad S_{\ell k} \leq U_\ell b_{\ell k}.$$

The formulation is exact, with no quality discretization. With integer flow bounds, an integral flow is a polynomial-size threshold certificate; (12) checks its quality feasibility directly. With rational bounds, a branch LP vertex provides a rational certificate instead. Thus this endpoint-specification threshold class belongs to **NP**, while the number of support branches can still grow exponentially.

2 Algebraic complexity and exact certificates

The quadratic equations of pooling give an existential-real upper bound (Section 1.2). We show that the network structure does not make exact decision easier: one attribute, binary source qualities, and small local degrees suffice for $\exists\mathbb{R}$ -completeness. A second construction puts the entire nonlinear interaction in one pool, using bypasses and a growing number of attributes. We then explain why these results coexist with NP certificates when suitable dimensions are fixed. All results concern the standard, finite-capacity model of Section 1.1.

Earlier hardness proofs are combinatorial. Alfaki and Haugland [4] reduce from maximum independent set with one pool and many attributes, and Haugland [24, Theorems 3–5] establishes NP -hardness with one upper-bounded attribute under several restrictions on inputs and products. Our reductions instead realize the arithmetic operations of an $\exists\mathbb{R}$ -complete problem through exact product quality specifications. To the best of our knowledge, the pooling literature reviewed here contains no previous $\exists\mathbb{R}$ -completeness result for either the bounded-degree, one-attribute class of Theorem 2.5 or the one-pool class of Theorem 2.7. These are the precise novelty claims; the source problem and its algebraic universality are established results.

2.1 The algebraic source problem and saturation

An ETR-INV instance consists of variables in $[1/2, 2]$ and equations of the forms

$$x = 1, \quad x + y = z, \quad xy = 1. \quad (13)$$

Variable names may repeat. Abrahamsen, Adamaszek, and Miltzow proved this problem $\exists\mathbb{R}$ -complete [2, Definition 5 and Theorem 7 of the arXiv version]. Replacing $x = 1$ by $x^2 = 1$ preserves its solutions because all variables are positive. We henceforth use only addition and inversion equations. Write n, m for their numbers of variables and equations, and $V(\Phi)$ for the solution set. An extra variable satisfying $u^2 = 1$ may be added to ensure $n \geq 1$ without changing satisfiability or introducing a choice in any solution.

The reductions need several prescribed positive totals, although the threshold instances have zero flow lower bounds. The following elementary device supplies them. A *marked* input, pool, or product is a node whose upper capacity is to be attained. Marking is part of the construction, not an extra type of physical constraint.

Lemma 2.1 (Saturation by a profit threshold). *Let F_I, F_L, F_J be marked nodes with upper capacities b_u . Assign an arc cost of minus one for each of the following groups containing it: arcs leaving a marked input, arcs leaving a marked pool, and arcs entering a marked product. With*

$$\zeta = \sum_{u \in F_I \cup F_L \cup F_J} b_u,$$

every capacitated flow has profit at most ζ , and it reaches this threshold exactly when all marked nodes attain their capacities. Every arc cost lies in $\{0, -1, -2\}$.

Proof. Summing the contributions of each group gives profit $\sum_{i \in F_I} s_i + \sum_{\ell \in F_L} T_\ell + \sum_{j \in F_J} t_j$. Each summand is bounded above by its capacity, so equality in their sum is equivalent to equality in every bound. A standard arc belongs to at most two groups, including when it is a bypass. Pool throughput is counted using outgoing arcs only. $\square$

2.2 One attribute without bypasses

Theorem 2.2. *The pooling threshold problem is $\exists\mathbb{R}$ -complete with all the following restrictions: one attribute; source qualities in $\{0, 1\}$; lower and upper product quality bounds; no bypasses; zero flow lower bounds; input out-degree at most two; pool and product in-degree at most two; and out-degree one at every unmarked pool. Costs lie in $\{0, -1, -2\}$. Node capacities and quality bounds have numerators and denominators polynomially bounded in the size of the source instance.*

We specify the complete construction and prove its correspondence with $V(\Phi)$. Every node introduced for a use of a gadget is fresh except for the named connection nodes. Throughout this construction, a *vacuous* product has quality interval $[0, 1]$. A *storage pool* is a marked pool of capacity B , with one quality-one incoming stream, its own unmarked quality-zero diluent input of capacity B , and its own unmarked vacuous slack product of capacity B . The pool has arcs from both inputs and an arc to its slack product. If its quality-one intake is a , saturation gives diluent intake $B - a$ and quality a/B . The positive integer B will be fixed after counting its other outlets.

For a variable v , create a marked quality-one input s_v of capacity $5/2$, feeding two storage pools $P_v, \bar{P}_v$. Let v also denote the flow on (s_v, P_v) , and put $\bar{v} = 5/2 - v$. At saturation the other arc carries $\bar{v}$, and the two pool qualities are v/B and $\bar{v}/B$. The notation anticipates, but does not yet assume, $v \in [1/2, 2]$.

Lemma 2.3 (Reciprocal emission). *Suppose a storage pool Q has quality-one intake a . Add a marked product t of capacity 2 and exact quality $1/(2B)$, with inputs from Q and a new relay pool H of capacity 2. A marked quality-zero input r of capacity 2 feeds H and has exactly one other outgoing arc, to be connected by the calling gadget. Pool H has only the outlet to t . In any saturated feasible flow,*

$$f_{Qt} = f_{r,\text{other}} = 1/a, \quad f_{rH} = f_{Ht} = 2 - 1/a, \quad a > 0. \quad (14)$$

In particular, $a \geq 1/2$. Conversely, for $a \in [1/2, 2]$ these flows satisfy the new source, relay, and product constraints.

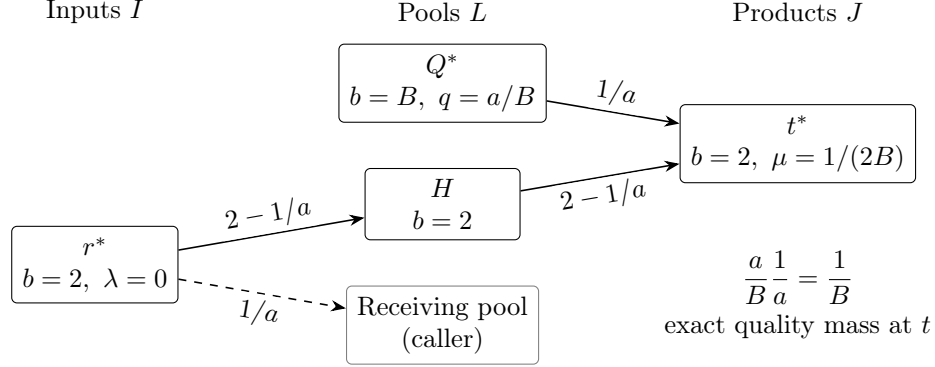
Proof. The quality mass from H is zero, including when its throughput is zero. The exact quality and saturated total at t give $(a/B)f_{Qt} = 2/(2B)$, so $af_{Qt} = 1$. Its capacity implies $1/a \leq 2$. Product conservation fixes f_{Ht} ; relay conservation fixes f_{rH} ; saturation of r fixes its remaining arc. Conversely, $1/a \in [1/2, 2]$ makes all four quantities in (14) nonnegative and at most 2, with the required totals and quality mass. This checks the gadget conditional on the specified flow on its connecting arc; the receiving node is checked by the calling gadget. $\square$

Thus a product's exact quality turns an intake a into a copy of $1/a$ on another input's outgoing arc. We also need to change that arc's source quality without changing its amount. To convert a delivered quality-zero amount $d \in [0, 2]$ to quality one, route it into a new pool H_0 of capacity 2, whose only outlet enters a marked vacuous product t_0 of capacity 2. A second pool H_1 of capacity 2 has only an outlet to t_0 and is fed by a new marked quality-one input r_1 of capacity 2; r_1 has one other outgoing arc. Conservation and saturation force

$$f_{H_0t_0} = d, \quad f_{r_1H_1} = f_{H_1t_0} = 2 - d, \quad f_{r_1,\text{other}} = d. \quad (15)$$

These nonnegative flows respect all new capacities, and the product is vacuous. This proves both directions of the conversion. The conversion uses no pool–pool arc: all physical arcs keep the standard input–pool–product form. Figure 1 shows the two constructions and their connecting source arcs.

(a) Reciprocal emission



(b) Conversion of a quality-zero amount d to quality one

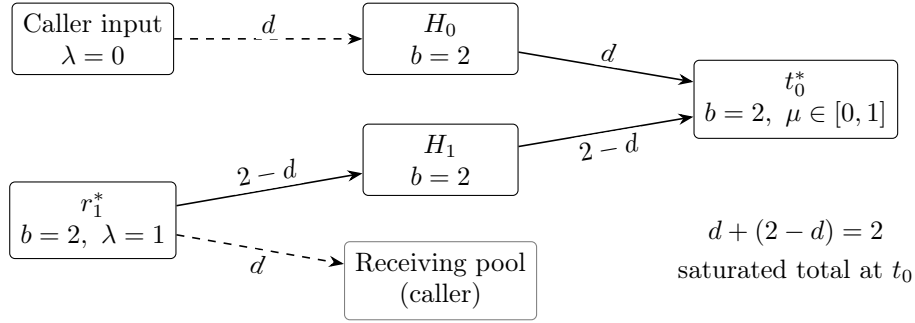


Figure 1: Physical arcs in reciprocal emission and quality conversion. A star marks a node whose upper capacity b is attained at the profit threshold. Here λ is an input quality, q a pool quality, and μ a product quality specification; $[0, 1]$ is vacuous. Arrow labels give the flows at saturation. Dashed arrows are physical input–pool arcs whose other endpoint is supplied by the calling gadget; they carry the logical continuation. No arrow joins two pools. The storage pool Q has quality-one intake a ; its feed and diluent arcs, slack outlet, and other uses are omitted. The receiving pools' other arcs and markings are determined by the caller.

Create two more storage pools $R_v, \bar{R}_v$ for every variable. Emit from P_v and $\bar{P}_v$, convert each delivered stream to quality one by (15), and use these as the quality-one feeds of R_v and $\bar{R}_v$. Lemma 2.3 applied to the first two pools gives $v, \bar{v} \geq 1/2$, hence

$$v \in [1/2, 2], \quad \begin{array}{c|cccc} Q & P_v & \bar{P}_v & R_v & \bar{R}_v \\ \hline Bq_Q & v & \bar{v} & 1/v & 1/\bar{v} \\ \text{amount delivered by an emission} & 1/v & 1/\bar{v} & v & \bar{v} \end{array}. \quad (16)$$

These initial emissions are created even for unused variables. Thus every variable has the required range independently of its equations.

For an inversion equation $xy = 1$, emit from $\bar{R}_y$ and route its quality-zero delivery $\bar{y}$ into a new pool H_c of capacity $5/2$. Its only outlet enters a marked product t of capacity $5/2$ and exact quality $2/(5B)$. Add the other product input (P_x, t) . Saturation and conservation give

$$f_{H_c t} = \bar{y}, \quad f_{P_x t} = 5/2 - \bar{y} = y.$$

The product's quality equation is then

$$(x/B)y = (5/2)2/(5B) = 1/B, \quad (17)$$

which is precisely $xy = 1$. Conversely, if the equation holds and its variables lie in $[1/2, 2]$, the stated flows are nonnegative and within the new capacities, and satisfy the exact quality. Nothing changes for $x = y$.

For an addition $x + y = z$, make separate emissions from R_x and R_y , delivering x, y at quality zero to a new pool H_+ of capacity 4. Even if $x = y$, the two emissions use distinct relay sources and products. The only outlet of H_+ enters a marked product t of capacity $5/2$ and exact quality $2/(5B)$. Add the second input $(\bar{R}_z, t)$. Since H_+ carries only quality zero, saturation and the product quality equation imply

$$f_{H_+ t} = x + y, \quad \frac{5/2 - x - y}{B\bar{z}} = \frac{1}{B}, \quad \text{hence } x + y = z. \quad (18)$$

Conversely, for a solution of the addition equation, these outlet amounts are $z \leq 2$ and $\bar{z} \in [1/2, 2]$. They sum to $5/2$, meet its quality pin, and satisfy the capacity 4 of H_+ . Coincidences between the result variable and a summand cause no ambiguity in this reasoning; such an equation has no solution in the positive variable range.

It remains to check that many gadget uses can coexist. Count, at each storage pool, its arcs to products other than its slack product, including direct arcs in (17) and (18). Let M be the maximum of these counts and set

$$B = 2M + 3. \quad (19)$$

Initial emissions give $M \geq 1$, while $M \leq 2m + 1$. In a solution of Φ , every one of these non-slack outlets carries at most 2. Every quality-one intake in (16) lies in $[1/2, 2]$. Thus each storage pool can receive its diluent $B - a \in [0, B]$ and send the residual amount

$$B - \sum_{t \neq \text{slack}} f_{Qt} \in [3, B] \quad (20)$$

to its slack product. Every other capacity was checked locally above. All product bounds not designated as exact are $[0, 1]$; they hold because every positive stream is a mixture of source qualities zero and one. At a zero-flow relay choose quality zero or its source quality; its quality mass is zero either way.

Proof of Theorem 2.2. Use Lemma 2.1 with precisely the nodes designated as marked in the construction. Any feasible flow meeting its threshold saturates these nodes. The variable feeds then belong to $[1/2, 2]$ by (16), and (17)–(18) recover every equation of Φ . Conversely, a solution of Φ determines all emissions, conversions, equation gadgets, diluents, and slack flows by the displayed formulas. The local checks and (20) prove feasibility; saturation gives profit ζ .

There are $O(n + m)$ nodes and arcs. Capacities belong to $\{2, 5/2, 4, B\}$, exact quality values are $1/(2B)$ and $2/(5B)$, and $B \leq 4m + 5$. The threshold is a sum of $O(n + m)$ capacities, hence has polynomial magnitude and encoding length. Each input has at most two outlets. Each pool has at most two inputs (the addition pool has two zero-quality inputs); each product has at most two inputs; each unmarked pool has one outlet. Costs have the values in Lemma 2.1. Constructing the graph and its rational data takes polynomial bit time. This proves hardness. Membership is the quadratic existential formulation in Section 1.2, with the linear threshold inequality appended. $\square$

The proof establishes a property stronger than a decision reduction.

Lemma 2.4 (Exact correspondence of flows). *The threshold-feasible arc flows of the construction are in bijection with $V(\Phi)$. Both maps use rational functions with rational coefficients whose denominators are nonzero on their domains.*

Proof. A solution determines every arc in the construction: variable feeds, their complements, reciprocal emissions, relay and conversion flows, equation flows, diluents, and slacks. The preceding proof shows that any threshold-feasible flow must have exactly these values. The inverse map reads the arcs (s_v, P_v) . The only divisions are by v and $5/2 - v$, both at least $1/2$. These maps are continuous, so they also give a homeomorphism of the two flow solution sets. The assertion concerns arc flows, not the irrelevant quality assigned to a zero-flow pool. $\square$

2.3 Fixed finite data and bounded degrees

Theorem 2.5. *The hardness in Theorem 2.2 persists when every node has in-degree at most two and out-degree at most three, all capacities belong to $\{2, 5/2, 4, 7\}$, and all quality bounds belong to $\{0, 1, 1/14, 2/35\}$. All instance numbers except the profit threshold belong to fixed finite sets; the threshold is a half-integer of magnitude $O(n + m)$. The flow correspondence of Lemma 2.4 still holds.*

Proof. Replace repeated uses of a storage pool by a chain of storage pools, all of capacity $B = 7$. A link consists of an emission, followed by the quality conversion (15), feeding the next storage pool. Each new pool has its own capacity-seven diluent and slack product. Starting with quality-one intake v , these links create intakes

$$v, 1/v, v, 1/v, \dots; \quad \bar{v}, 1/\bar{v}, \bar{v}, 1/\bar{v}, \dots$$

on two chains rooted at P_v and $\bar{P}_v$, which share only their initial variable input. These are chains of logical dependence; the physical graph remains three-layered and has no pool–pool arcs.

Allocate each equation’s requested storage-pool outlet to a distinct position of the required type on its chain. The requests for an inversion are one P_x -type outlet and one $\bar{R}_y$ -type emission; those for an addition are two R -type emissions and one $\bar{R}_z$ -type outlet. At a chain position permit at most one equation outlet, besides its possible link and slack outlet. Process requests sequentially. If the current position has the wrong type, one new link reaches the right type. If it already served a request of the right type, two new links reach a fresh position of that type. Thus each request needs at most two links. Advance each root at least once, even if its variable has no equations; this

enforces $v, \bar{v} \geq 1/2$ exactly as before. There are at most $3m$ requests and at most $2n$ additional root links, so at most $6m + 2n$ links suffice.

In a saturated flow, applying Lemma 2.3 and (15) successively along a chain proves the displayed alternation of intakes. The equation gadgets therefore recover the same equations, even with repeated variable names. Conversely, for a solution of Φ , all chain intakes and all non-slack outlets are in $[1/2, 2]$. Each storage pool has at most two non-slack outlets, so its slack is at least $7 - 4 = 3$ and its diluent is at least $7 - 2 = 5$. These choices satisfy every capacity. Each link has constant size, and the rest of the construction is unchanged, giving $O(n + m)$ nodes and arcs.

For completeness, the local degrees are as follows. Variable, emission, and conversion inputs have out-degree two; diluents have out-degree one. Storage pools have in-degree two and out-degree at most three. Relay and conversion pools have in-degree one and out-degree one; an addition pool has in-degree two and out-degree one. Every non-slack product has in-degree two and every slack product has in-degree one. Substitution of $B = 7$ gives the stated finite sets of capacities and quality bounds. Lemma 2.1 gives the cost set and the threshold bound. Every new link is uniquely determined by its initial intake, so the rational bijection is preserved. $\square$

Corollary 2.6. *The preceding threshold classes are $\exists\mathbb{R}$ -complete also with two attributes and upper quality bounds only, while keeping binary source qualities and the same degree restrictions. If exact positive node throughput contracts are allowed, the corresponding feasibility problems are $\exists\mathbb{R}$ -complete. In the bounded-degree feasibility construction all numerical data belong to fixed finite sets.*

Proof. For a source quality $\lambda_i \in \{0, 1\}$ add the complementary attribute $1 - \lambda_i$. At a positive-throughput pool the complementary quality is $1 - q_\ell$. At a product the complementary quality mass is $t_j - m_j$, including at zero flow. Thus a lower bound $m_j \geq \underline{\mu}_j t_j$ becomes the upper bound $t_j - m_j \leq (1 - \underline{\mu}_j)t_j$. Zero-flow pools contribute zero and can receive complementary qualities consistently. No flow or degree changes, and the set of quality constants remains finite (with their complements added). For feasibility, impose lower node bounds equal to the upper capacities precisely at the marked nodes. This implements the same saturated solution set without an objective or a threshold. Membership again follows from the quadratic formulation. $\square$

Allowing bypasses or generalized arcs contains these networks as special cases. In contrast, their zero-lower-bound feasibility problem is trivial because zero flow is feasible. The threshold construction is not a feasibility-hardness claim for that zero-lower-bound class. Nor does complementing an attribute establish $\exists\mathbb{R}$ -hardness with *one* upper-bounded attribute. Exact pins supply both directions of (17) and (18); retaining only an upper inequality loses that implication. This one-attribute upper-bound-only $\exists\mathbb{R}$ boundary remains unresolved here.

2.4 One pool with bypasses

Theorem 2.7. *The pooling threshold problem with a single pool and bypasses is $\exists\mathbb{R}$ -complete. Hardness holds with source qualities in $\{0, 1\}$, zero flow lower bounds, input out-degree at most two, product in-degree at most two, costs in $\{0, -1, -2\}$, and rational numbers of polynomial magnitude and encoding length. The attribute count and pool degrees grow with the instance. Lower and upper quality bounds may be replaced by upper bounds on twice as many attributes. The threshold-feasible arc flows are again rationally equivalent to the source solution set.*

Proof. Start with ETR-INV after replacing $x = 1$ by $x^2 = 1$. Replace each repeated-summand addition $x + x = z$ by

$$xu = 1, \quad ux' = 1, \quad x + x' = z,$$

using fresh variables u, x' . These uniquely equal $1/x, x$, within $[1/2, 2]$, so the replacement preserves solutions by rational maps and leaves no addition with identical summand names. The construction below uses this normalized instance.

There is one pool p . Each variable v has an unmarked input s_v of capacity 2 with its only arc to p . For each auxiliary index h specified below, add a marked input a_h of capacity 2, with arcs to p and to its own marked product j_h of capacity 2; the only other input to j_h is (p, j_h) . Let N be the total number of variable and auxiliary inputs and set $B = 4N$. Mark the pool with capacity B . Add an unmarked filler input of capacity B , whose only arc goes to p , and an unmarked slack product of capacity B , whose only incoming arc is from p .

For every non-filler input s introduce an attribute A_s , with quality one at s and zero at every other input. For every addition $x + y = z$ introduce an attribute A_{x+y} supported with quality one on s_x, s_y only. The normalization makes these binary qualities even when the original equation had repeated summands. The filler has zero quality in every attribute. All unspecified product bounds are $[0, 1]$. A *pin* (j_h, A, κ) means exact quality κ for attribute A at j_h .

An inversion $xy = 1$ uses two fresh indices h_1, h_2 and the three pins

$$(j_{h_1}, A_{s_x}, 1/(2B)), \quad (j_{h_2}, A_{a_{h_1}}, 1/(2B)), \quad (j_{h_2}, A_{s_y}, 1/(2B)). \quad (21)$$

An addition $x + y = z$ uses one index h and the two pins

$$(j_h, A_{s_z}, 1/B), \quad (j_h, A_{x+y}, 1/B). \quad (22)$$

For every variable not appearing in either role of an inversion, add one more index h and the range pin $(j_h, A_{s_v}, 1/(2B))$. This includes unused variables and variables appearing only in additions.

Choose costs by Lemma 2.1. If there are H auxiliaries, the profit threshold is $\zeta = B + 4H$. For clarity, the costs on (s_v, p) and the filler arc are zero, those on (a_h, p) and the slack outlet are -1 , and those on (a_h, j_h) and (p, j_h) are -2 . In any flow meeting the threshold, put $X_v = f_{s_v p}$ and $\tau_h = f_{a_h p}$. Source and product saturation give

$$f_{a_h j_h} = 2 - \tau_h, \quad f_{p j_h} = \tau_h. \quad (23)$$

The pool has throughput B , so $q_p^{A_{s_v}} = X_v/B$, $q_p^{A_{a_h}} = \tau_h/B$, and $q_p^{A_{x+y}} = (X_x + X_y)/B$. At each pin the bypass input a_h has quality zero in the pinned attribute: in (21) the auxiliary attribute is that of a_{h_1} at the distinct product j_{h_2} . Consequently its pin equation is

$$q_p^A \tau_h = 2\kappa. \quad (24)$$

The three inversion pins yield

$$X_x \tau_{h_1} = 1, \quad \tau_{h_1} \tau_{h_2} = 1, \quad X_y \tau_{h_2} = 1.$$

They force $\tau_{h_1} = X_y$, $\tau_{h_2} = 1/X_y$, and $X_x X_y = 1$. Since every variable and auxiliary intake lies in $[0, 2]$, these identities also force $X_x, X_y \in [1/2, 2]$, including when $x = y$. The addition pins give $X_z \tau_h = (X_x + X_y) \tau_h = 2$, so $\tau_h > 0$ and $X_x + X_y = X_z$. Each range pin gives $X_v \tau_h = 1$, hence $X_v \geq 1/2$. Thus all variables have the intended range and solve the source instance.

Conversely, given a source solution x , set $X_v = x_v$; set $\tau_{h_1} = x_y$ and $\tau_{h_2} = 1/x_y$ for an inversion; set $\tau_h = 2/x_z$ for an addition; and set $\tau_h = 1/x_v$ for a range pin. In an addition, $x_z = x_x + x_y \in [1, 2]$, so $2/x_z \in [1, 2]$. All auxiliary intakes therefore lie in $[1/2, 2]$. Use (23), filler intake $B - \sum_v X_v - \sum_h \tau_h$, and slack outlet $B - \sum_h \tau_h$. The sum of all non-filler intakes is at most $2N = B/2$, so filler and slack are both in $[B/2, B]$. The pool conserves flow and all marked totals are attained. Equations (24)

hold by substitution. Every unpinned attribute has average in $[0, 1]$, because all source qualities are binary; its bounds therefore hold, including at the slack product. This proves threshold feasibility.

Each source solution fixes every flow just listed, and every saturated flow is forced to have them. Along with the normalization this is the claimed rational bijection. The graph and attribute count have size $O(n + m)$; a dense listing of source qualities and product bounds has size $O((n + m)^2)$, still polynomial. The only nonconstant numerical values are B , $1/B$, $1/(2B)$, and ζ , all with polynomial encoding length. External degrees follow directly from the arc list. Membership in $\exists\mathbb{R}$ follows from the quadratic formulation. Finally, with bypasses as well, complementary quality mass is exactly $t_j - m_j$; the proof of Corollary 2.6 converts the lower bounds to upper bounds on binary complementary attributes. $\square$

As before, imposing the marked node totals as exact contracts gives $\exists\mathbb{R}$ -complete one-pool feasibility. This construction makes no fixed-attribute or fixed-pool-degree claim. Its many independent attribute coordinates are essential to the proved scope: the fixed dimension certificate result below places one pool with a fixed attribute count in NP.

2.5 Algebraic degree and the meaning of exactness

Theorem 2.8. *For every irrational real algebraic number α , there is a rational pooling instance and a rational profit threshold with exactly one threshold-feasible arc flow. A designated arc has value $(\alpha - b)/a$ for rationals $a \neq 0, b$, and the field generated by all its flow values is $\mathbb{Q}(\alpha)$. The instance can be chosen either with the finite-data and degree restrictions of Theorem 2.5, or with one pool and bypasses as in Theorem 2.7. In particular, rational instances can have only irrational optimal flows and can require arbitrarily high algebraic degree.*

Proof. Choose a nonzero integer polynomial p vanishing at α and rational $c < \alpha < d$ such that α is its only root in $[c, d]$. The formula $p(x) = 0$, $c \leq x \leq d$ defines the compact singleton $\{\alpha\}$. We use the compact *basic closed* case of the ETR-INV transformations of Abrahamsen and Miltzow [1, Lemmas C–G, underlying Theorem 1]. Here “basic closed” means a conjunction of polynomial equations and weak inequalities; no elimination of strict inequalities or disjunctions is needed. To spell out the relevant preservation property, introduce slacks $s = x - c \geq 0$, $t = d - x \geq 0$ by equations, and evaluate the polynomials through addition and multiplication gates. Each new value is uniquely determined by x . Clear rational coefficients to obtain integer polynomial equations; the added slacks and gates retain a compact singleton solution set. This is an ETR-COMPACT instance in the sense required by their Lemma C, so the compactification step of their Lemma B, which need not preserve the solution set, is not used. Their subsequent transformations scale all these coordinates into a small interval, shift the computational coordinates near one, replace multiplication by squaring, and replace squaring by addition and reciprocal gates. All introduced gate values are uniquely determined rational functions; the chosen intervals keep their denominators away from zero. Each reverse transformation recovers the previous equations. Original coordinates are retained up to rational shifts and nonzero scalings. Thus the resulting ETR-INV set is a singleton $\{x^*\}$ with coordinates in $\mathbb{Q}(\alpha)$ and a replacement coordinate u satisfying $au + b = \alpha$ for rational $a \neq 0, b$. The auxiliary interval promises already hold for all solutions and can be discarded. In particular, $u = (\alpha - b)/a$ has the same degree as α . Only this basic-closed singleton consequence of the transformations is used here. It is also the algebraic-field consequence illustrated by their Corollary 2; the additional step here is its realization by the restricted physical pooling networks above.

Apply either pooling construction. Its rational bijection produces exactly one threshold-feasible flow, with all coordinates in $\mathbb{Q}(\alpha)$, and reads u on a variable feed arc. The chain refinement retains

the initial variable feed; the one-pool normalization only adds uniquely determined variables. Thus the field generated by the flows contains $au + b = \alpha$ and is exactly $\mathbb{Q}(\alpha)$. No subfield of $\mathbb{R}$ omitting α can contain all flow values. The threshold is an attained global upper bound on profit by Lemma 2.1, so this unique threshold flow is the unique optimal arc flow. Real algebraic numbers have unbounded degree, for example the positive roots of $X^d - 2$ for integers $d \geq 2$, whose minimal polynomials are irreducible by Eisenstein's criterion at 2. $\square$

The result concerns required *flow coordinates*; the forcing objective itself has the rational optimum ζ . Irrational optimal values already occur in much smaller networks, with one upper quality bound and every local degree at most two. The following exact example also illustrates why writing a concentration without its throughput denominator changes the problem.

Example 2.9 (A quadratic irrational optimum). There are inputs A, B, C , pools P, Q , and products j_1, j_2 . Input qualities are $3/4, 1/2, 0$ and their capacities are $1, 2, 1$, respectively. Pool capacities are $2, 1$. Each product has capacity one; its sole quality upper bound is $1/4$ at j_1 and 1 at j_2 . All lower flow bounds are zero. The complete arc and cost list is

arc	AP	BP	CQ	Pj_2	Pj_1	Qj_1
cost	1	2	1	-4	-3	-1

Its minimum cost is $-(5 + \sqrt{3})/2$.

Proof. Put $a = f_{AP}$, $b = f_{BP}$, $t = a + b$, $z = f_{Pj_1}$, and $c = f_{Qj_1} = f_{CQ}$. The other pool outlet is $t - z$ and the cost is

$$a + 2b + c - 4(t - z) - 3z - c = -3a - 2b + z. \quad (25)$$

For $t > 0$, the quality constraint at j_1 is $(3a + 2b)z \leq t(z + c)$. Its capacity gives $z + c \leq 1$, while the capacity at j_2 gives $z \geq \max\{0, t - 1\}$. If (a, b, z, c) is feasible, replacing z by $z_0 = \max\{0, t - 1\}$ and c by $1 - z_0$ preserves feasibility: $0 \leq z_0 \leq z \leq 1$, both product capacities hold, the new clean flow is in $[0, 1]$, and $(3a + 2b)z_0 \leq (3a + 2b)z \leq t(z + c) \leq t$. It weakly improves (25). At $t = 0$ the mixing flow is zero and its cost is zero, with no division required.

For $t \leq 1$, the best profit is at most $3a + 2b \leq 3t \leq 3$, attained at $a = 1, b = 0$. For $t > 1$, the normalized choices give profit $a + t + 1$ and quality condition $(a + 2t)(t - 1) \leq t$. Thus

$$0 \leq a \leq \min\{1, h(t)\}, \quad h(t) = \frac{t}{t - 1} - 2t.$$

Because $1 < t \leq 2$, $a \leq 1$ ensures $a \leq t$, and $b = t - a \leq 2$; there are no further intake restrictions. Feasibility is equivalent to $h(t) \geq 0$, or $t \leq 3/2$. The function h is strictly decreasing and equals one at $t_0 = (1 + \sqrt{3})/2$. For $1 < t \leq t_0$ the best profit is $t + 2$, strictly increasing. For $t_0 \leq t \leq 3/2$ it is $2 - t + 1/(t - 1)$, strictly decreasing. The global maximum is therefore $(5 + \sqrt{3})/2$, attained with $a = 1$, $b = z = (\sqrt{3} - 1)/2$, and $c = 1 - b$. In this family the quality condition at j_1 is $b(3/4 + b/2)/(1 + b) \leq 1/4$, equivalently $2b^2 + 2b - 1 \leq 0$. All displayed flows respect their original capacities. $\square$

Since $\text{NP} \subseteq \exists\mathbb{R}$, completeness implies that general pooling threshold decision lies in NP only if $\text{NP} = \exists\mathbb{R}$. This obstruction concerns all polynomial-size certificates and comes from completeness, not from the irrational example: a certificate need not list rational feasible flows. The containment $\exists\mathbb{R} \subseteq \text{PSPACE}$ gives a polynomial-space upper bound [1, Section 1.1].

There is also a precise distinction between exact feasibility and rational approximate descriptions. For a bounded rational standard instance with an attained optimum, choose pool qualities in the

input quality box as in Section 1.2. All lifted coordinates, coefficients, and first derivatives of the degree-two constraint polynomials on a unit enlargement of this box have magnitude at most $2^{\text{poly}(N)}$, where N is input length. This follows by bounding each coordinate and coefficient by its rational encoding length and summing the polynomially many monomials. Consequently a bound $L = 2^{\text{poly}(N)}$ controls the change in every constraint residual and in the objective when all coordinates change by at most δ . Round an optimal lifted point to a rational grid of mesh $\delta \leq \epsilon/L$. The rounded description has encoding length $\text{poly}(N + \log(1/\epsilon))$, its objective differs from the optimum by at most ϵ , and every equality residual and inequality violation is at most ϵ . This is an existence statement for absolute residuals in the specified polynomial formulation, allowing *all* constraints to have that tolerance. It neither constructs such a point in polynomial time nor promises an exactly feasible rational approximation. Rescaling a constraint changes this residual metric.

2.6 Linear fibers with a fixed number of parameters

We now give the certificate argument used for the ordinary NP-complete pooling subclasses later in the paper. Its nonlinear dimension is fixed, but the remaining linear program can have arbitrarily many coupled variables and rows. The ingredients are linear programming vertex theory, Cramer's rule, and fixed-dimensional real-algebraic decision; no novelty is claimed for these ingredients.

Theorem 2.10 (A basis-index certificate). *Fix r . Consider deciding whether*

$$\eta \in D \subseteq \mathbb{R}^r, \quad A(\eta)x \leq b(\eta), \quad x \in \mathbb{R}^n, \quad (26)$$

where D has an explicit rational polynomial description and the entries of A, b are explicitly represented rational polynomials. Assume all defining and entry polynomial degrees are polynomially bounded in input length and every nonempty fiber in x is bounded. Then feasibility is in NP. A threshold on an objective linear in x , with such polynomial parameter coefficients, can be included among the rows.

Proof. Suppose first $n > 0$ and (26) is feasible at η^* . Its fiber is a nonempty compact polyhedron and therefore has a vertex x^* . One elementary justification is to minimize its coordinates successively, retaining each minimizing face; after n steps the retained point is unique, hence extreme. The normals of active inequalities at that vertex span $\mathbb{R}^n$. Otherwise a nonzero vector perpendicular to all active normals would give a sufficiently short feasible segment in both directions, since the finitely many inactive inequalities have positive slack. This contradicts extremality, even if the polyhedron is lower-dimensional. Choose n linearly independent active rows.

The certificate consists only of their row indices. Let $C(\eta)$ be the resulting square coefficient matrix and $d(\eta)$ its right-hand vector. Form the polynomial determinant and Cramer numerators

$$\Delta(\eta) = \det C(\eta), \quad U(\eta) = \text{adj}(C(\eta))d(\eta).$$

Accept the certificate exactly when the following system in the fixed r real variables is feasible:

$$\eta \in D, \quad \Delta(\eta)^2 > 0, \quad [A_j(\eta)U(\eta) - b_j(\eta)\Delta(\eta)]\Delta(\eta) \leq 0 \quad \text{for every row } j. \quad (27)$$

At $\Delta \neq 0$, this is exactly the original row inequality for $x = U/\Delta$ multiplied by the positive number Δ^2 . It is valid for either determinant sign and excludes the singular locus. Acceptance therefore produces a feasible point. Conversely, the selected active basis at (η^*, x^*) has nonzero determinant and satisfies (27), so some certificate is accepted.

We verify the bit bound as well as the algebra. Let d_0 bound the degrees of entries in A, b . Determinants and Cramer numerators have degree at most nd_0 , and the row polynomials in (27) have

degree at most $(2n + 1)d_0$. For fixed r , dense coefficient arrays at these degrees have polynomially many entries. Clear input rational denominators first; their product has polynomial encoding length. If entry coefficients have magnitude at most H and each entry has at most M_0 monomials, each determinant coefficient is a sum of at most $n!M_0^n$ products of n coefficients. Its bit length is thus $O(n \log H + n \log M_0 + n \log(n + 1))$, polynomial in the input. The same bound applies to Cramer numerators and, with an additional polynomial factor, to the verification rows.

These arrays can actually be computed in polynomial time, rather than merely having polynomial size. Evaluate each determinant on the tensor grid $\{0, \dots, nd_0\}^r$, compute its rational determinants by exact linear algebra, and interpolate one variable at a time. Both the grid size and the evaluated integer bit lengths are polynomial. Vandermonde interpolation uses rational linear algebra on polynomial-size integer matrices with polynomial bit lengths, so its intermediate and output encoding lengths are polynomial. The same procedure for the n column-replacement determinants computes U . Constant polynomials cause no difficulty.

Finally, fixed-dimensional real-algebraic decision checks (27) in polynomial bit time. In particular, Basu et al. [7, Theorem 1.3.3 and the bit-size convention in Section 1.3] give a bound polynomial in the number of polynomials, their degree, coefficient length, and formula length when the dimension r is fixed. Strict inequalities such as $\Delta^2 > 0$ are included in that sign-condition formulation. The certificate length is $O(n \log(M + 1))$ for M rows. When $n = 0$, use the same fixed-dimensional decision procedure directly, without a basis. The objective threshold is included before taking a vertex, so intersecting the original fiber with its halfspace does not affect boundedness or the argument. $\square$

Corollary 2.11 (Pooling certificate scopes). *For standard pooling with rational data and finite flow upper bounds, feasibility and the linear-cost threshold problem belong to NP whenever any one of the following quantities is bounded by a fixed constant:*

- (i) *the number of pools times the attribute count;*
- (ii) *the number of pools times the affine rank of the input quality vectors;*
- (iii) *the number of pools times the number of products.*

These statements permit arbitrary bypass graphs, flow lower bounds, exact contracts, and arbitrary rational linear arc costs. They also give NP membership for fixed pool and input counts.

Proof. For (i), use all pool qualities as the parameter vector η . With these fixed, flow conservation, blending, product quality constraints, all bounds, and the cost threshold are linear in the arc flows. Their coefficient polynomials have degree at most one. Each active pool quality lies in the coordinatewise input-quality box, and inactive pools may be assigned qualities in that box without changing feasibility. The parameter domain is therefore a rational box of fixed dimension, and the flow fibers are bounded.

For (ii), use the rational affine coordinates $\lambda_i = \lambda_0 + Ba_i$ of Section 1.2 and the pool coordinates η_ℓ from (7). At fixed coordinates, that equation and every original product specification remain linear in flows. The number of parameters is the pool count times the affine rank, regardless of the number of original attributes. The affine transformation has polynomial encoding length; coordinatewise minima and maxima of the a_i bound the active mixtures, and inactive coordinates can be chosen in the same box. At rank zero this is a linear problem with no parameters. Empty-input instances have only zero flow and are checked directly. Since affine rank is at most the input count minus one, fixed pool and input counts are a special case.

For (iii), parameterize each pool by its outlet fractions $\theta_{\ell j} \geq 0$, with sum one over its allowed outlets. Remove a pool with no outlet only after checking its pool and incident arc lower bounds against zero; preserve all external-node constraints. For a remaining pool let $T_\ell = \sum_i y_{i\ell}$. Substitute

$$v_{\ell j} = \theta_{\ell j} T_\ell, \quad \text{mass of attribute } k \text{ from } \ell \text{ to } j = \theta_{\ell j} \sum_i \lambda_{ik} y_{i\ell}. \quad (28)$$

The remaining variables are intake and bypass flows. With fractions fixed, input totals, pool totals, product totals, product quality masses, feed and outlet bounds, and the threshold are all linear expressions in these variables. The parameter domain is a product of simplices with at most $|L||J|$ coordinates, and its coefficient degrees are at most one. A positive-throughput physical pool gives these fractions by dividing its outlets by T_ℓ . Conversely, for a solution of the substituted system with $T_\ell > 0$, assign $q_{\ell k} = \sum_i \lambda_{ik} y_{i\ell} / T_\ell$; (28) then gives exactly its physical mass deliveries. When $T_\ell = 0$, all intakes and reconstructed outlets are zero and any split vector is valid. A pool without outlets was already checked separately. Finite intake and bypass bounds make the fibers bounded. Theorem 2.10 proves every case. $\square$

There is one useful certificate scope independent of these products of dimensions.

Proposition 2.12 (One pool without bypasses). *One-pool pooling without bypasses belongs to NP even with unrestricted input, product, and attribute counts, and with the finite bounds and linear objectives allowed above.*

Proof. Guess a subset J^+ of products allowed to receive positive pool flow. Set all other outlets to zero. Write $T = \sum_i y_i$ and impose the linear inequalities

$$\underline{\mu}_{jk} T \leq \sum_i \lambda_{ik} y_i \leq \bar{\mu}_{jk} T \quad (j \in J^+, k = 1, \dots, K),$$

along with all original linear flow bounds, conservation, and the threshold. This is a rational linear program decidable in polynomial bit time. For a physical solution choose J^+ to be its positive outlets; when $T > 0$, each such product receives the pool's common quality, giving exactly these inequalities. If $T = 0$, all flows are zero and the empty choice applies. Conversely, any solution of the linear program is physical: assign the intake average when $T > 0$. Every positive outlet belongs to J^+ and meets its specification; zero outlets impose no quality condition. At $T = 0$, choose an arbitrary pool quality. This proves soundness even though the guessed set need not have every one of its outlets positive. The argument is the active-product linearization in Haugland's proof of Theorem 2 [24], with lower quality bounds and flow contracts included explicitly. The one-pool, no-bypass model with both quality directions also has the earlier MILP representation of Dey and Gupte [15, electronic companion, Appendix A]. $\square$

These certificate results supply the NP upper bounds for the two-pool, two-product hardness construction and the one-pool fixed-quality bypass constructions below. One physical quality with lower and upper specifications, its two-attribute upper-bound encoding, and the later single-upper-quality and constant-physical-data constructions all fit (i); strong hardness requires their separate reductions and does not follow from this certificate lemma. Enumerating bases gives no polynomial algorithm, since there can be exponentially many candidate row sets. Nor do these results give a general NP certificate when both the pool count and the relevant quality or outlet dimension grow. The product-fraction proof uses the standard three-layer model and makes no claim for pool-pool networks.

3 Hardness under degree, size, and data restrictions

This section shows that three restrictions that might suggest tractability do not suffice: degree two at every layer, two pools with two products, and a single scalar mixing variable with bounded local data. The restrictions must be read together. A network with few mixing arcs can still have many external sources linked by bypasses, and small physical coefficients can encode precise relations through a large network.

Theorem 3.3 combines scalar qualities with all four layer degrees equal to two, restrictions that Haugland [24, Sections 4–6] treats separately, and answers the degree questions of Boland et al. [10, author manuscript, Section 4, p. 8]; section 3.1 develops the two pure modes and section 3.2 imposes all degree bounds. Theorem 3.9 answers the product-bounded case of the two-pool question of Haugland [24, Section 6, p. 214] and Boland et al. Bypasses change the one-pool boundary [26, Section 4.5, p. 607]: theorem 3.19 uses many bypass sources to encode a linear system while the pool has two actual feeds. Broad one-pool capacity hardness was already asserted by Baltean-Lugojan and Misener [6, Remark 4.6], whose polynomial result assumes unbounded feed availability and pool capacity and fixed product demands; we claim no priority for general one-pool hardness or for the positive-product source problem. To the best of our knowledge, neither the finite-data restriction nor the simultaneous scalar all-degree-two restriction was established in the cited pooling complexity literature.

After the degree constructions, two products realize a positive-product objective, copy networks replace its attribute rows by physical bypass constraints, an explicit penalty removes contracts, and binary circuits with a saturation threshold give the finite-data and five-exception results. All networks are acyclic and have no pool-to-pool arcs. Degrees count physical arcs, including bypasses where present.

Approximation ratios refer to profit, the negative of cost; a ratio for a possibly negative minimum cost would not express these guarantees. The NP membership statements for fixed mixing parameters invoke corollary 2.11 and do not assume rational optimal flows. For input qualities in $[0, 1]$ and only upper product quality bounds in $\{0, 1\}$, the disjunctive certificate (12) gives NP membership for threshold decision even with many pools.

3.1 Orientation and the first degree restrictions

A capacitated orientation of an undirected graph assigns each edge to one of its endpoints. The load at a vertex is the sum of the weights assigned to it. We refine the satisfiability construction of Asahiro et al. [5, Section 5] with private weight-two triangles; an occurrence restriction and a bipartite subdivision then give maximum degree three.

Lemma 3.1. *Deciding whether every vertex load can be at most its capacity is NP-complete on simple bipartite graphs of maximum degree three, with all weights and capacities in $\{1, 2\}$. All vertices of degree three can be required to belong to one side of the bipartition.*

Proof. Start with a 3-SAT instance whose clauses each use three distinct variables, a standard NP-complete restriction. For a variable occurring $r \geq 2$ times, replace its occurrences by distinct variables $x_1, \dots, x_r$ and add $x_i \vee \neg x_{i+1}$ cyclically. These clauses force all copies equal: if one is false, its successor must be false, and the cycle propagates this value. A variable occurring once needs no cycle. Each resulting literal occurs at most twice, since a copy occurs once in an original clause, once positively in a cycle clause, and once negatively in a cycle clause. The clauses have two or three literals on distinct variables. Thus satisfiability remains NP-hard under these restrictions.

Make two vertices for each variable's literals and join them by a weight-two edge. Make a vertex for each clause and join it by weight-one edges to its literals. A two-literal clause also has a weight-one edge to a vertex of a new private triangle whose three edges have weight two. Every clause vertex now has degree three. Literal vertices have degree at most three and triangle vertices have degree at most three. Set every capacity to two. The graph is simple.

Given a satisfying assignment, assign each variable edge to its false literal. Orient each triangle cyclically, so every triangle vertex has load two. For each clause choose one true literal, assign its clause edge to that literal, and assign the other two clause edges to the clause. A true literal receives at most two unit edges. Thus every load is at most two. Conversely, the weight-two edges in a triangle must enter three different vertices, filling all three capacities. Its clause edge must therefore enter the clause. The endpoint receiving a variable edge is called false; it too cannot receive a clause edge. Each clause must assign at least one of its three unit edges away from itself, necessarily to a true literal. This gives a satisfying assignment.

Finally subdivide every edge $e = uv$ once, giving both half-edges weight w_e and the new vertex m_e capacity w_e . An edge assigned to v extends by assigning um_e to m_e and m_ev to v . Conversely, both half-edges cannot enter m_e . If one does, assign the original edge to the other original endpoint; if neither does, choose either endpoint. No original load increases. The subdivision is bipartite, keeps original degrees, and gives all new vertices degree two. Membership in NP follows by listing an orientation. $\square$

Theorem 3.2. *One-attribute pooling threshold decision without bypasses is strongly NP-complete under either of these restrictions: every input has out-degree one and every product has in-degree at most three; or every product has in-degree one and every input has out-degree at most three. In both cases every pool has exactly two inputs and two products, capacities lie in $\{1, 2\}$, input qualities and product upper bounds lie in $\{0, 1\}$, and costs lie in $\{-2, -1, 0, 1\}$. Only upper product quality specifications are imposed, and there are no positive lower flow bounds.*

Proof. Let $(U \cup W, E)$ be an instance of lemma 3.1, with capacities T_v and weights w_e . In the first construction, edge $e = uv$ has a private quality-zero input a_e , a private quality-one input d_e , and a pool ℓ_e , all of capacity w_e . Vertex v is a product of capacity T_v ; its upper quality is zero for $v \in U$ and one for $v \in W$. The four arcs at ℓ_e , in order from the clean input, from the dirty input, to u , and to w , have costs $1, 0, -2, -1$.

Write these flows as a, d, s, b , with throughput $t = a + d = s + b \leq w_e$. Their cost is

$$a - 2s - b = a - s - t. \quad (29)$$

If $s > 0$, the strict product's upper bound gives zero pool quality: all terms in its quality-mass inequality are nonnegative. Hence $d = 0$ and $a = t \geq s$. If $s = 0$, $a \geq s$ is immediate. The edge cost is therefore at least $-w_e$. Equality in the total lower bound $-\sum_e w_e$ forces $t = w_e$ and $a = s$ at every pool. If $s > 0$, the pool uses $a = s = w_e, d = b = 0$; otherwise it uses $d = b = w_e, a = s = 0$. Assign e to the corresponding product endpoint. Its load is exactly that product's inflow and respects T_v . Conversely a feasible orientation routes w_e in the corresponding pure mode and attains the bound.

For the second construction, each graph vertex is instead an input of capacity T_v , quality zero on U and one on W . Every edge has a capacity- w_e pool and two private capacity- w_e products, with bounds zero and one. Use the same four arc costs. Formula (29) and its equality conditions are unchanged. Now the load at an endpoint is the total flow taken from its input. The same orientation equivalence proves hardness. The graph degrees give the asserted physical degrees, and all data are bounded constants apart from the linear-size threshold. Membership in NP follows from the endpoint disjunction (12). $\square$

The proofs extend in two ways. First, the pool capacity can be omitted. In the private-input construction, $s > 0$ forces $d = 0$, so $\text{cost} = -s \geq -w_e$; when $s = 0$, $\text{cost} = -d \geq -w_e$. Equality in the first case sends w_e to the strict endpoint; in the second it takes w_e from the dirty input and sends at least w_e to the lax endpoint. Thus assigning the edge accordingly never exceeds its product's load. Conversely the earlier orientation flows remain feasible. In the private-product construction, the same cost expressions apply; $s \leq w_e$ and, when $s = 0$, $d \leq b \leq w_e$. At equality the selected input supplies at least w_e . This again gives the orientation. Additional cost-neutral cross-routing may now occur, so equality need not make every physical flow integral.

Second, apply either construction to two vertices joined by parallel edges of positive integer weights $s_1, \dots, s_n$, with both capacities B and $\sum s_i = 2B$. A feasible orientation is precisely a partition of the weights into equal sums. This proves ordinary NP-hardness with two products and private inputs, or two inputs and private products, respectively; it does not establish strong hardness for these two-node specializations. Parallel edges in the source create distinct pools, not parallel physical arcs.

At the other end of the degree boundary, Haugland [24, Proposition 3] shows that if every pool has in-degree one or out-degree one, a network without bypasses has an LP formulation. An in-degree-one pool has fixed quality. For any remaining pool, which has one outlet, substitute its incoming quality mass directly in the unique product's quality inequality. Keep its linear throughput capacity and mass balance. A feasible LP point recovers the pool quality by the usual mixture at positive throughput and arbitrarily at zero throughput. This works coordinate by coordinate. Thus $(2, 2)$ is the smallest common pool-degree pattern that permits the hardness just proved.

3.2 All four degrees two: exact modes and approximation

We use the established constant-gap hardness of maximum independent set on simple cubic graphs supplied with a proper three-edge-coloring. The explicit construction of Chlebík and Chlebíková [13, author manuscript, Section 5(A), pp. 25–26] produces edge-three-colored three-regular graphs, with the coloring as part of the instance, and preserves the independent-set objective and gap. No particular gap constant is needed. The coloring is essential: cubic degree alone would not supply the three disjoint resource matchings used below.

Theorem 3.3. *One-attribute pooling threshold decision is strongly NP-complete when all four degree counts (input out-degree, pool in-degree, pool out-degree, and product in-degree) are exactly two, all vertex capacities are one, qualities and upper specifications lie in $\{0, 1\}$, and costs lie in $\{-2, -1, 0, 1\}$. Only upper product quality specifications are imposed. There are no bypasses or positive lower flow bounds. Its nonnegative profit objective on the constructed family has no PTAS unless $P = NP$. In the constructed family, every feasible solution has an integral pure-mode replacement of at least the same profit.*

Proof. Let $G = (V, E_1 \dot{\cup} E_2 \dot{\cup} E_3)$ be the source graph, $n = |V|$. Each color class is a perfect matching. Make a pool ℓ_v for each vertex. A clean input a_e for $e \in E_1$ feeds its two endpoint pools, and a dirty input d_e for $e \in E_3$ feeds its two endpoint pools. A strict product s_e for $e \in E_2$ receives from its two endpoint pools. Initially give each pool a private lax product b_v . All capacities are one. Qualities are zero for clean inputs and one for dirty inputs; strict and lax bounds are zero and one. The four arcs at each pool have costs $1, 0, -2, -1$ as above.

Write the flows at v as a_v, d_v, s_v, b_v . Mass balance shows that its profit is $s_v + d_v \geq 0$. If $s_v > 0$, its quality must be zero and $d_v = 0$. Keep only the clean-to-strict flow $a'_v = s'_v = s_v$, deleting the other flows. If $s_v = 0$, keep only $d'_v = b'_v = d_v$. In either case every arc flow decreases or stays fixed,

profit is preserved, and the new qualities, respectively zero and one, satisfy all products. Thus any feasible flow can be made pure, with at most one allowed mode retained per pool.

Fix those mode choices, assigning inactive pools arbitrarily. Replace each allowed mode by an edge between its input and product. Feasible throughputs are precisely fractional matchings in this bipartite graph; the unit edge bound also enforces the corresponding pool capacity. The bipartite matching polytope is integral: orient its incidence edges from the input side to the product side to obtain a network incidence matrix, and add integral bounds. A maximum matching therefore has size at least the old fractional total. Route one unit in its selected modes. This proves the claimed replacement and is algorithmic.

Let H be obtained by subdividing each edge of E_3 twice. For $uv \in E_3$ write its new path $u - d_u - d_v - v$. An original vertex v represents its pool's clean mode and d_v its dirty mode. Clean modes conflict along E_1 through a shared input and along E_2 through a shared product. Dirty modes conflict on $d_u d_v$ through a shared input; v and d_v conflict because they are modes of one pool. These are all conflicts. Consequently integral mode selections are independent sets of H , with profit equal to cardinality.

The following identity includes a constructive recovery:

$$\alpha(H) = |E_3| + \alpha(G) = n/2 + \alpha(G). \quad (30)$$

Indeed, extend any independent set of G by one internal vertex of each subdivided edge, avoiding its selected endpoint if present. For the reverse direction, if an independent set of H contains both original endpoints u, v of an E_3 edge, it contains neither internal vertex. Delete u and insert d_u . This preserves cardinality and independence. Since E_3 is a matching, the changes do not interfere. Afterwards the selected original vertices form an independent set of G , and at most one internal vertex is present per edge. Their number is at least the original cardinality minus $n/2$. Combining this with integral replacement proves that the pooling optimum is exactly $n/2 + \alpha(G)$. Threshold $n/2 + k$ gives strong hardness.

A cubic graph has $\alpha(G) \geq n/4$, by successively choosing a vertex and deleting it and its at most three neighbors. Therefore $P^* = n/2 + \alpha(G) \leq 3\alpha(G)$. A feasible profit $P \geq (1 - \varepsilon)P^*$ yields in polynomial time an independent set of size at least

$$P - n/2 \geq \alpha(G) - \varepsilon P^* \geq (1 - 3\varepsilon)\alpha(G). \quad (31)$$

A PTAS would contradict the source gap. Here and in the approximation recovery below, we use the ordinary bit model: an approximation algorithm returns feasible rational arc flows in binary, with output length bounded by its polynomial running time. Cleanup uses only those flows; its exact rational operations and the subsequent matching computations take time polynomial in their encoding length. At positive throughput, pool quality is recovered from the incoming quality mass; at zero throughput it may be chosen arbitrarily.

Finally merge the two private lax products of the endpoints of every E_3 edge into one capacity-one lax product. Now there are n inputs, n pools, and n products, all with the required degree counts exactly two. Cleanup still only decreases resource use. In pure modes the new constraint is $d_u + d_v \leq 1$, already imposed by their shared dirty input. Thus it changes neither the integral selections nor the optimum or recovery. Each of the two undirected layer-incidence graphs is now a disjoint union of cycles. The endpoint disjunction (12) gives NP membership for the full stated zero-tolerance restriction class. $\square$

Hard selection of sparse active flows was already established by Haugland [25, Definition 2.2 and Proposition 3.1, p. 96], who imposes restrictions on the number of positive inlet or outlet

flows at each pool and proves NP-hardness of all four single-flow variants. In theorem 3.3, integral pure-mode replacement is instead a proved property of an ordinary continuous pooling family under the simultaneous degree and data restrictions. The next proposition extends that replacement argument to linked weighted costs.

Proposition 3.4 (A weighted integral-mode family). *Allow arbitrary sharing degrees, zero flow lower bounds, and nonnegative integer source, pool, product, and (if imposed) arc capacities. Each pool still has one quality-zero input, one quality-one input, one strict product with bound zero, and one lax product with bound one. Only upper product quality specifications are imposed. For arbitrary rational α_v, β_v , give its clean intake, dirty intake, strict outlet, and lax outlet costs $\beta_v, 0, -(\alpha_v + \beta_v), -\beta_v$. An optimal solution exists with integral throughputs and only pure modes. With unit capacities this optimization is exactly maximum-weight rainbow matching in a bipartite multigraph: pool v contributes a clean-input/strict-product edge of weight α_v and a dirty-input/lax-product edge of weight β_v . These two edges have disjoint endpoints and the common color v ; at most one edge of each color may be selected.*

Proof. Mass balance makes cost $-\alpha_v s_v - \beta_v d_v$. The cleanup in the preceding proof preserves s_v and d_v separately, and thus preserves cost even for signed rewards. Once one mode per pool is fixed, maximize weighted flow between the input and product vertices, with their integer capacities. Bound each retained mode edge by the minimum of its pool capacity and its two physical arc capacities, omitting any arc bound that is not imposed. These edge bounds are integral. This is a capacitated bipartite flow polytope with an integral optimum. It improves or preserves the previous objective and restores a feasible integral pooling point. The full feasible set is compact in bounded quality coordinates, so an optimum exists to which this replacement applies. For unit capacities, the two edges of a color have disjoint endpoints because their input and product types differ; the only further restriction is to choose at most one of them. This is precisely the stated rainbow matching problem. The conclusion concerns these four linked arc costs, not arbitrary pooling costs. $\square$

Proposition 3.5 (Positive strict-product concentration bounds). *There is a fixed rational $\delta > 0$ for which replacing the strict bounds in theorem 3.3 by δ preserves absence of a profit PTAS unless $P = NP$, and exact optimization remains strongly NP-hard. The same conclusion holds with input qualities $\{1, 2\}$ and product bounds $\{1 + \delta, 2\}$.*

Proof. For a feasible flow with profit P and any $\eta > 0$, separate the pools by whether their quality p_v is at most η . For rational η , this is the exact rational test $d_v \leq \eta(a_v + d_v)$ at nonzero throughput; empty pools may be put in either group. In the first group keep only clean-to-strict flow $\min(a_v, s_v)$ and set quality zero. Its profit loss is at most $2d_v = 2p_v t_v \leq 2\eta$, since $s_v - a_v = d_v - b_v \leq d_v$. In the other group keep only dirty-to-lax flow $\min(d_v, b_v)$ and set quality one. Its loss is at most $2s_v$, since $d_v - b_v = s_v - a_v \leq s_v$. Every arc flow weakly decreases, so the new solution is feasible with zero strict bounds. Summing the old strict quality inequalities gives $\sum_v p_v s_v \leq \delta \sum_v s_v \leq \delta n$, whence its profit is at least $P - 2n(\eta + \delta/\eta)$. Empty pools cause no change.

Let $\gamma > 0$ be a fixed independent-set approximation gap: a guarantee $(1 - \gamma)\alpha(G)$ is NP-hard. Choose a positive rational $\eta < \gamma/32$ and put $\delta = \eta^2$. Then $P_0^* \leq P_\delta^* \leq P_0^* + 4\eta n$. A PTAS for the positive concentration bound, with fixed $\varepsilon < \gamma/6$, followed by cleanup, integral replacement, and recovery, returns a set of size at least

$$(1 - \varepsilon)P_0^* - 4\eta n - n/2 \geq (1 - 3\varepsilon - 16\eta)\alpha(G) > (1 - \gamma)\alpha(G),$$

a contradiction. These are fixed rational data, so the same gap reduction also proves strong hardness of exact optimization. It does not preserve (30) or assert integral optima at a positive bound. Here

the bound limits concentration exactly; the argument does not allow arbitrary numerical residuals in the flow or quality constraints. Adding one to all qualities and bounds leaves the flow constraints unchanged by mass conservation and gives the final assertion. $\square$

Example 3.6 (A positive concentration bound can destroy integral optimality). Take K_4 on vertices $0, 1, 2, 3$, with matchings $E_1 = \{01, 23\}$, $E_2 = \{02, 13\}$, and $E_3 = \{03, 12\}$. Use the exactly-two-degree construction with strict bound $0 < \delta \leq 1/2$. At pools 0 and 3 take clean flow $1 - \delta$ and dirty flow δ , and send one unit to the respective strict product. At pool 1 take one dirty unit and send it to the lax product; leave pool 2 inactive. The shared dirty input for 03 uses $2\delta \leq 1$, and the other uses one. Each clean input uses $1 - \delta$, each strict product receives one unit of quality δ , and the used lax product receives one unit of quality one. Thus this is feasible and has profit $3 + 2\delta$. At zero tolerance, (30) gives optimum $2 + \alpha(K_4) = 3$. Any integral unit-capacity flow at $\delta < 1$ uses only pure pools, and a strict product receives at most one unit; it cannot receive a dirty unit. Every such integral flow is therefore also zero-tolerance feasible and has profit at most three. Hence the positive-tolerance feasible point strictly exceeds every integral one.

3.3 A positive-product source and two pools on a tree

The next reductions use the positive-product construction of Matsui [35, METR95-13, Sections 2–3, pp. 2–7]. We record the arithmetic needed later and prove its separation estimate directly, because Theorem 2.1 of the report, stated for arbitrary p , is stronger than the estimate its hardness reduction needs (remark 3.8).

Lemma 3.7 (The explicit positive-product family). *For a zero-one matrix M with $n \geq 5$ columns, put $p = n^{n^4}$ and let $\Omega(M)$ consist of $(r, s) \in [0, 1]^{n+n^2}$ satisfying*

$$Mr = \mathbf{1}, \quad s_{ii} = r_i, \quad s_{ij} \leq r_i, \quad s_{ij} \leq r_j, \quad s_{ij} \geq r_i + r_j - 1 \quad (i \neq j).$$

Write

$$\begin{aligned} X &= \sum_i p^i r_i, & Y &= \sum_{i,j} p^{i+j} s_{ij}, \\ U &= 2p^{4n} - p + Y + 2p^{2n} X, & V &= 2p^{4n} - p + Y - 2p^{2n} X, & K &= 4p^{8n}. \end{aligned} \quad (32)$$

Both factors are strictly positive on $\Omega(M)$, and deciding whether $UV \leq K$ somewhere on it is NP-hard. All coefficients have polynomial binary encoding length.

Proof. The zero-one feasibility question $Mr = \mathbf{1}$ is NP-complete (it is exact cover in incidence-matrix form). Adding zero columns permits $n \geq 5$ without changing that answer. Let $N = n + n^2 \leq n^3$ and $h = (n^3)^{-n^3}$. At every vertex of $\Omega(M)$, an invertible active N -row system has integral coefficients of absolute value at most one. Cramer's rule and the determinant bound $N^{N/2} \leq (n^3)^{n^3}$ imply that a nonintegral coordinate lies in $[h, 1 - h]$.

At a vertex where r is not binary, let k be the largest fractional index. If $i > k$ or $j > k$, the McCormick constraints give $s_{ij} = r_i r_j$, so those terms cancel in $D = Y - X^2$. The diagonal term at k is $p^{2k} r_k (1 - r_k) \geq p^{2k} h/2$. Each of the other at most n^2 signed terms with indices at most k is bounded below by $-p^{2k-1}$. Hence

$$D \geq p^{2k-1} (ph/2 - n^2) > p. \quad (33)$$

For the strict inequality, $ph = n^{n^3(n-3)}$ and $n \geq 5$ imply $ph/2 > n^2 + 1$. If $Mr = \mathbf{1}$ has no binary solution, every vertex has fractional r ; the concavity of $Y - X^2$ and convex decomposition into

vertices extend $D > p$ to the whole nonempty polytope. If the polytope is empty the product decision is already negative.

The exact product identity is

$$UV - K = (Y - p)^2 + 4p^{4n}(D - p). \quad (34)$$

Thus the no case has $UV > K$. In the yes case a binary r extends by $s_{ij} = r_i r_j$, giving $D = 0$. Since $0 \leq Y = X^2 \leq n^2 p^{2n}$,

$$(Y - p)^2 \leq \max\{p^2, n^4 p^{4n}\} < 4p^{4n+1},$$

where $n^4 < p$. Equation (34) now gives $UV < K$. Finally $U \geq 2p^{4n} - p > 0$ and $V \geq 2p^{4n} - p - 2np^{3n} > 0$, since $p^n > 2n + 1$. The largest exponents in (32) require $O(n^5 \log n)$ bits. If M has m rows, the polytope has $N = n + n^2$ variables and $O(m + n^2)$ rows. Thus the row count is polynomial in the source size; no bound on m relative to n is needed. $\square$

Remark 3.8. Estimate (33) proves exactly the large- p case used here, without the report's printed estimate for arbitrary positive integers p . That estimate can fail: for example, $n = 5, p = 2$, $r_i = 1/2, s_{ii} = 1/2, s_{ij} = 0$ for $i \neq j$, and $\epsilon = 1/4$ satisfy that estimate's hypotheses but give $Y - X^2 = -279$, less than its claimed lower bound $-99/2$. The positive-product hardness source and all subsequent reductions rely only on the corrected argument above.

Theorem 3.9. *Pooling threshold decision is NP-complete with exactly two pools and two products, arbitrary input and attribute counts, upper quality bounds only, upper flow capacities in $\{1, 2\}$, no bypasses, and no positive lower flow bounds. Every input has out-degree one and the underlying undirected network is a tree. One pool has one input and one product. Suppressing that pool gives the same conclusion with one mixing pool, two products, and exactly one bypass arc. These are ordinary NP-completeness claims.*

Proof. Begin with a bounded positive-product instance $\min_{w \in P_0} U(w)V(w) \leq K$, $K > 0$, from lemma 3.7. If P_0 is empty, map it to a fixed no instance: take one variable input, the two pools, the anchor, and the two products with the arc pattern below, zero rewards, quality zero everywhere, and profit threshold one. Otherwise compute $u_{\min} = \min_{P_0} U > 0$ by rational LP and replace U, V by $2U/u_{\min}, u_{\min}V/2$. Their product is unchanged and $U \geq 2$. Intersect P_0 with $V \leq K$. This keeps all yes witnesses, because $UV \leq K$ implies $V \leq K/2$. An empty intersection is again mapped to a fixed no instance. These operations have polynomial bit complexity.

For $w \in [0, 1]^s$, embed the remaining polytope in a simplex by $z_i = w_i/s$ and $z_0 = 1 - \sum_{i=1}^s z_i$, adding $z_i \leq 1/s$. Denote its linear description by

$$P = \{z \geq 0 : \mathbf{1}^T z = 1, A_r z \leq d_r \ (1 \leq r \leq R)\}.$$

Affine functions become linear on the simplex. Write $a(z) = U(z) - 1 = \sum_i a_i z_i$ and $b(z) = K - V(z) = \sum_i b_i z_i$. Then $a \geq 1, b \geq 0$ on P , although individual coefficients may have either sign.

For each coordinate make an input i feeding only pool L . Give these inputs and arcs capacity two and pool L capacity two. One anchor input feeds only pool H , with all three capacities one. Pool L feeds products 1 and 2, and H feeds only product 1. All outlet arcs and both products have capacity one. Reward intake iL by b_i ; all other rewards are zero. There is one distinguished attribute and one attribute per polytope row:

Attribute	Input i	Anchor	Product 1 bound	Product 2 bound
Distinguished	a_i	0	1	$\max(0, \max_i a_i)$
Row r	A_{ri}	d_r	d_r	d_r

For positive throughput T at L , its intake proportions are $z_i = x_i/T$. Write its outlet flows y_1, y_2 and the anchor flow h ; total product flows are $D_1 = y_1 + h, D_2 = y_2$. At product 1 the row- r inequality is $(A_r z)y_1 + d_r h \leq d_r(y_1 + h)$; product 2 gives $(A_r z)y_2 \leq d_r y_2$. Since $y_1 + y_2 = T > 0$, at least one enforces each row, and therefore $z \in P$. The distinguished inequality is $a(z)y_1 \leq D_1$, so

$$b(z)T = b(z)(D_2 + y_1) \leq b(z)(D_2 + D_1/a(z)) \leq b(z)(1 + 1/a(z)).$$

Conversely, for any $z \in P$, choose

$$y_1 = 1/a(z), \quad y_2 = 1, \quad h = 1 - 1/a(z), \quad T = 1 + 1/a(z), \quad x_i = Tz_i. \quad (35)$$

Both products are filled, all flows respect their bounds, and every attribute constraint holds. Inactive L has zero profit. Compactness and $a \geq 1, b \geq 0$ therefore give the exact identity

$$P^* = \max_{z \in P} b(z)(1 + 1/a(z)). \quad (36)$$

For each $z \in P$,

$$b(z)(1 + 1/a(z)) \geq K \iff (K - V)U \geq K(U - 1) \iff UV \leq K. \quad (37)$$

This proves hardness.

The topology is shown in figure 2. The undirected edges form the path L –product 1– H –anchor, with all variable inputs and product 2 as leaves at L . Suppressing H creates exactly the anchor bypass. There are three actual pool-outlet arcs in this construction and at most four in the full two-pool, two-product class. Fixing their outlet fractions gives bounded linear fibers even with unbounded attributes. The bounded-flow certificate of corollary 2.11, which also covers the suppressed version’s two outlet fractions and bypass, therefore gives NP membership in both versions. $\square$

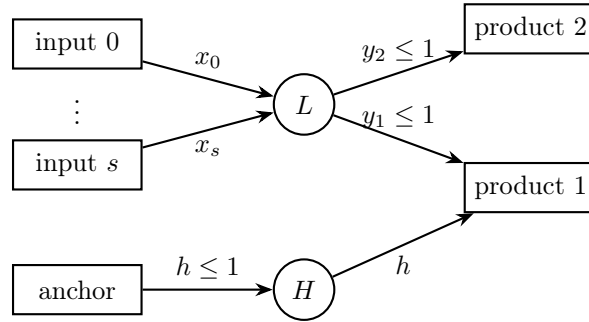


Figure 2: The two-pool tree in theorem 3.9. Pool L mixes the variable inputs; H carries the fixed-quality anchor to product 1. Suppressing H produces one mixing pool and one bypass. The primary product capacities, together with the distinguished quality bound at product 1, give the radial limit $T \leq 1 + 1/a(z)$.

Each attribute in this construction can be shifted by a common rational constant at all inputs and products to make its data nonnegative, then scaled into $[0, 1]$. Conservation preserves its inequalities and the encoding remains polynomial. Nonnegative input costs and product revenues also suffice: choose $B \geq \max(0, \max_i b_i)$, charge $B - b_i$ at variable inputs and B at the anchor, and pay B at either product. All common B terms cancel. Source and pool upper bounds are redundant given the two product capacities and may be omitted. Alternatively, exact unit demands at both products preserve (36), since (35) attains every mixture’s bound with both products filled. None of these modifications makes the large quality and reward coefficients bounded integers.

Proposition 3.10 (An FPTAS for the represented subfamily). *Given a nonempty bounded rational polytope P and rational affine functions $a \geq 1, b \geq 0$ on P , a feasible point of value at least $(1 - \varepsilon) \max_P b(1 + 1/a)$ can be found with at most $\lceil 1/\varepsilon \rceil + 1$ LPs, for rational $0 < \varepsilon < 1$. This gives an FPTAS for the pooling subfamily supplied with the representation used in theorem 3.9.*

Proof. Set $N = \lceil 1/\varepsilon \rceil$. For $g = k/N$, $0 \leq k \leq N$, maximize b over P with $a \leq 1/g$ if $g > 0$; impose no additional row for $g = 0$. Skip infeasible LPs and evaluate the true objective at one rational optimal vertex of every remaining LP. Retain the best. Let z^* attain the original optimum and put $t^* = 1/a(z^*) \in (0, 1]$. The grid value $g = \lfloor Nt^* \rfloor / N$ keeps z^* feasible and satisfies $t^* - g < 1/N$. The LP optimizer z_g has $b(z_g) \geq b(z^*)$ and $1/a(z_g) \geq g$. Consequently

$$b(z_g)(1 + 1/a(z_g)) \geq b(z^*)(1 + g) \geq \text{OPT} - b(z^*)/N \geq (1 - 1/N) \text{OPT}.$$

This also covers $\text{OPT} = 0$. All LP encodings and their rational vertex solutions have bit length polynomial in the input and $\log N$. Recover pooling flows by (35). The number of LPs is polynomial in $1/\varepsilon$. The algorithm neither recognizes arbitrary pooling representations of this form nor gives an FPTAS for general two-pool, two-product pooling. $\square$

Table 2 summarizes the three copy-based hardness routes of sections 3.4 to 3.7. They share gadgets, which retain private midpoint supplies and use separate full and half cycles, but they do not form one implication chain.

Route and dependencies	Restrictions obtained
Value-preserving penalty: theorem 3.13, then lemmas 3.14 and 3.15, then theorem 3.17.	One pool, two outlets, one upper scalar quality, input degree at most two and product degree at most three; zero lower flow bounds. The reduction proves ordinary threshold hardness and preserves the optimum up to an explicit offset; large data remain.
Finite-data threshold and completion: the cycle gadgets give lemma 3.18; theorem 3.19 encodes the source threshold physically before removing contracts.	Two actual feeds and outlets, the same degree bounds, fixed finite physical data. Contracted feasibility first represents the source decision; completion then gives strong threshold hardness with zero lower flow bounds and bounded costs and revenues.
Contracted-feasibility refinement: theorem 3.20 starts from the threshold-encoding circuit inside theorem 3.19, before completion.	The same two-feed, two-outlet and finite-data restrictions, with at most three input and two product exceptions to exact contracts, and a redundant common pool bound. Strong feasibility hardness uses the already encoded threshold, without an economic objective.

Table 2: Dependencies of the copy-based refinements. Degree bounds count all physical arcs. Each route retains the hypotheses of its cited results.

3.4 Copying bounded signals through physical bypasses

We next replace the many attribute rows by ordinary sources and products. A *port* is an input-to-product arc whose source is assigned when the gadgets are connected. A source assigned several ports has their common quality and exactly those outgoing arcs. Equal quality labels never identify sources unless the construction explicitly joins their supply equations. Every requested occurrence

gets its own gadget, so repeated terms never create parallel arcs or reuse one physical flow. A *signal* is the common number encoded by these distinct flows, not an extra physical flow that can be sent to several destinations. Source supplies and product demands are ordinary node totals as in equation (1).

Lemma 3.11 (Copy and conversion). *For distinct rational endpoint qualities α, β , there is a pool-free gadget whose four endpoint ports have flows $x, x, 2 - x, 2 - x$, for an arbitrary $x \in [0, 2]$, with each pair consisting of one α port and one β port. Chains of such gadgets can copy a bounded signal to arbitrarily many ports and convert it to an actual pool intake of any nonzero prescribed quality.*

Proof. Put $\gamma = (\alpha + \beta)/2$. Make products A, B , each with exact demand four and exact quality γ . Each receives one endpoint port of each quality, of capacity two, and one capacity-four arc from a common quality- γ source of exact supply four. If endpoint flows at A are u, v and middle flow is m , subtracting γ times mass balance from quality balance gives $(\alpha - \gamma)u + (\beta - \gamma)v = 0$, so $u = v = x$ and $m = 4 - 2x$. At B the same argument gives $u = v = x'$ and $m = 4 - 2x'$. Exact middle supply implies $x + x' = 2$. Conversely these formulas fill every constraint for any $x \in [0, 2]$.

Join the B - α port of one gadget to the A - α port of the next with a quality- α source of exact supply two. It forces $(2 - x) + x' = 2$, so the signals agree. Ordinary gadgets with endpoints $(0, 2)$ provide repeated positive and complementary quality-two ports. Unused ports have distinct private sources with upper supply equal to the port capacity and lower supply zero. End a chain with a $(0, a_i)$ gadget. Its B - a_i port and a new pool intake share a quality- a_i source of exact supply two. As the former carries $2 - x_i$, the actual intake is x_i . The unused A - a_i port is supplied privately. All reverse extensions follow from the displayed formulas. $\square$

Lemma 3.12 (Bounded-coefficient rows). *A homogeneous system $Cx \leq 0$, with integer coefficients and $x \in [0, 2]^m$, has a copy-network realization of size polynomial in m and $\sum_{r,i} |C_{ri}|$. Its projection onto the original signals is exactly that system.*

Proof. For a row c , allocate c_i quality-two positive ports x_i when $c_i > 0$ and $-c_i$ complementary ports $2 - x_i$ when $c_i < 0$. Assign all these ports to one quality-two source with upper supply $B = 2 \sum_{c_i < 0} (-c_i)$ and lower supply zero. Its total outflow is $c^T x + B$, so its supply bound is precisely $c^T x \leq 0$. Omit zero rows. Allocate enough copies for every occurrence and fill unused ports privately. The copy identities prove one projection direction; filling the chains at any feasible signal vector proves the other. Two rows encode an equality. $\square$

Theorem 3.13 (One pool and fixed attribute dimension). *With bypasses, pooling threshold decision is NP-complete with one pool, one physical quality having lower and upper product specifications, and two pool outlets. Exact supplies and demands are allowed. Equivalently, two attributes with upper specifications only suffice. All qualities may be normalized into $[0, 1]$. The hardness is ordinary, not strong, NP-completeness.*

Proof. Use lemma 3.7 without the LP normalization of theorem 3.9. Put $s = n + n^2$ and use simplex coordinates for $w = (r, s_{ij}) \in [0, 1]^s$ as before. If $T = \sum_i x_i > 0$, homogenizing the simplex polytope at $z = x/T$ gives integer rows $Cx \leq 0$ of polynomially bounded coefficients. For example,

$$sx_h \leq T, \quad sx_{ij} \geq sx_i + sx_j - T, \quad s \sum_i M_{ri} x_i = T.$$

Each row has coefficient magnitude at most $s+1$ and there are polynomially many rows, so lemma 3.12 has polynomial size.

At the unrestricted simplex generators $w = 0$ and $w = se_h$, write the values of the factors as U_i, V_i . With $u_0 = 2p^{4n} - p$,

$$u_0 \leq U_i \leq u_0 + sp^{2n} + 2sp^{3n}, \quad V_i \leq u_0 + sp^{2n} < K. \quad (38)$$

The lower bound follows because all nonconstant coefficients of U are nonnegative; the upper bounds follow by evaluating its r or s coefficient. Since $s < p^{2n}$ and $p \geq 2$, the last strict inequality holds. Thus the simplex coefficient vectors $a_i = U_i - 1 > 1$ and $b_i = K - V_i > 0$ have polynomial bit length, although large magnitudes.

Copy every signal x_i and convert it to a quality- a_i intake of one capacity-two pool. Enforce all cone rows by lemma 3.12. The pool has two capacity-one outlets to capacity-one primary products. The first also receives a capacity-one quality-zero anchor bypass and has upper quality one; the second has redundant upper quality $\max_i a_i$. Reward the actual intake x_i by b_i . All other arcs have zero reward.

For $T > 0$ the copied rows enforce $z = x/T$ in the simplex image of $\Omega(M)$, and the pool quality is $a(z)$. The argument of theorem 3.9 gives profit at most $b(z)(1 + 1/a(z))$; conversely (35) extends to the exact copy network. The zero-intake assignment satisfies every homogeneous row and also fills the copies, so the exact optimum, even for empty $\Omega(M)$, is

$$\max(\{0\} \cup \{b(z)(1 + 1/a(z)) : z \in P\}). \quad (39)$$

Threshold $K > 0$ and (37) prove hardness. The fixed pool-quality parameter gives NP membership by corollary 2.11.

Exact scalar qualities use both a lower and an upper specification. Adding the negative of the scalar as a second coordinate expresses the same restrictions by upper bounds only. Coordinate shifts and positive scaling normalize them into $[0, 1]$ without changing any feasible flow. $\square$

For an input-cost realization of the same profit, reward the private input serving the conversion gadget's A - a_i port by b_i , and give the actual intake zero reward. The copied port carries x_i , so the objective is unchanged. A common charge and revenue at every input and product cancel by mass conservation and make all input costs and product revenues nonnegative. After contracts are relaxed, the two flows need not be equal, so this realization needs an additional error estimate (theorem 3.17).

Lemma 3.14 (Full and half ports and bounded degrees). *The construction of theorem 3.13 can have input out-degree at most three, product in-degree at most three, and all flow upper bounds at most four. Exact supplies and demands and exact scalar qualities remain allowed. A full-port-only variant has input out-degree at most four.*

Proof. First, with ordinary endpoint qualities $(0, 2)$, enforce $2v = u + w$ through an exact-supply-four source on four full ports $v, v, 2 - u, 2 - w$. For a row from lemma 3.12, list its m port values $f_j \in [0, 2]$ and upper sum bound B . Pad by zeros to $N = 2^{\lceil \log_2 m \rceil}$ leaves and impose pairwise averages in a complete binary tree. A zero signal is fixed by a zero-upper-supply source on its positive port. Induction gives root $v_{\text{root}} = \sum_j f_j/N$; a private positive root port with upper supply B/N enforces the row. When $m = 1$, use the leaf itself; omit $m = 0$. Since $m \leq N < 2m$ and $0 \leq B/N \leq 2$, this construction has linear size per row, upper capacities at most four, and input degree at most four. Each gadget product still has three inputs.

The full-port construction above uses only its ordinary row ports in the averaging trees; its conversion gadgets remain at the original signals. For degree three, use full gadgets with endpoints zero and three, midpoint quality $3/2$, demand four at each product, and middle supply four. Their zero and quality-three port pairs are both $(x, 2 - x)$. A half gadget has endpoints zero and three,

middle quality one, demand three at each product, and middle supply three. Give its zero ports capacity two and its quality-three ports capacity one. At one product the equations $u + v + m = 3$ and $3v + m = 3$ give $u = 2v$. Summing both products and the middle supply gives the port formulas

$$(A_0, B_0) = (x, 2 - x), \quad (A_3, B_3) = (x/2, 1 - x/2). \quad (40)$$

Their middle flows are $3 - 3x/2$ and $3x/2$, so all $x \in [0, 2]$ are feasible. The zero-port formulas coincide with those of full and conversion gadgets, so exact-two zero-quality links copy the same signal along mixed chains.

Use an exact-supply-two quality-three source on $v, 1 - u/2, 1 - w/2$. Its three-port equation is exactly $2v = u + w$. If a child is $2 - x$, the needed complementary half port is $x/2$. Replace every averaging source in the same padded trees by this gate, retaining the root bound B/N . Every feasible original row assignment extends by its arithmetic averages; conversely induction recovers the row from every feasible physical network. Allocate a distinct gadget for each occurrence. Middle, chain, and conversion sources have degree two, private and root sources degree one, and averaging sources degree three. Products have degree three, except the two primary products of degrees two and one. The pool retains its two outlets. Size and bounds are as claimed. The dyadic root capacities have polynomial bit length; large factor qualities and rewards still prevent a strong-hardness claim. $\square$

Lemma 3.15 (Closed cycles with one upper quality). *The degree-three realization can use one scalar quality and upper product specifications only. It can further have input out-degree at most two and product in-degree at most three, with flow upper bounds at most four. Exact flow contracts remain in this lemma.*

Proof. A full gadget with endpoints $0, \beta, \beta > 0$, retains demands four and middle supply four but only the upper quality $\beta/2$. Writing zero and other endpoint flows as u_j, v_j , its inequality is $v_j \leq u_j$. For one signal, place all full gadgets, including its conversion gadget if present, in a closed zero-port cycle: each link has quality zero, exact supply two, and joins the second zero port to the next gadget's first zero port. For r gadgets, links supply total zero-port flow $2r$. Product demands minus middle supplies give total endpoint flow $4r$, so total other-port flow is $2r$. Every $v_j \leq u_j$ is therefore tight. The full formulas follow, and each link forces the common signal to agree with its successor. This works when different gadgets have different positive β , because their normalized inequalities are identical. A one-gadget cycle joins two distinct products and creates no parallel arc.

Put the half gadgets for a signal in a separate closed zero-port cycle. Here the upper inequality is $2v_j \leq u_j$. Links supply total zero flow $2r$, while total endpoint flow is $3r$. Thus other-port flow is r and every inequality is tight, giving (40). Do not mix full and half gadgets in one cycle. Instead, couple their signals by a quality-three exact-supply-two source on one extra full positive port and two distinct complementary half ports:

$$x_f + (1 - x_h/2) + (1 - x_h/2) = 2.$$

It forces $x_f = x_h$. Every signal with a half occurrence gets these three reserved occurrences, creating its full cycle if needed. Signals without half occurrences need no coupling. Every common signal in $[0, 2]$ extends to both cycles and the coupling source by the formulas, so the earlier row projection and objective identity remain exact.

The only degree-three inputs are now averaging and coupling sources. Each has quality three, exact supply two, and port capacities $(c_1, c_2, c_3) = (2, 1, 1)$. Replace it by three same-quality inputs of exact supplies c_h . Input h feeds its original port with flow f_h and a new common collector with flow $c_h - f_h$, both arc capacities c_h . Give the collector exact demand $\sum_h c_h - 2 = 2$ and redundant upper quality three. This is feasible exactly when $\sum_h f_h = 2$, which was the original source equation.

It reduces input degree to two and creates a product of degree three. All other inputs already have degree at most two. The transformation has linear size and preserves all original signal and intake flows. All quality values are nonnegative and can be scaled globally into $[0, 1]$. $\square$

Figure 3 shows the physical full and half gadgets and their separate zero-port cycles.

The full-only degree-four construction, the full/half degree-three construction, and the cyclic input-degree-two construction are exact projection refinements of the same row realization, so each keeps the earlier guarantees: one exact scalar quality or two upper attributes suffice before the cyclic refinement, and one upper scalar quality suffices afterwards. All constructions so far use positive flow contracts; removing them is a separate step.

3.5 An explicit exact penalty for removing contracts

The penalty below preserves the optimal value of the contracted hard family up to an explicit additive constant. The constant-data construction of section 3.6 instead encodes threshold feasibility and needs no value-preserving penalty.

Lemma 3.16 (A rational polyhedral error bound). *Let $Q = \{v \in \mathbb{R}^N : Rv \leq s\}$ be nonempty, $N \geq 1$, and let each row be scaled by a positive factor to have integral coefficients of absolute value at most $K_0 \geq 1$. Let R, s denote the resulting scaled system. In this representation,*

$$\text{dist}_1(w, Q) \leq H \max_j (R_j w - s_j)_+, \quad H = N(NK_0)^{N-1}. \quad (41)$$

Proof. Let v be the Euclidean projection of w onto the closed convex set Q , and put $h = w - v$. If $h = 0$ the result is immediate. The projection optimality condition puts h in the cone of the active row normals. A conic representation can be chosen with linearly independent normals: if a supported collection is dependent, move its nonnegative coefficients along a dependence until one coefficient becomes zero and repeat. Thus $h = B^T \lambda$, where B has $r \leq N$ linearly independent active integral rows and $\lambda \geq 0$. Set $\eta = \max_j (R_j w - s_j)_+$. Since these rows are tight at v ,

$$\|h\|_2^2 = \lambda^T B h \leq \|\lambda\|_1 \eta, \quad \|\lambda\|_1 \leq \sqrt{N} \|h\|_2 / \sigma_{\min}(B).$$

Since B has full row rank and integral entries, $\det(BB^T)$ is a positive integer and therefore at least one. All singular values of B are at most $\|B\|_F \leq NK_0$. Their product is at least one, so $\sigma_{\min}(B) \geq (NK_0)^{-(r-1)} \geq (NK_0)^{-(N-1)}$. Cancel $\|h\|_2 > 0$ and use $\|h\|_1 \leq \sqrt{N} \|h\|_2$. $\square$

This is a coarse explicit form of a polyhedral error bound in the sense of Hoffman [30, Section 2]. Its proof provides the coefficient control needed here without computing a best error-bound constant.

Theorem 3.17 (Upper flow bounds only). *Pooling threshold decision is NP-complete with one pool, one upper-bound scalar quality, no positive lower flow bounds, input out-degree at most two, product in-degree at most three, and two pool outlets. All flow upper bounds are at most four and all quality data can lie in $[0, 1]$. Nonnegative input production costs and product unit revenues suffice. The hardness is ordinary NP-completeness; removing contracts shifts the optimal value by an explicit constant.*

Proof. Use the Matsui-specific construction of theorem 3.13, followed by lemmas 3.14 and 3.15, before any final quality normalization. Every actual intake has quality $a_i > 1$ and reward $b_i > 0$. Let w contain all copy-network bypass flows and all actual intakes x_i ; exclude the anchor and the two primary outlets. Include $T = \sum_i x_i \leq 2$. The exact copy subsystem is a bounded rational polytope

$$Q = \{w : Aw \leq d, Ew = e\}.$$

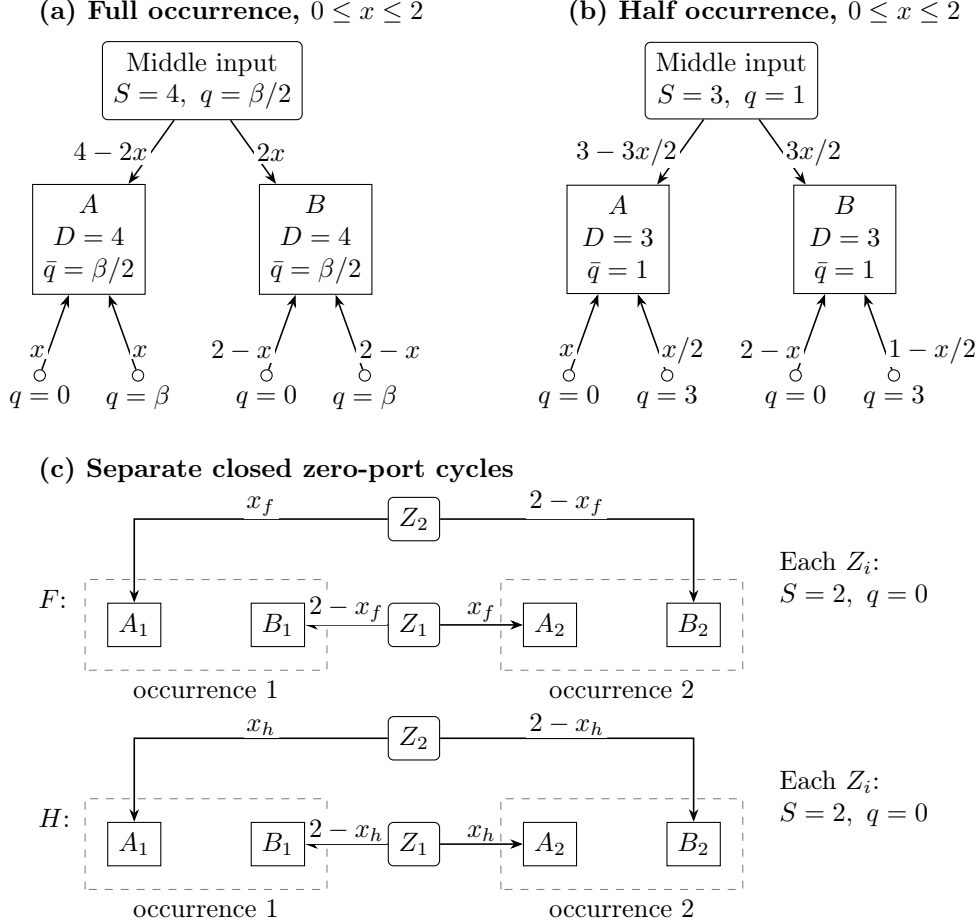


Figure 3: Physical copy gadgets and their zero-port links (lemmas 3.11, 3.14 and 3.15). Here S and D are exact node supply and demand, q is an input quality, and $\bar{q}$ is a product upper bound. Open circles mark ports whose sources are assigned when connecting gadgets; every arrow is an input-to-product physical arc. In (a), $\beta > 0$; ordinary full gadgets use $\beta = 3$. Endpoint capacities are two, except that the half gadget's quality-three ports have capacity one; middle-arc capacities are S . The displayed flows attain the quality bounds. Exact product qualities enforce them locally; upper bounds alone enforce them after closing the separate cycles in (c). That panel shows two occurrences per cycle and only their zero-port arcs; middle and positive-quality arcs are omitted. Additional occurrences are inserted in the same manner, with distinct products at every occurrence. The dashed boxes group products and are not physical nodes or arcs. Full and half signals are coupled by the separate source equation $x_f + (1 - x_h/2) + (1 - x_h/2) = 2$, using two distinct half occurrences, as in the proof of lemma 3.15.

Here $Aw \leq d$ contains all upper bounds, nonnegativity, and the linear quality inequalities at copy products; $Ew = e$ lists all exact source supplies and gadget-product demands *after* the cycles and source splitting. Its rows have zero-one coefficients and integral right sides in $\{0, 1, 2, 3, 4\}$, and $Ew \leq e$ is included in $Aw \leq d$. No primary outlet quality inequality or pool-quality parameter occurs in this polytope. It is nonempty: the all-zero original intake signals satisfy every homogeneous cone row, and their complementary copies and auxiliary averages fill all exact contracts, even if the source polytope has no positive-throughput point.

Delete the positive lower flow bounds of the physical network. Any resulting feasible solution has $Aw \leq d$ and deficits $\Delta = e - Ew \geq 0$, $\delta = \mathbf{1}^T \Delta$. Apply lemma 3.16 to the rows $A, E, -E$. Clear denominators in A if needed, leaving the integral contract rows unchanged. All scaled A residuals remain nonpositive and the largest positive contract residual is at most δ . Thus some $w' \in Q$ satisfies

$$\|w' - w\|_1 \leq H\delta, \quad \|x' - x\|_1 \leq H\delta. \quad (42)$$

The cleared coefficient bound K_0 and hence H have polynomial binary encoding length; row denominator products suffice to compute such a bound. Computing a nearest point is not part of the reduction.

Restoring the copy subsystem may violate the primary outlet constraints. To repair this, set

$$T = \mathbf{1}^T x, \quad S = a^T x, \quad F(x) = S(T - 1) - T.$$

For $x \geq 0, T \leq 2$, the primary subsystem can accommodate x exactly when $F(x) \leq 0$. To see this for $T > 0$, its quality is $S/T \geq 1$; its maximal throughput along that ray is $1 + T/S$, by the two unit products and anchor dilution. Equivalently $T \leq 1 + T/S$, or $F \leq 0$. For a direct feasible realization when $T \leq 1$, send everything to product 2. When $T > 1$, send one unit to product 2 and $T - 1$ to product 1, adding anchor $(S/T - 1)(T - 1)$. The total at product 1 is $(S/T)(T - 1) \leq 1$ and anchor flow is between zero and one. Zero throughput is feasible separately.

Let $a_{\max} = \max_i a_i$ and $L = 3a_{\max} + 1$. On the convex set $x \geq 0, \mathbf{1}^T x \leq 2$,

$$\frac{\partial F}{\partial x_i} = a_i(T - 1) + S - 1, \quad \left| \frac{\partial F}{\partial x_i} \right| \leq L.$$

The original x is feasible, so $F(x')_+ \leq L\|x' - x\|_1$. If $F(x') \leq 0$, set $x'' = x'$. Otherwise $T' > 1, S' > 1$, and set

$$\theta = \frac{1 + T'/S'}{T'} < 1, \quad x'' = \theta x'.$$

Its throughput is exactly the feasible radial maximum and $F(x'') = 0$. Moreover,

$$\|x' - x''\|_1 = T' - 1 - T'/S' = F(x')/S' \leq L\|x' - x\|_1 \leq LH\delta. \quad (43)$$

Scaling preserves every homogeneous cone row and the signal box. Rebuild the exact copy network from x'' and its determined averages and complementary ports, then fill the primary products as above. This produces a feasible contracted pooling solution.

Let $b_{\max} = \max_i b_i$ and $C = b_{\max}(1 + L)H$. Equations (42)–(43) give the uniform bound

$$b^T x \leq b^T x'' + C\delta. \quad (44)$$

Choose $M_0 = C + 1$, and add reward M_0 per unit at each originally contracted source and gadget product. The new profit is $b^T x + M_0 \mathbf{1}^T Ew$. Put $B_0 = M_0 \mathbf{1}^T e$. For every relaxed feasible solution,

$$P_{\text{new}} - B_0 = b^T x - M_0 \delta \leq b^T x'' - (M_0 - C)\delta \leq \text{OPT}_{\text{contract}}.$$

Conversely every contracted feasible point is relaxed-feasible with zero deficit. Therefore

$$\text{OPT}_{\text{new}} = B_0 + \text{OPT}_{\text{contract}}. \quad (45)$$

A positive deficit is strictly suboptimal relative to its repair. Threshold $B_0 + K$ proves NP-hardness.

The constants $a_{\max}, b_{\max}, K_0, H, L, C, M_0, B_0$ are explicit rationals of polynomial bit length. Each penalty term is an ordinary linear source or product throughput reward, so it is a legal arc objective. No new arc or quality row is introduced and all upper flow bounds are unchanged. The fixed-quality certificate gives membership in NP.

For the stated economics, use instead the private conversion-port input rewards described after theorem 3.13. Write that objective as $c^T w$, with $\|c\|_{\infty} \leq b_{\max}$. In the relaxed instance it need not equal $b^T x$. Projection to w' changes it by at most $b_{\max} H \delta$; at $w' \in Q$ it equals $b^T x'$. Rebuilding at x'' changes this value by at most $b_{\max} L H \delta$. Thus the same C proves (45). Base and source rewards are negative input production costs, and gadget-product penalties are revenues. Add a common charge $D_0 \geq M_0 + b_{\max}$ at all inputs and the same revenue at all products. Conservation cancels the offset and all costs and revenues are nonnegative. Finally divide the original nonnegative qualities by their global maximum; all quality inequalities are preserved and data lie in $[0, 1]$. The penalty may be computed in the unscaled coordinates. The penalty has large magnitude, and the reduction gives neither strong hardness nor an approximation barrier. $\square$

3.6 Constant physical data and two actual feeds

Large source coefficients can instead be stored in the topology of a linear circuit. This yields a stronger decision result whose local physical data all belong to fixed finite sets.

Lemma 3.18 (Constant-data linear circuits). *Every rational system of linear equations and weak inequalities on variables in $[0, 2]^m$ is the exact projection of a polynomial-size pool-free direct blending network with one upper-bound scalar quality, input out-degree at most two, and product in-degree at most three. Flow bounds lie in $\{0, 1, 2, 3, 4\}$, with exact integer supply and demand contracts allowed. Qualities and upper specifications belong to $\{0, 1, 3/2, 3\}$. Each original coordinate is the flow on a designated full positive port.*

Proof. Use the full and half closed cycles of lemma 3.15, with ordinary endpoint qualities zero and three and middle qualities $3/2$ and one. Every requested port occurrence has a distinct gadget. Copies needed by gates below are allocated first; their zero ports are then joined into the closed cycles, so gate fan-out never increases a source degree. The following quality-three sources implement gates on signals in $[0, 2]$:

Equation or inequality	Requested port flows	Source supply
$v = (u + w)/2$	$v, 1 - u/2, 1 - w/2$	exactly 2
$w = u + v$	$u, v, 2 - w$	exactly 2
$u = v$	$u, 2 - v$	exactly 2
$u \leq v$	$u, 2 - v$	at most 2
$u = 0$	u	exactly 0
$u = 1$	u	exactly 1

Each row follows directly from the source's supply constraint. Couple full and half cycles by the exact-two source already proved. Privately supply every unused other-endpoint port up to its capacity.

For a dyadic coefficient $c/2^L \in [0, 1)$, with binary bits $c = \sum_{k=0}^{L-1} c_k 2^k$, implement

$$t_0 = 0, \quad t_{k+1} = (t_k + c_k x)/2 \quad (0 \leq k < L). \quad (46)$$

The bits are used from least to most significant. Induction gives $t_L = \sum_{k=0}^{L-1} c_k x/2^{L-k} = cx/2^L$. Every intermediate signal remains in $[0, 2]$ and there are $O(L)$ gates. Use the zero signal for $c_k = 0$; coefficients zero and one may instead use zero or equality gates.

For a rational row, clear denominators to obtain $\sum_i c_i x_i \leq d$ with integer c_i, d . Choose a power of two $B > \sum_i |c_i| + |d|$. Compute each $(|c_i|/B)x_i$ and the constant $|d|/B$ from the unit signal. Put the positive terms on one side and negative terms on the other, placing the constant according to its sign, and sum with addition gates. Each total and every partial sum is at most $(2 \sum |c_i| + |d|)/B < 2$, so no hidden restriction is introduced by the signal box. One comparison enforces the row; an equality gate between the sums enforces an equality. Empty sides use zero. All these circuits have size polynomial in the row's binary length, including large denominators after clearing. Conversely the gate equations force the stated row in every physical solution.

A three-port average or coupling source has capacities $(2, 1, 1)$ and supply two; a three-port addition source has capacities $(2, 2, 2)$ and supply two. Apply the complement-flow splitting of lemma 3.15 to each, with collector demand $\sum c_h - 2$, respectively two or four. The new input supplies and arc capacities are at most two. Collectors have redundant upper quality three and degree three. All inputs now have degree at most two. Full and half product demands and middle supplies are four and three, and every other flow bound is an integer between zero and four. Reserve a full positive port for each original coordinate and supply it privately; this adds no constraint on its signal. The two projection directions proved above complete the result. $\square$

Lemma 3.18 is a representation result for linear polyhedra, not a hardness claim about LP feasibility. Unlike lemma 3.12, it represents binary-encoded coefficients. Dividing qualities by three gives the fixed normalized alphabet $\{0, 1/3, 1/2, 1\}$.

Theorem 3.19. *Pooling threshold decision is strongly NP-complete with one pool, one upper-bound scalar quality, exactly two actual feed arcs and two actual outlet arcs, input out-degree at most two, and product in-degree at most three. Every flow lower bound is zero and every upper bound belongs to $\{0, 1, 2, 3, 4\}$. Qualities and upper specifications belong to*

$$\{0, 1/66, 1/33, 1/22, 1/11, 1/2, 1\}. \quad (47)$$

Input unit production costs belong to $\{0, 1\}$, product unit revenues to $\{1, 2\}$, and the profit threshold is an integer at most four times the number of inputs and products.

Proof. Take the source family of lemma 3.7. Write $P_4 = p^{4n}$, $s = n + n^2$, and $u_0 = 2P_4 - p$. Let D_0 be the largest power of two at most $u_0/2$, so $2D_0 \leq u_0 < 4D_0$. The generator bounds (38), together with $s < p^n$, imply

$$U_i < u_0 + P_4 + 2P_4 < 5P_4, \quad u_0 > P_4, \quad D_0 > P_4/4.$$

Thus $2 \leq U_i/D_0 < 20$. Also $D_0 < P_4$ and $V_i \leq u_0 + sp^{2n} < 3P_4$, so $D_0 V_i < 3p^{8n} < K$. Normalize the factors by $U \mapsto U/D_0, V \mapsto D_0 V$, and define

$$a_i = U_i/D_0 - 1 \in [1, 19), \quad b_i = K - D_0 V_i > 0, \quad \rho_i = (a_i - 1)/32 = (U_i - 2D_0)/(32D_0) \in [0, 1). \quad (48)$$

Each b_i is a positive integer and each ρ_i is dyadic. All binary encodings remain polynomial; their large values will appear only in circuit descriptions.

Let $x_i \in [0, 2]$ be the source simplex signals. Use lemma 3.18 to enforce the homogeneous source cone $Cx \leq 0$, and introduce signals

$$T = \sum_i x_i, \quad h = \sum_i \rho_i x_i, \quad \ell + h = T. \quad (49)$$

All these signals lie in $[0, 2]$, in particular enforcing $T \leq 2$. They can be built either as the general row circuits or as direct nonnegative partial sums and dyadic multipliers. In the latter form all partial sums respect two whenever the intended assignment does, since $0 \leq \rho_i \leq 1$ and $\sum_i x_i \leq 2$. Only signals ℓ and h get actual intake conversion gadgets, with endpoint pairs $(0, 1)$ and $(0, 33)$ respectively. Insert each in its signal's full cycle. Their positive endpoint *qualities* permit lemma 3.15; their flows may be zero. Their exact-two conversion sources force actual intakes ℓ, h . These are the only two pool feeds. The pool's quality mass is

$$\ell + 33h = T + 32h = \sum_i a_i x_i. \quad (50)$$

The pool again has capacity two and feeds two unit-capacity outlets to unit-capacity primary products. The first has upper quality one and a quality-zero anchor bypass of capacity one. The second has redundant upper quality 33. For $T > 0$ the source mixture $z = x/T$ has pool quality $a(z) \geq 1$, and its maximal radial throughput is $1 + 1/a(z)$. Finally encode the affine row

$$\sum_i b_i x_i \geq K \quad (51)$$

by the same binary circuits. This is implemented by sources, flows, and quality rows, not retained as an external objective or side constraint. It forces $T > 0$ because $K > 0$. A feasible contracted network therefore gives $z \in P$ and $b(z)T \geq K$, and hence $b(z)(1 + 1/a(z)) \geq K$. By (37) applied to the normalized factors, this is equivalent to $U(z)V(z) \leq K$ in the original source. Conversely, a source yes witness uses (35), satisfies (49), and fills every circuit at its prescribed signal values. Thus contracted feasibility is exactly the source decision. Unlike the penalty construction, this contracted network need not have a zero-intake feasible point.

Apply the three-port source splitting after all gates and both intake conversions have been specified. It preserves both actual feeds and outlets and gives input degree two and product degree three. The unscaled quality alphabet, including the conversion middle qualities, is

$$\{0, 1/2, 1, 3/2, 3, 33/2, 33\}.$$

All upper quality specifications also lie in this set. Divide by 33 to obtain (47). All flow bounds are integers from zero to four.

It remains to remove positive lower flow bounds. List every positive exact input-supply and product-demand contract of this final network as $Ew = e$. Each $e_j \in \{1, 2, 3, 4\}$. Retain the upper bounds and delete all positive lower bounds. Maximize the contract-completion profit

$$P_{\text{comp}} = \sum_j E_j w, \quad S_0 = \sum_j e_j. \quad (52)$$

Every term is at most its own contract, so a feasible relaxed network reaches S_0 exactly when every original contract holds. No cancellation of deficits is possible. Consequently it reaches this threshold exactly when the original contracted network is feasible. Each arc occurs in at most one contracted-input term and one contracted-product term, so arc rewards are at most two and S_0 is at most four times the number of external nodes.

For the asserted economics, initially charge minus one at each contracted input and zero at other inputs; give revenue one to each contracted product and zero to other products. This realizes (52). Add one to every input cost and every product revenue. Conservation cancels this offset and gives precisely the sets $\{0, 1\}$ and $\{1, 2\}$. The entire transformation has polynomial size, and even unary data encoding has polynomial size: only the threshold grows, linearly in network size. This proves strong NP-hardness. One scalar pool-quality parameter gives NP membership by corollary 2.11. $\square$

Strong NP-completeness here concerns exact threshold attainment. Binary averaging chains can distinguish extremely close flow values using constant local coefficients. The proof gives no inverse-polynomial completion gap, approximation barrier, or robustness to a fixed feasibility tolerance; bounded physical data alone do not imply them.

3.7 Feasibility with five contract exceptions

Call an input exceptional if it has no exact total supply. Call a product exceptional if it lacks either an exact demand or an exact quality. The positive result in theorem 5.10 assumes bypass degree at most two, a fixed number of exceptions, exact ordinary contracts, and redundant common pool bounds. The hard construction below keeps the last three conditions but has three bypass inlets at ordinary products. This compares bypass degrees; total physical degree also counts pool arcs, and total product degree remains at most three, as in theorem 3.19. Positive contracts are part of its feasibility model.

Theorem 3.20. *Pooling feasibility is strongly NP-complete with one pool, one scalar quality, two actual feeds and two actual outlets, input out-degree at most two, and product in-degree at most three. At most five external nodes are exceptional: three inputs and two products. All other inputs have exact supplies and all other products have exact demands and qualities. Flow bounds lie in $\{0, 1, 2, 3, 4\}$ and quality data in (47). The pool has lower bound zero and redundant upper bound two. The exceptional products use only upper quality specifications. There is no economic objective or external threshold row.*

Proof. Start from the contracted circuit in theorem 3.19, including the circuit implementing (51), before the completion objective and before splitting its gate sources. Its only nonlinear interface consists of the feeds ℓ, h , the two primary products, and the anchor. We make exact projection transformations to remove the many auxiliary exceptions.

First replace each comparison gate $u \leq v$, previously represented by an upper-supply-two source on $u, 2 - v$, by an exact-supply-two source on $u, s, 2 - v$, with a new signal $s \in [0, 2]$. Its equation is $u + s = v$. A feasible extension exists exactly when $u \leq v$, since then $s = v - u \in [0, 2]$. Give this slack its usual copy cycle. Equality, addition, averaging, coupling, zero, and unit gates already use exact supplies. This includes every comparison in the source and threshold circuits.

Second group the unused positive-quality ports by the gate that requested their complementary ports. For an exact quality-three source requesting flows f_h with capacities c_h and total supply S , the unused mates carry $c_h - f_h$. Assign them to one new quality-three source of exact supply $\sum_h c_h - S$. This introduces no further condition, since the requested source already has $\sum_h f_h = S$. The possible supplies are listed below:

Gate	Capacities	S	Complement supply
Average or full/half coupling	2, 1, 1	2	2
Addition or slack comparison	2, 2, 2	2	4
Equality	2, 2	2	2
Zero	2	0	2
Unit	2	1	1

There are no unused zero ports because their cycles are closed. The proof of copying does not depend on private port supplies: it uses only exact middle supplies, exact product demands, and exact zero-quality links. The port identities therefore hold before any consequence of the grouped supplies is used, which justifies the grouping.

Third split every three-port exact source, including the newly grouped ones of supply four. A source of supply S and capacities c_h becomes three same-quality sources of exact supplies c_h , each feeding its original port f_h and a new collector with $c_h - f_h$. The collector has exact demand $\sum_h c_h - S$ and exact quality equal to that source quality. Its demand enforces exactly $\sum_h f_h = S$. For both supply-two and supply-four gates, collector demands are two or four. Every new source has degree two and the collector degree three. Sources with one or two ports already respect the input-degree bound.

All full and half cycle products already attain their upper quality bounds by lemma 3.15, including conversion gadgets with positive endpoint qualities one and 33. We may therefore require their qualities exactly, without removing any feasible point. Collector qualities are also automatic, because every collector receives a single source quality. All these products already have exact demands.

After grouping, the only unused positive-endpoint ports that still need variable private inputs are the positive ports in the two intake conversion gadgets. Their complementary ports share exact-supply-two sources with the actual pool feeds. Thus exactly two private conversion fillers and the anchor remain variable inputs. The two primary products retain variable demands and upper qualities. Every other external node has the required exact contracts, giving five exceptions. Original source signals need no extra designated output ports for this reduction; each ordinary port is allocated to a gate or its full/half coupling. The final external-node inventory is as follows. Degrees count all physical arcs, and quality values here are before division by 33.

External node	Final contract	Degree
Middle inputs	Exact supply 4 (full) or 3 (half)	2
Zero-link inputs	Exact supply 2, quality 0	2
Gate and complement inputs after splitting	Exact supply in $\{0, 1, 2\}$, quality 3	≤ 2
Two conversion feed inputs	Exact supply 2, quality 1 or 33	2
Two private conversion fillers	Supply in $[0, 2]$, quality 1 or 33	1
Anchor	Supply in $[0, 1]$, quality 0	1
Full and conversion products	Exact demand 4, exact midpoint quality	3
Half products	Exact demand 3, exact quality 1	3
Collectors	Exact demand 2 or 4, exact quality 3	3
Two primary products	Demands in $[0, 1]$; upper qualities 1 and 33	2, 1

All input qualities are fixed. Ordinary products have three bypass inlets and no pool inlet; the primary products receive the two actual pool outlets.

The transformations preserve all signal constraints and the physical interface in both directions. In particular the threshold circuit still excludes zero original intake, and feasibility remains equivalent to $UV \leq K$. Each change has size linear in the affected ports. Allocate a separate gadget for every

requested gate port before grouping its unused mate; this makes the grouping disjoint and avoids parallel physical arcs. The quality alphabet is unchanged, normalized by division by 33, and all flow bounds remain in $\{0, 1, 2, 3, 4\}$. The two unit-capacity outlets alone imply throughput at most two, making the pool bound redundant. The reduction has polynomial size with fixed finite data and is strongly NP-hard. The same fixed scalar parameter gives NP membership. $\square$

3.8 Interpretation of the restrictions

The two reduction families expose different obstacles. In the all-degree-two family every feasible flow has an integral pure-mode replacement, so the hard part is choosing compatible modes, and the unit-data family has a constant approximation gap. In the fixed-pool family the nonlinear interface has constant dimension, but the surrounding linear constraints encode a hard positive-product problem; hence a fixed number of nonlinear parameters gives an NP certificate but not necessarily a polynomial algorithm. Its represented subfamily has an FPTAS, and the finite-data bypass construction proves strong hardness of exact threshold attainment without a gap. The five-exception version concerns feasibility under exact auxiliary contracts; the zero-lower-bound version always has the zero solution and concerns an economic threshold.

4 Structural exact algorithms

The hardness results of section 3 do not make every large pooling network intractable. Two decompositions give exact algorithms. The first keeps a fixed number of nonlinear variables and aggregate rows and partitions all other flows into independent polyhedra of fixed dimension (sections 4.1 to 4.3). The second eliminates long bypass paths and keeps a fixed number of actual boundary flows (sections 4.4 to 4.6). Both allow arbitrarily many variables; their hypotheses control different sources of coupling. Throughout this section pooling means the standard three-layer model of Section 1.1, with finite rational flow bounds, arbitrary rational lower and upper throughput and attribute specifications, and no pool–pool arcs. Unless restricted explicitly, objectives are rational linear costs on actual arcs. All algorithmic conclusions concern binary input size and exact algebraic output as in Section 1.2.

The two decompositions retain different information, and this determines the objective scope of each result. The block construction retains both aggregate masses and the objective value, and therefore permits costs on every arc. The path construction retains only boundary feasibility; it permits arbitrary costs only when exact equalities reduce them to the retained coordinates.

4.1 A fixed nonlinear core and independent polyhedral blocks

Fix integers $r, k \geq 0$ and $d, D \geq 1$. Let $C \subset \mathbb{R}^r$ be a compact set specified by a quantifier-free rational polynomial formula of degree at most D . For $j = 1, \dots, N$, let

$$P_j(x) = \{z \in \mathbb{R}^{d_j} : B_j(x)z \leq b_j(x)\}, \quad 1 \leq d_j \leq d.$$

Every coordinate has an explicit finite rational lower and upper box bound among these inequalities. Empty blocks and lower-dimensional blocks are allowed. All entries of $B_j, b_j, A_j, b, c_j, c_0$ below are

rational polynomials in x of degree at most D , with $A_j(x) \in \mathbb{R}^{k \times d_j}$ and $b(x) \in \mathbb{R}^k$. Consider

$$\begin{aligned} \min \quad & c_0(x) + \sum_{j=1}^N c_j(x)^T z_j, \\ \text{subject to} \quad & x \in C, \quad z_j \in P_j(x) \quad (1 \leq j \leq N), \\ & \sum_{j=1}^N A_j(x) z_j = b(x). \end{aligned} \tag{53}$$

The row counts of the local systems and the number of blocks are part of the input and may grow.

Theorem 4.1 (Fixed core, local dimension, and aggregate dimension). *For every fixed (r, d, k, D) , feasibility and global optimization of (53) are solvable in polynomial bit time. If feasible, the algorithm returns an exact optimum and optimizer of polynomial encoding length. All optimizer coordinates can be represented in one real-algebraic extension of $\mathbb{Q}$ of polynomial degree. A fixed number of aggregate inequalities can also be included.*

The geometric ingredients are established: support directions select compatible faces of Minkowski summands, and sums of polytopes in fixed dimension have polynomial complexity [18, 21]. These results concern specified polytopes. Here every summand changes with a nonlinear parameter, so computing a Minkowski sum separately at each parameter would not give a finite global algorithm. The additional construction is a polynomial list of support formulas valid uniformly over the whole core, including parameters where a local basis becomes singular or a block loses dimension. The required sign enumeration, quantifier elimination, and algebraic sampling are the algorithms of Basu et al. [7, Theorem 1.3.1 and Sections 3.1.3–3.2].

Proof. If $N = 0$, append a scalar block fixed to zero, with zero contribution and cost. A block with no coordinates in an application is likewise padded by one coordinate fixed to zero. We first prove the equation form.

Local vertices and their validity. Let M_j be the number of rows in block j . For every d_j -row subset I , form

$$\delta_{jI}(x) = \det B_{jI}(x), \quad p_{jI}(x) = \text{adj}(B_{jI}(x))b_{jI}(x).$$

An identically zero determinant can be discarded. Otherwise the candidate $v_{jI} = p_{jI}/\delta_{jI}$ is defined when $\delta_{jI} \neq 0$. For each row a , put

$$e_{jIa} = B_{ja}p_{jI} - b_{ja}\delta_{jI}. \tag{54}$$

The candidate is feasible exactly when $\delta_{jI} \neq 0$ and $e_{jIa}/\delta_{jI} \leq 0$ for every a . These tests use signs of polynomials of degree at most $(d+1)D$; the determinant and numerator degrees are at most dD .

Every nonempty bounded polyhedron has a vertex. At any vertex its active row normals span the ambient space: otherwise a nonzero orthogonal direction, with sufficiently small positive and negative steps, gives a feasible segment through that point. Thus some d_j active rows are independent, even when the polyhedron has smaller affine dimension or consists of one point. The candidates therefore include every vertex, and a block is nonempty if and only if at least one candidate passes its validity tests.

Support choices without a Cartesian product. Set $h = k + 1$ and define

$$W_j(x) = \begin{pmatrix} A_j(x) \\ c_j(x)^T \end{pmatrix}, \quad T_j(x) = W_j(x)P_j(x), \quad w(x, v) = \begin{pmatrix} b(x) \\ v - c_0(x) \end{pmatrix}.$$

Here v denotes an objective value. For a support direction $u \in \mathbb{R}^h$, the score of a valid candidate is

$$\rho_{jI}(x, u) = \frac{n_{jI}(x, u)}{\delta_{jI}(x)}, \quad n_{jI} = u^T W_j(x) p_{jI}(x).$$

Comparison of candidates I, I' in the same block uses the determinant signs and the sign of

$$n_{jI} \delta_{jI'} - n_{jI'} \delta_{jI}. \quad (55)$$

Let $\mathcal{P}$ contain these comparison polynomials, all determinants, and all polynomials in (54). Identically zero polynomials have a remembered zero sign and need not be stored. Since $\binom{M_j}{d_j} \leq M_j^d$, the number of polynomials is polynomial in the input size. Their degrees are bounded in terms of d, D : a score comparison has degree at most $(2d + 1)D + 1$, counting the support coordinates as variables. Their coefficient lengths are polynomial, and they involve only the fixed number $r + h$ of coordinates (x, u) .

Enumerate all realizable sign conditions of $\mathcal{P}$, including zero signs, using fixed-dimensional real-algebraic sign enumeration. There are polynomially many such conditions, and the enumeration has polynomial bit complexity. A sign condition determines all valid candidates and all score comparisons. If a block is empty, discard the condition. Otherwise choose in every block the first maximizing candidate in a fixed order. This choice is constant throughout the realization of the sign condition; connectedness of that realization is unnecessary. In particular, degeneracies and ties do not require enumerating all tuples of local bases.

For a retained condition σ , abbreviate its selected determinants and numerators to δ_j, n_j and put

$$H_\sigma(x) = \prod_{j=1}^N \delta_j(x)^2, \quad (56)$$

$$F_\sigma(x, v, u) = u^T w(x, v) H_\sigma(x) - \sum_{j=1}^N n_j(x, u) \delta_j(x) \prod_{i \neq j} \delta_i(x)^2. \quad (57)$$

On this condition $H_\sigma > 0$, so $F_\sigma \leq 0$ is equivalent to

$$u^T w(x, v) \leq \sum_{j=1}^N \max_{z \in P_j(x)} u^T W_j(x) z. \quad (58)$$

Squared denominators preserve the inequality for either determinant sign. The degree of F_σ grows as $O(N)$, with constants depending on the fixed parameters; it is not bounded independently of N . Nevertheless it has polynomially many monomials because its number of variables is fixed. Expanding the products takes polynomial bit time: there are polynomially many multiplication steps and monomials, and the coefficient lengths grow by at most the sum of the factor lengths plus the logarithms of the numbers of terms accumulated. This also bounds the coefficients of the sum in (57).

A formula in a fixed number of variables. Write $S_\sigma(x, u)$ for the conjunction of signs defining σ , and set

$$\Phi(x, v, u) = \bigvee_{\sigma \text{ retained}} (S_\sigma(x, u) \wedge F_\sigma(x, v, u) \leq 0).$$

We claim that the objective value v is attainable at a specified x if and only if

$$\forall u \in \mathbb{R}^h \quad \Phi(x, v, u). \quad (59)$$

If some block is empty, all conditions realized at (x, u) are discarded, for every u , so this formula is false. If all blocks are nonempty, their image sum $T(x) = \sum_j T_j(x)$ is nonempty compact convex. Its support function is the sum of their support functions. A point belongs to a nonempty compact convex set exactly when it satisfies every support inequality, by separating any point outside the set. Thus (59) says precisely that $w(x, v) \in T(x)$. This is equivalent to the existence of the block vectors with the required aggregate and objective values, proving the claim.

Let $E(x, v)$ abbreviate $x \in C$ and (59). The formula

$$\exists x E(x, v)$$

defines the attainable objective values. Its total variable count is $r + h + 1$, and its degree, length, and rational coefficient encoding are polynomial. Fixed-dimensional quantifier elimination therefore produces a univariate formula in polynomial bit time, even with the degree growth noted above. The original feasible set is closed in the product of the compact core and the finite block boxes; hence it is compact. The objective is continuous, so its nonempty value set is compact and has a least element. Univariate root isolation decides emptiness and finds that element exactly.

For joint optimizer recovery one can instead eliminate quantifiers in

$$E(x, v) \wedge \neg \exists x', v' [E(x', v') \wedge v' < v]. \quad (60)$$

Rename the bound support variables in each occurrence of E . The total number of variables remains fixed. Algebraic sampling gives an optimal pair (x^*, v^*) with polynomial-size rational univariate representation: there is one real algebraic α of polynomial degree, with isolating data, and rational functions of α representing all these coordinates. Equivalently, they lie in the common represented field $F = \mathbb{Q}(\alpha)$.

Recovering all blocks. Fix $x = x^*$ in the local inequalities and aggregate equations, and append

$$\sum_j c_j(x^*)^T z_j = v^* - c_0(x^*).$$

This is a feasible bounded linear system of polynomial dimension over F . Its coefficients have polynomial descriptions, because evaluating the input polynomials uses bounded degree and the sampled coordinates have polynomial descriptions in the same field. The algebraic linear programming result of Adler and Beling [3, Section 5, Remark 1], which outlines the rational-machine implementation of their algorithm, solves this system in time polynomial in the LP dimension, coefficient encoding, and degree of a common extension. That degree is polynomial here; separate small representations of unrelated algebraic coefficients would not suffice. The explicit coordinate-box rows ensure full column rank of the recovery constraint matrix. All recovered blocks may be chosen in F , because a vertex is obtained from a nonsingular subsystem by field linear algebra. Determinant and field-arithmetic bounds give polynomial encoding length, consistent with the algorithm's output guarantee. Explicitly, represent each coefficient in the power basis of F and clear its rational denominators. A determinant of order at most the recovery dimension has polynomial degree and coefficient bit length before reduction modulo the minimal polynomial of α : its factorial number of expansion terms contributes only polynomially many bits. Polynomial division and inversion in F preserve polynomial encoding at these dimensions. This bounds the Cramer ratios for an entire vertex, rather than assuming that an arbitrary sequence of field operations has small intermediate bit length. Together with (x^*, v^*) these blocks give the promised optimizer.

Aggregate inequalities. A row $\sum_j a_j(x)^T z_j \leq \beta(x)$ is converted to an equation by adding its nonnegative slack as a scalar block. A polynomial-bit rational upper bound for that slack can be

computed, even if the core formula does not explicitly give a box. First decide whether C is empty. If nonempty, eliminate all but one coordinate from its formula, for each of the fixed number of coordinates. Each compact coordinate projection has finite endpoints, and each endpoint is a root of a nonzero polynomial in the resulting univariate formula: on an interval containing no such root every atomic sign, and hence membership, is constant. A Cauchy root bound for these polynomially encoded polynomials gives a rational box $[-R, R]^r$ containing C , with polynomial bit length. When $r = 0$ no such step is needed. Interval arithmetic over this box and the supplied block boxes bounds $\beta(x) - \sum_j a_j(x)^T z_j$ by a polynomial-bit rational number in absolute value. The coefficient degree is fixed, and the polynomial number of summands adds only polynomial encoding. A fixed number of such slack blocks preserves all dimension hypotheses. This completes the proof. $\square$

The block LP algorithms of Cslovjecsek et al. [14, Theorem 1] also exploit a fixed number of linking rows and independent local optimization. Their matrices and polyhedra are fixed linear data. The additional step above is global optimization over a nonlinear core that changes those local data; fixing a few nonlinear variables in an otherwise unrestricted LP family would not establish this theorem.

Remark 4.2 (What the theorem requires). A fixed number of design binaries may be included in the core by $x_i(x_i - 1) = 0$, provided the rest of the model retains the displayed form. The theorem does not cover an unbounded family of local binaries: with no core, blocks $z_j \in \{0, 1\}$ and one aggregate equation already encode subset sum. Nor can arbitrary compact convex nonlinear blocks replace the polyhedra without changing the output guarantee. Blocks $0 \leq z_j \leq a_j + 1$, $z_j^2 \leq a_j$, with objective $-\sum_j z_j$, have optimum $-\sum_j \sqrt{a_j}$. For distinct primes a_j this value has degree 2^N . Indeed the multiquadratic extension has independent sign automorphisms (the prime square classes are independent), and its distinct square-root monomials are a rational basis. No nontrivial sign change fixes the displayed sum, so its orbit has size 2^N . This is an exact-output obstruction, not an approximation hardness claim. Polyhedral support values avoid it because they are rational functions of one common fixed-dimensional core. The theorem also gives no strongly polynomial or fixed-parameter runtime bound.

4.2 Two basic pooling consequences

Corollary 4.3 (Fixed numbers of inputs and pools). *For fixed m, p , pooling with at most m inputs and p pools admits polynomial-bit exact global optimization and optimizer recovery. The numbers of products and qualities, and the bypass arcs, are unrestricted.*

Proof. If there are no inputs, check the all-zero flow directly. Otherwise, remove a pool with no incoming or no outgoing arc after forcing its incident flows to zero and checking that zero belongs to its throughput interval and every incident-arc interval. Keep every adjacent external-node contract. For each remaining pool introduce inlet fractions $\theta_{i\ell} \geq 0$ with $\sum_i \theta_{i\ell} = 1$ and $\theta_{i\ell} = 0$ on missing inlet arcs. They form a compact core of dimension at most mp .

The block for product j contains $(v_{\ell j})_\ell$ and $(z_{ij})_i$, of dimension at most $p + m$, with missing arcs fixed to zero. Its flow boxes and product-throughput bounds are linear. For attribute k , the upper quality row is

$$\sum_{\ell} \left(\sum_i \lambda_{ik} \theta_{i\ell} - \bar{\mu}_{jk} \right) v_{\ell j} + \sum_i (\lambda_{ik} - \bar{\mu}_{jk}) z_{ij} \leq 0. \quad (61)$$

The lower row replaces these coefficients by their lower-bound counterparts $\underline{\mu}_{jk} - \sum_i \lambda_{ik} \theta_{i\ell}$ and $\underline{\mu}_{jk} - \lambda_{ik}$. Thus an arbitrary attribute count increases the number of local rows only.

Let $T_\ell = \sum_j v_{\ell j}$. The remaining interval bounds apply to

$$T_\ell, \quad y_{i\ell} := \sum_j \theta_{i\ell} v_{\ell j}, \quad s_i = \sum_j \left(\sum_\ell \theta_{i\ell} v_{\ell j} + z_{ij} \right). \quad (62)$$

These impose pool capacities, inlet-arc bounds, and input withdrawal bounds, respectively, using at most $2(p + mp + m)$ aggregate inequalities. A feed-arc cost $c_{i\ell} y_{i\ell}$ contributes $\sum_i c_{i\ell} \theta_{i\ell}$ to the coefficient of every $v_{\ell j}$; all other costs already belong to their product blocks. Theorem 4.1 applies with affine coefficient data.

For exactness, a positive-throughput physical pool has fractions $y_{i\ell}/T_\ell$. A zero-throughput pool permits any simplex vector on its nonempty incoming-arc set. Conversely, $y_{i\ell} = \theta_{i\ell} T_\ell$ gives mass balance and a pool quality $\sum_i \lambda_i \theta_{i\ell}$. Equations (61)–(62) are then exactly the physical quality and throughput conditions, at positive and zero throughput. The objective is preserved in both directions. $\square$

Corollary 4.4 (Fixed pools and attributes without bypasses). *For fixed p, K , pooling with at most p pools and K attributes, without bypasses, admits polynomial-bit exact global optimization and optimizer recovery. Inputs and products may be arbitrarily numerous. A fixed number of bypass arcs is also permitted.*

Proof. If there are no inputs, all flows vanish and feasibility is checked directly. Otherwise the pK pool qualities form the core, bounded coordinatewise between the smallest and largest input qualities. For each input, use the block $(y_{i\ell})_\ell$ with its arc and withdrawal bounds. For each product, use the block $(v_{\ell j})_\ell$ with its arc, delivery, and quality bounds. All local dimensions are at most p , and each product quality row is affine in the core. The only linking equations are

$$\sum_i y_{i\ell} - \sum_j v_{\ell j} = 0, \quad \sum_i \lambda_{ik} y_{i\ell} - q_{\ell k} \sum_j v_{\ell j} = 0. \quad (63)$$

There are $p(K + 1)$ equations; pool throughput bounds add at most $2p$ aggregate inequalities. Costs separate over the blocks. These are the physical concentration equations, with valid arbitrary boxed qualities at inactive pools. Theorem 4.1 proves the claim.

For a fixed number of bypasses, put their bounded flows in the core. Subtract their input withdrawals, product deliveries, and known quality masses from the appropriate local right-hand sides, and put their costs in the core objective. The dimensions remain fixed and the reformulation remains exact. Unbounded bypasses require a further structural argument, supplied next. $\square$

The piecewise-parametric pooling analysis of Baltean-Lugojan and Misener [6, Assumption 2.2] is a related decomposition method. It uses one quality, removes feed and pool capacities, and fixes positive product demands, unlike the capacitated, arbitrary-attribute model of Corollary 4.3.

These corollaries extend the one-pool, fixed-input algorithms of Boland et al. [10, Theorem 1] and Haugland and Hendrix [26, Theorem 4.1], whose models omit bypasses; the latter discusses this convention explicitly. For its final no-bypass model, Haugland [24, Section 6, p. 214] asks about a fixed pool count together with a bound on inputs, products, or qualities. Corollaries 4.3 and 4.4 answer the input-bounded and quality-bounded cases, respectively. For the remaining case, in which only pools and products are bounded and the numbers of inputs and qualities grow, Theorem 3.9 gives ordinary NP-completeness with two pools and two products, without bypasses. The two-source/two-product hardness results in the same Haugland article allow the pool count to grow, so they do not contradict the fixed-input conclusion here. The earlier one-pool algorithms and the fixed-dimensional Minkowski and algebraic algorithms are prior ingredients; the uniform block theorem and its pooling mappings are the additional results proved here.

4.3 Affine quality rank and bypass vertex integrity

The preceding block decompositions fail for unrestricted bypass graphs, because a bypass can couple an input block to a product block. A small set of bypass endpoints can absorb these couplings if the remaining components are small. Low affine rank of the input qualities lets us retain all product specifications with only a fixed number of aggregate masses.

For a graph G , define its vertex integrity by

$$\iota(G) = \min_{W \subseteq V(G)} \left(|W| + \max_{H \text{ component of } G-W} |V(H)| \right),$$

where the maximum over no components is zero. We use this established graph parameter only through its definition. For the polynomial-time results with fixed parameters here, a bounded value is equivalent to fixed integers c, h such that deleting at most c vertices leaves components of at most h vertices. A vertex cover of size c gives $h = 1$.

Theorem 4.5 (Controlled bypass structure). *Fix the pool count p , the affine rank t of the rational input-quality vectors, and a bound on the vertex integrity of the bypass graph. Pooling feasibility and global optimization then admit polynomial-bit exact algorithms, with a polynomial-size algebraic optimizer when feasible. Inputs, products, attributes, and bypass arcs may all grow. The product specification vectors are unrestricted.*

Proof. The empty-input instance has zero flow and is checked directly. Otherwise use the exact affine coordinates of Section 1.2:

$$\lambda_i = \lambda_0 + B a_i, \quad q_\ell = \lambda_0 + B \eta_\ell, \quad B \in \mathbb{Q}^{K \times t}. \quad (64)$$

The columns of B are independent and all rational encoding lengths are polynomial. Put $\min_i a_{is} \leq \eta_{\ell s} \leq \max_i a_{is}$ for every coordinate. Active pools have these averaged coordinates; inactive pools may be assigned arbitrary coordinates in the same box. The pool equations are mass balance and $\sum_i a_{is} y_{i\ell} = \eta_{\ell s} \sum_j v_{\ell j}$. Their equivalence to all original attribute balances follows by multiplying by B and using its column independence.

Choose deleted inputs $A \subseteq I$ and products $B_0 \subseteq J$, with $c_I = |A|$, $c_J = |B_0|$ and $c_I + c_J \leq c$, leaving components (I_β, J_β) of at most h vertices. This can be found by enumerating all vertex subsets of size at most c and checking their components, in polynomial time for fixed c . Failure of this search means only that the structural hypothesis is absent, not that the pooling instance is infeasible. Isolated vertices are components.

The core contains all pt coordinates $\eta_{\ell s}$, all existing $y_{i\ell}$ with $i \in A$, all $v_{\ell j}$ with $j \in B_0$, and all z_{ij} with $(i, j) \in A \times B_0$. It also contains, for each $j \in B_0$, a total delivery variable τ_j and coordinate-mass variables M_{js} , $1 \leq s \leq t$. Each component block owns

$$\begin{aligned} & y_{i\ell} \ (i \in I_\beta), \quad v_{\ell j} \ (j \in J_\beta), \\ & z_{ij} \ ((i, j) \in I_\beta \times J_\beta), \quad z_{ij} \ ((i, j) \in A \times J_\beta), \quad z_{ij} \ ((i, j) \in I_\beta \times B_0), \end{aligned} \quad (65)$$

restricted to existing arcs. Every flow is assigned exactly once: no bypass joins two distinct remaining components. Each block has at most $ph + ch + h^2$ coordinates. For $h = 1$, the sharper bound is $p + c$.

All arc bounds belong to the owning core or block. For each input in I_β , all its incident flows are in block β , so its withdrawal bounds are local. For each product in J_β , the same is true of its delivery and attribute rows. For example, its upper row is

$$\sum_\ell (\lambda_{0\ell} + (B\eta_\ell)_k - \bar{\mu}_{jk}) v_{\ell j} + \sum_i (\lambda_{ik} - \bar{\mu}_{jk}) z_{ij} \leq 0. \quad (66)$$

The lower row reverses the corresponding quality differences. Given the core these are linear inequalities in the block coordinates. An arbitrary attribute count creates only more local rows.

The remaining pool mass and coordinate equations are

$$\sum_{\beta} \left(\sum_{i \in I_{\beta}} y_{i\ell} - \sum_{j \in J_{\beta}} v_{\ell j} \right) = \sum_{j \in B_0} v_{\ell j} - \sum_{i \in A} y_{i\ell}, \quad (67)$$

$$\sum_{\beta} \left(\sum_{i \in I_{\beta}} a_{is} y_{i\ell} - \eta_{\ell s} \sum_{j \in J_{\beta}} v_{\ell j} \right) = \eta_{\ell s} \sum_{j \in B_0} v_{\ell j} - \sum_{i \in A} a_{is} y_{i\ell}. \quad (68)$$

There are $p(t+1)$ equations, with affine block coefficients and core right-hand sides of degree at most two. Pool bounds apply to $\sum_{j \in B_0} v_{\ell j} + \sum_{\beta} \sum_{j \in J_{\beta}} v_{\ell j}$. For a deleted input $i \in A$, withdrawal bounds apply to

$$\sum_{\ell} y_{i\ell} + \sum_{j \in B_0} z_{ij} + \sum_{\beta} \sum_{j \in J_{\beta}} z_{ij}.$$

These contribute at most $2p + 2c_I$ aggregate inequalities.

Deleted products need a separate step. Leaving all of their original attribute rows as aggregate inequalities would make their number grow with K , even when t is fixed. Instead define their retained totals by

$$\sum_{\beta} \sum_{i \in I_{\beta}} z_{ij} = \tau_j - \sum_{\ell} v_{\ell j} - \sum_{i \in A} z_{ij}, \quad (69)$$

$$\sum_{\beta} \sum_{i \in I_{\beta}} a_{is} z_{ij} = M_{js} - \sum_{\ell} \eta_{\ell s} v_{\ell j} - \sum_{i \in A} a_{is} z_{ij}. \quad (70)$$

These use $c_J(t+1)$ aggregate equations. All their delivery and quality requirements now lie in the core: delivery bounds apply to τ_j , and the attribute rows are

$$(\lambda_{0k} - \bar{\mu}_{jk})\tau_j + \sum_s B_{ks} M_{js} \leq 0, \quad (\underline{\mu}_{jk} - \lambda_{0k})\tau_j - \sum_s B_{ks} M_{js} \leq 0. \quad (71)$$

No rank hypothesis on the product specification vectors was used. To bound these variables, let H_j be the sum of the finite upper bounds on all arcs entering j , and $A_s = \max_i |a_{is}|$. Valid bounds are $0 \leq \tau_j \leq H_j$ and $|M_{js}| \leq A_s H_j$, because all incoming coordinate qualities lie in $[-A_s, A_s]$. The bounds are rational of polynomial encoding length. The core domain, formed from these boxes, core arc bounds, and (71) and delivery bounds, is compact.

For exactness, map every physical flow to its uniquely owned coordinates, choose its active pool averages, and set the retained product totals to their actual values. All constructed equations hold. Conversely, the local rows impose every undeleted external-node bound, (68) imposes the exact pool balances, and the pool and deleted-input aggregate inequalities impose their remaining throughput bounds. Equations (70) identify the actual deleted-product masses, so (71) gives exactly their original attribute requirements. At zero pool throughput nonnegative incident flows vanish, and no division is needed. All physical arc costs belong to their owning core or block objective and are unchanged. Thus both formulations have the same feasible flows and cost values.

The parameters of Theorem 4.1, after adding bounded scalar slacks, satisfy

$$\begin{aligned} r &\leq pt + pc + c^2 + c_J(t+1), \\ d &\leq \max(1, ph + ch + h^2), \\ k &\leq p(t+1) + c_J(t+1) + 2p + 2c_I, \quad D = 2. \end{aligned} \quad (72)$$

All are fixed. If no blocks remain, append a zero block. The theorem therefore supplies exact optimization and recovery; the rational map (64) recovers every original pool attribute in polynomial bit time as well. $\square$

Corollary 4.6 (Specializations and elementary cases). *Theorem 4.5 includes each of the following cases:*

- (i) *Fixed pools, fixed attribute count, and a fixed-size bypass vertex cover, allowing arbitrarily many bypasses at fixed input or product hubs.*
- (ii) *Fixed pools and fixed affine quality rank, with bypass components of bounded size, including an unbounded matching.*
- (iii) *Fixed pools and fixed affine quality rank without bypasses. A fixed number of distinct input-quality vectors also suffices in place of the rank bound.*
- (iv) *Fixed inputs and pools with arbitrary bypasses and attributes, as in Corollary 4.3.*

With no pools, or with affine quality rank zero, the model is an LP without any bypass-structure restriction; in the rank-zero case the pool count can also grow.

Proof. For (i), $t \leq K$ and a vertex cover leaves isolated vertices. For (ii) take $c = 0$. For (iii), a graph with no edges has singleton components, and q distinct vectors have affine rank at most $q - 1$. For (iv), all inputs form a vertex cover and $t \leq |I| - 1$. With no pools only known input-quality bypasses remain. At rank zero all input vectors equal λ_0 ; set every pool quality to λ_0 . Conservation then makes all mixing equations redundant and all product conditions linear. Inactive pools cause no exception. $\square$

The covered-product totals in (70) are what permit affine rank in place of the attribute count: replacing the attribute count by rank in a row-count bound alone would overlook the arbitrary specification rows at a covered product. Graph degree and treewidth alone do not bound vertex integrity, since paths already have growing vertex integrity. The next subsections treat path geometry directly, with a different retained boundary and a more restricted objective.

4.4 Composition of planar relations

A bypass path can be described by its consecutive scalar arc flows. Each internal node gives a compact planar relation between its two incident flows. We show that projecting a whole path onto its two endpoint flows has polynomial description size. This projection statement is stronger than deciding a fixed linear instance, and it retains no accumulated objective coordinate.

Lemma 4.7 (Two planar relations). *Let $P \subset \mathbb{R}^2$ in coordinates (x, y) and $Q \subset \mathbb{R}^2$ in coordinates (y, z) be compact rational polyhedra, given by m_P and m_Q weak linear inequalities. Their composition*

$$P \circ Q = \{(x, z) : \exists y, (x, y) \in P, (y, z) \in Q\}$$

has a rational inequality description with at most $64(m_P + m_Q + 2)$ rows. It can be constructed with polynomially many rational operations. If the input row coefficients have bit lengths at most b_P, b_Q , the output coefficients can be chosen with bit length at most $C(b_P + b_Q + 1)$ for a universal constant C . Empty sets, segments, and points are included.

Proof. First compute the interval $J = \text{proj}_y Q$ and clip P by $y \in J$. This adds at most two rows. Empty cases are immediate. Suppose initially that the resulting x domain and J are nondegenerate intervals. Write the vertical slices of clipped P as $[l(x), u(x)]$ and those of Q as $[q(y), r(y)]$. The functions l, q are convex piecewise affine, and u, r are concave piecewise affine. Each of l, u has at most $m_P + 2$ affine pieces, and each of q, r has at most m_Q pieces: each piece comes from a nonvertical supporting row of the relevant polygon. These statements also hold when the lower and upper boundaries coincide.

The minimum attainable z for a given x is $g(x) = \min_{l(x) \leq y \leq u(x)} q(y)$. Let $[a, b] \subseteq J$ be the minimizer interval of q , and $q_{\min}$ its minimum. Define on J

$$D(t) = \begin{cases} q(t) & t < a, \\ q_{\min} & t \geq a, \end{cases} \quad L(t) = \begin{cases} q_{\min} & t \leq b, \\ q(t) & t > b. \end{cases}$$

The function D is convex and nonincreasing; L is convex and nondecreasing. The interval $[l(x), u(x)]$ either lies to the left of $[a, b]$, meets it, or lies to its right. In the three cases its minimum of q is respectively $q(u(x))$, $q_{\min}$, or $q(l(x))$. Equivalently,

$$g(x) = \max\{L(l(x)), D(u(x))\}. \quad (73)$$

Both terms are convex: a convex nondecreasing outer function preserves convexity, and a convex nonincreasing outer function composed with a concave inner function is convex. These facts follow directly from the defining convexity inequality and the indicated monotonicity.

To count pieces of a composition, subdivide the inner domain at its own knots and at the boundary points of the preimages of every outer knot. A level set of a univariate convex or concave function has at most two boundary points: away from a minimum or maximum plateau there are at most two crossings, while a plateau contributes only its two endpoints. Thus, if the inner and outer functions have n and m affine pieces, this subdivision has at most $n + 2m + 2$ pieces, using a loose bound that also covers domain endpoints. On each open piece the composition is affine; flat preimages introduce no extra formula. Each of D, L has at most $m_Q + 1$ pieces. Consequently each term in (73) has at most $m_P + 2m_Q + 6$ pieces.

A convex piecewise-affine function on an interval equals the maximum of the affine extensions of its pieces: each such line supports the whole function. Therefore the maximum of two such functions uses at most the sum of their numbers of pieces. It follows that g has at most $2m_P + 4m_Q + 12$ pieces. Reflecting z gives the same bound for the concave upper boundary. The feasible z at fixed x form an interval, since they are the projection of a nonempty convex slice. Hence these two envelopes, with two vertical domain bounds, describe the composition. The resulting row count is at most $4m_P + 8m_Q + 26$, below the stated loose bound.

For degeneracies, if the x domain is a singleton, the composition is a vertical interval or point; minimize q and maximize r over the corresponding feasible y interval. If J is a singleton, clipping fixes y and the composition is the Cartesian product of an x interval and a z interval. These need at most four rows. A nonvertical segment has coincident affine lower and upper envelopes and was already covered. The empty relation can be described by $0 \leq -1$.

For a constructive algorithm and its coefficient bound, combine the row descriptions of $P(x, y)$ and $Q(y, z)$ and perform one Fourier–Motzkin elimination of y . Keep rows with zero coefficient on y ; combine every pair with opposite coefficient signs after positive scaling to cancel y . This gives at most $(m_P + m_Q)^2 + m_P + m_Q$ rows in (x, z) and is an exact projection. Each new row is formed from two input rows by a fixed number of rational additions and multiplications. Pairs may come from the same child as well as different children; either case satisfies the stated $C(b_P + b_Q + 1)$ bit bound.

Compute the bounded planar result's vertices by pairwise line intersections and row feasibility checks. A nonempty bounded polyhedron has vertices, even if its affine dimension is smaller. In full dimension, remove redundant rows one at a time, retaining one row for each facet; the envelope argument bounds their number. Testing all rows against the original list and deleting them simultaneously would be invalid in the presence of duplicate rows. Redundancy is decidable by optimizing each candidate row over the system with that row removed, using an exact planar LP (which may be unbounded). In dimension one, choose the two extreme endpoints from the vertices and describe their segment by a line equation and two endpoint inequalities; in dimension zero, fix the two coordinates. These exceptional descriptions also require only a constant number of rational operations on input-row intersections per coefficient. All cases therefore give the claimed output size, polynomial arithmetic work, and universal coefficient-bit bound. $\square$

Theorem 4.8 (Exact path endpoint projection and lifting). *For a chain of compact rational planar relations*

$$(x_{i-1}, x_i) \in P_i \quad (1 \leq i \leq n), \quad (74)$$

the exact projection onto (x_0, x_n) has a rational inequality description of polynomial size and bit length, computable in polynomial bit time. A feasible endpoint pair can be lifted to all internal coordinates using polynomially many arithmetic operations and comparisons. If the endpoints lie in a common represented real-algebraic field of polynomial degree and encoding length, the lifted coordinates stay in that field and have polynomial total encoding length; the lift takes polynomial bit time, including the input and output encoding.

Proof. Compose consecutive subchains in a balanced binary tree. Write $s(P) = m_P + 2$ for a relation's row count plus two. By Lemma 4.7, the parent size is at most a universal constant times the sum of its children's sizes. The depth is at most $\lceil \log_2 n \rceil$. At depth a the total size is at most a constant to the power a times the total leaf size. Therefore the maximum and total intermediate sizes are $n^{O(1)} \sum_i (m_i + 2)$. Each composition takes polynomial time in its children's sizes, so the total arithmetic work is polynomial. The same recurrence applies to rational coefficient bit length: $b_{\text{parent}} \leq C(b_{\text{left}} + b_{\text{right}} + 1)$. Logarithmic depth makes the resulting bit length polynomial in the original encoding. Rational comparisons, line intersections, and redundancy checks consequently all have polynomial bit cost. This argument uses balanced composition; no comparable bound for an arbitrary sequential elimination order is needed.

Store every child relation in the tree. At an internal node with feasible endpoints (x, z) , let $Y_L(x)$ be the left child's slice of shared y values and $Y_R(z)$ the right child's slice. These are closed bounded intervals, and their intersection is nonempty by the definition of composition. Choose the midpoint of that intersection and recurse into both children. This proves exact lifting, including zero-length slices.

For the representation guarantee, each endpoint of a slice is a maximum or minimum of rational affine functions of the prescribed outer coordinate. Rows whose coefficient on y is zero are feasibility tests; every division used to compute a finite slice bound is by a nonzero rational row coefficient. Exact comparisons select the active interval bounds. Their midpoint is a rational affine combination of the two parent endpoints. Recursing along the balanced tree therefore expresses every recovered coordinate as a rational affine combination of the original two endpoints, with polynomial coefficient length by the same depth bound. No new algebraic root or field extension is required. Exact sign comparisons in the supplied common field have polynomial bit complexity in its degree and representation size, proving the last claim. $\square$

Pairwise projection by resultants is established in the literature on two variables per inequality; see the revised manuscript of Simon et al. [42, Section 3]. The envelope and elimination methods of

Hochbaum and Naor [29, Section 2] are further algorithmic antecedents. The path proof above does not rely on a general polynomial-size closure claim for all such systems; it controls both the number of rows and the bit length for the two-endpoint projection that pooling needs.

4.5 Bounded pool attachments and arbitrary attributes

A bounded number of pool-incident arcs gives a fixed physical interface with the bypass graph, even if its path components are long. The input qualities on those paths are known, so their internal relations are rational and independent of the pool mixture.

Theorem 4.9 (Bounded attachments through degree-two bypasses). *Fix the total number a of input-pool and pool-product arcs. If the bypass graph has maximum degree at most two, pooling feasibility is decidable in polynomial bit time, with exact algebraic feasible-flow recovery. The number of attributes, distinct input-quality vectors, inputs, products, and bypass arcs is unrestricted. Multiple pools and all the rational throughput, arc, and quality bounds specified at the start of this section are allowed.*

Proof. A pool with no incoming or no outgoing arc has zero throughput and zero incident flows. Check that zero belongs to its throughput interval and every incident-arc interval, then remove these arcs and the pool. Preserve each adjacent input or product requirement, since other arcs may satisfy it. If no pool arcs remain, the instance is a rational blending LP. We may thus assume every remaining pool has an inlet and an outlet.

Call an external node adjacent to a pool an attachment. There are at most a distinct attachments, including shared nodes adjacent to several pools. Retain in the core all at most a pool-incident flows, all bypass flows incident to any attachment, and a fraction $\theta_{i\ell}$ for every existing inlet arc. At most $2a$ distinct bypass arcs are retained because each attachment has bypass degree at most two; an arc between two attachments is retained only once. There are at most a fractions. Thus the core has at most $4a$ scalar coordinates. Put all flow bounds in the core for retained arcs and require

$$\theta_{i\ell} \geq 0, \quad \sum_{i:i\ell \in A} \theta_{i\ell} = 1, \quad \sum_i y_{i\ell} = \sum_j v_{\ell j}, \quad y_{i\ell} = \theta_{i\ell} \sum_{i'} y_{i'\ell}. \quad (75)$$

Here the arc set A has its original meaning from Section 1.1. For each pool define its full quality vector by $q_\ell = \sum_i \lambda_i \theta_{i\ell}$; this is an expression, not K new variables. All input or product throughput rows at attachments involve retained flows only. So do their quality rows, after this substitution, and all pool throughput bounds. Their degrees are at most two, even when K grows. At positive throughput the equations force the actual mixture fractions. At zero throughput any legal simplex fraction gives the same zero transmitted mass.

Delete the attachment nodes from the bypass graph. Every remaining component is a path, a cycle, or an isolated node. A component incident to a deleted node is a path, possibly an isolated node, with at most two boundary arcs: a remaining vertex on a cycle already has its full two neighbors within that cycle, so such a cycle cannot meet the boundary. For a path, only its endpoints have room for a boundary arc. The flows on these boundary arcs have been retained in the core. Every nonattachment input has only bypass flows and hence has a withdrawal row in its at most two adjacent scalar arc coordinates. Every nonattachment product likewise has only known-quality bypass flows; its delivery and each attribute row involve its at most two adjacent coordinates, with rational constant coefficients. Together with the finite arc boxes, each node thus gives a compact planar relation between successive flows along the path.

For a component with two boundary arcs, apply Theorem 4.8 and insert its exact endpoint inequalities into the core. A component with one boundary arc has an interval projection; append a

dummy endpoint fixed to zero at its free end and express the terminal node's univariate rows as a planar relation with this dummy coordinate. The same theorem gives the required projection after substituting zero. Components with no boundary, including detached cycles, are rational LP feasibility problems and can be solved independently. A node with no usable arc and a positive withdrawal or delivery lower bound is rejected by these same local conditions. A bypass with both endpoints at attachments is already wholly in the core. A bypass cycle cut by an attachment can leave a path whose two boundary arcs meet that same attachment; the two arc coordinates are still distinct scalar endpoints, so the composition proof is unchanged.

The resulting core has fixed dimension, polynomially many polynomially encoded rows, and degree at most two. Its feasible set is exactly the projection of the bounded physical formulation with simplex fractions: the forward implication restricts a physical flow, and the converse lifts every projected path and combines it with the independently feasible detached components. All attachment rows and (75) remain, so this converse enforces every physical mixing, capacity, and quality requirement. The core is compact, and fixed-dimensional real-algebraic decision and sampling give a feasible core point or an infeasibility decision in polynomial bit time.

Represent the sampled coordinates in one polynomial-degree field. Lift all boundary-connected paths through their stored composition trees using Theorem 4.8; their rational affine lifts preserve that field and polynomial encoding. Detached components have polynomial-bit rational LP witnesses. Finally evaluate the affine expressions for all pool qualities. This recovers an exact physical feasible flow and proves the theorem. $\square$

Corollary 4.10 (Two feeds and two outlets). *With one pool, at most two actual inlet arcs, at most two actual outlet arcs, and bypass degree at most two, pooling feasibility and exact feasible-flow recovery are polynomial-time problems, even with arbitrarily many attributes.*

Proof. Theorem 4.9 applies with $a \leq 4$. A sharper core count is thirteen: four pool flows, at most eight incident bypass flows, and one inlet fraction $\theta \in [0, 1]$. With two feeds set $q = \theta\lambda_1 + (1 - \theta)\lambda_2$ and $y_1 = \theta(y_1 + y_2)$, together with mass balance. With one feed the fraction is fixed and the core is smaller. All zero-throughput and missing-arc cases are covered by the preceding proof. $\square$

4.6 Objectives on a fixed set of retained coordinates

Theorem 4.11 (Bounded-support exact optimization). *Under Theorem 4.9, fix also an integer s . Designate at most s further actual arc-flow coordinates. Any rational polynomial objective of fixed degree in these coordinates and the retained attachment core can be globally optimized in polynomial bit time, with exact optimum and physical optimizer recovery. Polynomially many additional closed polynomial equalities and weak inequalities of fixed degree on those same retained coordinates are allowed. In particular, the conclusion covers a linear cost supported on a fixed number of actual arcs.*

Proof. Retain every designated flow in the core and cut the scalar relation chains at it. A formerly eliminated path then splits into segments with at most two retained endpoints. In a detached cycle, cutting at one designated arc gives a path relation $R(x, x)$: the two endpoint occurrences refer to the same retained flow. Several designated arcs split the cycle into ordinary two-endpoint segments. Isolated nodes, one-endpoint segments, and components without retained coordinates are handled as in Theorem 4.9. This cuts relations at coordinates, so local node constraints remain attached to their correct pairs; none is discarded.

The projected system has at most $4a + s$ variables, polynomial encoding, and fixed degree, including the new constraints. It is the exact projection of a closed bounded feasible set, and hence compact. Its continuous polynomial objective attains an optimum if it is feasible. Introduce an

objective-value variable, eliminate quantifiers to describe attainable values, and sample an optimal fiber; equivalently use the optimality formula (60) with this fixed-dimensional system. Fixed-dimensional real-algebraic algorithms have polynomial bit complexity and return a polynomial-size common-field sample. Affine path lifting recovers all eliminated flows while preserving the objective, because every coordinate on which it depends was retained. Strict additional constraints are excluded from this attainment statement. $\square$

Corollary 4.12 (An objective reducible by exact equalities). *Let $Ef = e$ be rational linear conservation or contract equations valid for every feasible flow in an instance of Theorem 4.9. Suppose a supplied rational identity*

$$c = E^T \gamma + d \quad (76)$$

has polynomial encoding length, where d is supported on a fixed number of arcs. Then exact minimization of $c^T f$ is polynomial-time solvable, with value $\gamma^T e + \min d^T f$ and physical optimizer recovery.

Proof. Check (76) exactly by rational arithmetic. For every feasible f , $c^T f = \gamma^T e + d^T f$; apply Theorem 4.11 to d . The equality must hold throughout the feasible set, not just at selected candidates or for tight inequalities guessed to hold at an optimum. $\square$

For example, along a bypass path whose consecutive inputs have exact supplies S_i , write their two arcs to consecutive products as $S_i - u_i, u_i$. With product revenues R_{i-1}, R_i , the contribution of these arcs to revenue is

$$\sum_i (R_{i-1}(S_i - u_i) + R_i u_i) = \sum_i R_{i-1} S_i + \sum_i (R_i - R_{i-1}) u_i. \quad (77)$$

Their input production costs are constant under the exact supplies. Thus a fixed total number of revenue changes along these paths gives bounded support, with any attachment contributions already retained. The number of distinct revenues alone does not bound the number of changes: two values may alternate along an arbitrarily long path.

4.7 A sharp feasibility boundary and the remaining scope

Corollary 4.13 (Product degree two versus three). *Consider one-pool pooling with exactly two actual inlet arcs and two actual outlet arcs, total input out-degree at most two, and positive flow lower bounds permitted. If every product has total in-degree at most two, feasibility is decidable in polynomial bit time for arbitrary rational quality data. Allowing product total in-degree three makes feasibility strongly NP-complete, even with one upper-bound scalar quality, flow bounds in $\{0, 1, 2, 3, 4\}$, and quality data in (47).*

Proof. For the positive branch the bypass graph is a subgraph of the full external-node incidence structure, so its maximum degree is at most two. Corollary 4.10 applies. It actually allows some attachments of larger total degree when their bypass degree remains at most two.

For the negative branch, take the physical construction in the proof of Theorem 3.19 before the final contract-completion objective (52) is introduced. Retain its exact source supplies, collector demands, and all other designated flow contracts. The proof through (51) already shows that this contracted network is feasible exactly when the positive-product source instance is a yes instance. The source threshold is encoded by physical binary linear circuits, not by an extra objective row that would disappear when economics are removed.

The degree-two input and degree-three product bounds, the two actual feeds and outlets, and the fixed quality alphabet all hold before contract completion. Positive exact node contracts have values in $\{1, 2, 3, 4\}$, and all flow bounds lie in the stated alphabet. The source’s large rational data appear only in polynomial-size circuit topology, so even unary encoding of the physical data has polynomial size. This proves strong feasibility NP-hardness. With one pool and one scalar quality, the fixed-dimensional linear-fiber certificate in Corollary 2.11 gives NP membership. $\square$

The boundary concerns feasibility with prescribed nonzero requirements. When every flow lower bound is zero, the zero flow is feasible, and a profit threshold is a different decision problem. The positive result remains valid when its data are restricted to the negative branch’s fixed alphabets.

The path algorithm retains a fixed number of actual flow coordinates. A two-coordinate endpoint feasibility relation does not represent an unrestricted linear cost or a dense additional aggregate over eliminated arcs: interior flows with the same endpoints can have different aggregate values, and a threshold on that value cannot be checked after it is discarded. These theorems therefore do not resolve general profit optimization for one pool with two feeds, two outlets, and degree-two bypasses, and degree-two bypasses with unboundedly many attachments remain open (Section 6.3).

5 Exact contracts and their exceptions

Exact external contracts can remove the aggregate equations that obstruct path elimination. Throughout this section there is one pool, no pool–pool arc, and finite rational bounds on every actual flow. Write $C_i = \lambda_i$ for an input quality, y_i, v_j, z_{ij} for feed, outlet, and bypass flows, and q for pool quality. An absent feed or outlet has bounds $[0, 0]$. Exact supplies are a_i , and an exact product contract is demand b_j and quality B_j . The symbols C_i, B_j, q denote vectors except in explicitly scalar arguments. All lower bounds, including positive individual arc bounds, are retained. Exact quality at zero demand has the homogeneous meaning of Section 1.1.

The reductions use contracts in three ways. With a fixed set of outlets, the conserved input mass and attribute mass become quantities on a bounded boundary. Full contracts turn scalar quality equations into signed node divergences, whose feasibility cuts can be enumerated on paths and cycles. With a fixed number of exceptions, an active ordinary outlet confines the pool-quality vector to an affine plane, even if the ambient quality rank grows. Restrictive common capacity and two distinct full input vectors need separate arguments; neither case follows from the conservation identities alone.

The algorithms use rational LP, exact arithmetic in a common represented real-algebraic field, and fixed-dimensional real-algebraic decision and sampling, which allows polynomially growing degrees under dense encoding [7, Theorem 1.3.1 and Sections 3.1.3–3.2]. The algebraic LP step uses Adler and Beling [3, Section 5, Remark 1]. Polynomial time always means rational bit time; the exponent may depend on the specified fixed parameters.

Table 3 compares the physical classifications. An exact external contract is an exact source supply or an exact product demand and quality; exceptions may instead have the corresponding intervals. The bypass degree Δ counts only bypass arcs. T is pool throughput and E denotes standard input costs and product revenues. The witness column describes all original flows in one represented real field: F_{poly} has polynomial degree, $F_{\leq 2}$ has degree at most two, and $\mathbb{Q}$ means rational flows. Every row gives polynomial bit time and polynomial witness encoding for fixed values of its parameters. All individual arc intervals are allowed, except that the last row requires zero outlet lower bounds. “Redundant” common bounds mean lower bound zero and a redundant upper bound; “arbitrary” permits restrictive lower and upper bounds.

Table 3: Scope of the one-pool physical algorithms. The cited statements give the full hypotheses.

Result / qualities	Contracts and fixed parameters	Bypass	Common bounds	Objective	Witness
Theorem 5.3; arbitrary rank	Exact sources; fixed r allowed outlets, whose products may have intervals; all other products exact	$\Delta \leq 2$	Arbitrary	E plus linear costs on the r outlets and T	F_{poly}
Theorem 5.6; scalar or affine rank ≤ 1	All external nodes exact	$\Delta \leq 2$	Redundant	Feasibility; E is constant	$F_{\leq 2}$
Theorem 5.10, Corollary 5.9; arbitrary rank	Fixed s exceptional external nodes; all ordinary nodes exact	$\Delta \leq 2$	Redundant	E ; a fixed number of additional arc costs	F_{poly}
Theorem 5.13, Corollary 5.14; scalar or affine rank ≤ 1	All external nodes exact	$\Delta \leq 2$	Arbitrary	E plus a rational linear cost on T	F_{poly}
Theorem 5.15, Corollary 5.16; at most two full input vectors	Exact products; source supply intervals; outlet lower bounds zero	Any	Lower zero; upper arbitrary	Feasibility only; E is $\mathbb{Q}$ constant if supplies are exact	

In the exception row, Corollary 5.9 gives the scalar and fixed-rank cases and Theorem 5.10 gives unrestricted rank. Additional designated arc costs are handled by a fixed enlargement of the exceptional set, as described after that theorem. The final row also permits the additional outlet resource rows of Corollary 5.16. No row covers arbitrary dense arc costs, and the restrictive-common-bound row gives no constant bound on the witness field degree.

5.1 Parameterized path relations

The two theorems in this subsection describe symbolic elimination with unknown parameters, relevant to the open boundary of Section 6.3. They retain endpoint relations, not accumulated costs or dense aggregates, and the later physical classifications do not use them.

Theorem 5.1 (One parameter and general planar relations). *Let ρ range over a compact rational interval and let $(x_{i-1}, x_i) \in P_i(\rho)$, $1 \leq i \leq h$, where each P_i is specified by weak inequalities linear in its two state coordinates, with explicitly encoded rational polynomial coefficients of fixed degree in ρ . Suppose every state has a uniform finite rational box bound. For input length N , the exact endpoint relation has a representation on $N^{O(\log(h+1))}$ parameter cells. Each cell is an open interval or singleton with polynomial-degree, polynomial-encoding algebraic endpoints. On each cell the relation has polynomially many linear endpoint inequalities with polynomial-degree, polynomial-bit integer-polynomial coefficients. This representation is constructible in quasipolynomial bit time. Existential feasibility has the same bound and admits a polynomial-encoding witness in one polynomial-degree real algebraic field.*

Proof. Clear rational denominators in the initial rows. Include both endpoints of the supplied parameter interval as singleton cells. Compose along a balanced binary tree, padding with bounded identity relations if necessary. Consider two children on a common open parameter cell, each with a fixed row list whose coefficients have degree at most D . In coordinates (x, y, z) , eliminate y by Fourier–Motzkin elimination. For rows with y coefficients $b_i > 0 > b_j$, use $-b_j$ times the first row plus b_i times the second. Retain all zero- y rows and all applicable pairs, including pairs from the

same child. There are quadratically many candidates in the total child row count, and coefficient degrees at most $2D$. Split first at roots of the b_i to fix the applicable pairs. At a root, treat a vanishing coefficient as zero, using its actual value.

For the resulting rows $A_r x + B_r z \leq C_r$, retain the four fixed box rows and split further at every root of every nonzero minor of orders one, two, and three in the augmented matrix (A_r, B_r, C_r) . For k candidates there are $O(k^3)$ such polynomials, of degree $O(D)$. Their signs fix whether any two boundary lines intersect uniquely and, by Cramer's rule, whether that intersection satisfies each row. Every nonempty bounded planar polyhedron, including a point or segment, has a vertex determined by two independent boundary lines.

On a sample in each open sign cell, delete redundant candidate rows one at a time, keeping the box. The same subset remains equivalent throughout the cell: each potential vertex of its bounded polyhedron has invariant feasibility and invariant satisfaction of every omitted row, so validity at all vertices persists. Emptiness also persists, since the existence of a feasible intersection pair is sign-determined. At every singleton, perform the same selection in the field of that parameter, recomputing zero coefficients and applicable elimination pairs there; specializing either neighboring open-cell description need not give the singleton projection. Store the selected symbolic rows instead of substituting unrelated algebraic coefficients into later steps. A row need represent the child only on its stored cell, because a parent uses it only on a common refinement. On an empty sign cell, store the constant contradiction $0 \leq -1$ together with the box, so an empty relation also has a constant-size description.

Lemma 4.7 bounds the number of rows needed in a nonempty composition by a universal constant times the sum of child sizes. This bound also holds for a subset of the candidate rows. In dimension two, keep the facet rows. For a segment, rows active at a relative interior point have normals perpendicular to its line; there must be normals of both signs, since otherwise a small displacement off the line would remain feasible. Two such rows define the affine hull, and at most two more rows delimit its endpoints. For a point, the active normals positively span the plane, since otherwise a nonzero feasible displacement would exist; at most four of them suffice: either three surround the origin or two opposite pairs do. Thus in each dimension a bounded-size subset of the original candidate rows gives the description. The four box rows add four to these bounds. Hence, with a larger universal C and the retained box,

$$m_{\text{parent}} \leq C(m_{\text{left}} + m_{\text{right}} + 2).$$

Logarithmic depth makes all row counts polynomial in N .

With one parameter, the common refinement of the two child partitions has at most the sum of their numbers of breakpoints. On each common cell, coefficient roots and minor roots add polynomially many cells. Consequently

$$L_{\text{parent}} \leq N^c(L_{\text{left}} + L_{\text{right}})$$

for a fixed c . Iterating over $O(\log(h+1))$ levels gives the quasipolynomial bound, including all intermediate work. This step requires a scalar parameter.

Each coefficient combination takes two polynomial products and a sum. Degree at most doubles per level, so it is $O(dh)$ for initial degree d . Coefficient bit lengths satisfy the same doubling recurrence with an additional logarithmic convolution term and remain polynomial in N . All minors therefore have polynomial degree and coefficient bit length. Exact univariate root isolation, comparison, and sign evaluation have polynomial cost per polynomial; separation bounds give polynomial-bit rational samples between distinct roots. Singleton arithmetic takes place in the field of the single chosen parameter and introduces no new parameter root.

It remains to prove the feasibility claim. On every cell of the final partition, nonemptiness of the endpoint relation is invariant: for $h \geq 2$ each composition step splits at the minor roots, and for $h = 1$ apply the same splitting to the rows of P_1 (equivalently, pad the chain with a bounded identity relation). Hence it suffices to test one sample of each open cell and each singleton. At a feasible parameter, solve the original bounded path LP over its field. A vertex basis expresses its coordinates as Cramer ratios of the original polynomial coefficients evaluated at that parameter. The determinants have polynomial degree and coefficient bit length: a determinant of dimension $O(N)$ has at most $O(N)!$ terms, so its coefficient bit length grows by only polynomially many bits. These expressions give a polynomial-size common-field witness, and the algebraic LP algorithm constructs one. Alternatively, store the balanced composition trees and lift interval slices as in Theorem 4.8. $\square$

Theorem 5.2 (Affine strips, trees, and fixed total cycle rank). *Let η have fixed dimension and range over a supplied quantifier-free rational polynomial domain, possibly all of its ambient space. Suppose a scalar path has finite polynomial coordinate bounds at every parameter and transitions*

$$\ell_i(\eta) \leq x_i \leq u_i(\eta), \quad a_i(\eta)x_{i-1} + b_i(\eta) \leq x_i \leq a_i(\eta)x_{i-1} + c_i(\eta).$$

All polynomials are densely encoded; degrees may grow. On each realizable sign condition of the gains a_i , the projection onto η and a fixed number of states has polynomial size, degree, and coefficient length. The same holds on a tree with arbitrarily oriented edges, and on a graph of fixed total cycle rank after retaining at most twice that many extra states. These constructions give polynomial-time feasibility and exact infimum and attainment analysis for polynomial objectives on the retained coordinates. An optimizer is returned when the infimum is attained; a nonempty compact closed original feasible set suffices.

Proof. First consider a path and a sign cell where every gain is nonzero. Put $A_0 = 1$, $A_i = \prod_{h=1}^i a_h$, and $y_i = x_i/A_i$. The transitions become $\alpha_i \leq y_i - y_{i-1} \leq \beta_i$, where the endpoints $b_i/A_i, c_i/A_i$ are ordered according to the known sign of A_i . The state bounds similarly become $L_i \leq y_i \leq U_i$. Include every lower-versus-upper consistency condition. Define directed distances

$$d(i, j) = \sum_{h=i+1}^j \beta_h, \quad d(j, i) = - \sum_{h=i+1}^j \alpha_h \quad (i < j), \quad d(i, i) = 0.$$

These are shortest-path distances: every walk on the bidirected path reduces to its unique simple path and two-edge excursions of lengths $\beta_h - \alpha_h \geq 0$. Feasibility is exactly

$$L_i \leq U_j + d(j, i) \quad \text{for every } i, j. \tag{78}$$

Necessity follows by summing differences. For sufficiency take $y_i = \min_j (U_j + d(j, i))$. This lies between L_i and U_i ; the triangle inequality yields $y_i \leq y_h + d(h, i)$ and hence both adjacent transition inequalities.

To retain a state, replace its interval by its prospective exact value x_i/A_i and keep its original bounds separately. This gives $O(n^2)$ inequalities on the retained coordinates. Multiple interval bounds are handled by taking all lower/upper pairs, so no active bound needs to be chosen. A zero gain gives simply $b_i \leq x_i \leq c_i$ and no restriction on x_{i-1} : split there, place that interval on its original head, and apply the construction to each part.

For a tree, delete zero-gain edges in the same way. On each remaining tree choose factors satisfying $A_v = a_e A_u$ for each original edge $u \rightarrow v$. Rooting assigns factors as products of gains or their reciprocals along root paths. Dividing by A_v gives the same two bounded differences on each

edge, with signs ordered using A_v . The unique simple path between any two vertices again gives distances and (78), with the same recovery proof. No zero-gain equation is reversed when an edge points toward the root.

Products of polynomially many densely encoded gains have polynomial degree and coefficient bit length in a fixed number of parameters. A common product of all nonzero gains, with a fixed sufficient multiplicity and the original coefficient denominators, clears all the products, reciprocals, and path sums just used. Its sign is known on the cell. Clearing denominators in (78) therefore gives polynomially encoded rows. Enumerating realizable gain sign conditions has polynomial complexity in fixed dimension; zero and lower-dimensional conditions are included. Recovery uses minima and rational functions in the same field as the retained sample.

Finally, remove c edges to form a spanning forest when the total cycle rank is c . Retain their at most $2c$ endpoints, project the forest, and restore every removed original strip row on these states. The remaining dimension is fixed. Quantifier elimination applied to the attainable objective values gives the stated feasibility, infimum, and attainment conclusions. Finite state bounds at each parameter do not by themselves imply bounded parameters or attainment, and the proof does not assume them. $\square$

Both sides of a strip must have the same gain. Arbitrary planar relations are covered instead by Theorem 5.1, whose bound is quasipolynomial and not claimed to be tight. Neither theorem retains an arbitrary dense objective. Two tempting shortcuts do not replace these projection proofs. First, with only scalar upper product quality bounds, lowering a fixed pool quality preserves each local quality row because its pool flow coefficient is nonnegative, so local path feasibility is nested in that parameter. This does not control changing facets or the global pool mixing equality, which is not a local upper-bound row. Second, inequalities $x_i \geq |x_{i-1} - a|$ do not enforce iterated absolute values, because existentially quantified intermediate states may exceed their lower bound.

5.2 A fixed number of pool outlets

Theorem 5.3 (Conserved boundary masses). *Suppose the bypass graph has maximum degree two, every source has exact supply a_i , and the set J_P of allowed pool outlets has fixed size r . Every product outside J_P has an exact product contract; products in J_P may have arbitrary demand and quality intervals. Feasibility and standard economic optimization are polynomial-time solvable, with an exact algebraic original optimizer when feasible. Arbitrarily many feeds and quality coordinates, arbitrary quality rank, restrictive common pool bounds, all individual arc bounds, and linear costs on the r outlets and on pool throughput are allowed.*

Proof. If $r = 0$, mass conservation forces every feed to zero. Check zero against all incident and common bounds and solve the remaining rational LP, retaining external contracts. Otherwise eliminate every feed by

$$Y_i = a_i - \sum_j z_{ij}.$$

Put its actual feed bounds on Y_i , or impose $Y_i = 0$ if absent. Each row uses at most two bypass coordinates. At $j \notin J_P$ impose $\sum_i z_{ij} = b_j$ and $\sum_i C_i z_{ij} = b_j B_j$. These also have at most two coordinates, regardless of the number of attributes.

Retain the bypass set H entering J_P , of size at most $2r$. Define

$$\begin{aligned} T &= \sum_i a_i - \sum_{j \notin J_P} b_j - \sum_{ij \in H} z_{ij}, \\ Q &= \sum_i C_i a_i - \sum_{j \notin J_P} b_j B_j - \sum_{ij \in H} C_i z_{ij}. \end{aligned} \quad (79)$$

Summing the exact source and nonreceiving-product equations gives, for every local bypass lift,

$$\sum_i Y_i = T, \quad \sum_i C_i Y_i = Q. \quad (80)$$

Their coefficients are polynomial-bit sums of rational products. Retain v_j and outlet fractions θ_j for all $j \in J_P$, and require

$$\theta_j \geq 0, \quad \sum_{j \in J_P} \theta_j = 1, \quad v_j = \theta_j T.$$

There are at most $4r$ retained coordinates. Keep all outlet and common pool bounds, every receiving-product demand bound on $t_j = v_j + \sum_i z_{ij}$, and every quality bound on the mass $\theta_j Q + \sum_i C_i z_{ij}$. These are quadratic rows in the core; additional quality coordinates add rows, not variables.

Delete J_P from the bypass graph. Every residual component with a boundary is a path with at most two retained endpoint arcs, and Theorem 4.8 gives its polynomial-size rational endpoint relation. A component with one boundary gives an interval; a component with no boundary, including a cycle or isolated node, is a bounded rational LP. Such a detached component may still feed the pool. Its arbitrary feasible lift does not change (80), so it needs no extra aggregate variable. Two boundary arcs may meet the same deleted product.

The resulting core is closed, bounded, fixed-dimensional, and defined by polynomially many quadratic or linear rows of polynomial bit length. If $T > 0$, (80) gives actual quality Q/T ; its mass into product j is $v_j Q/T = \theta_j Q$. If $T = 0$, nonnegative Y_i all vanish, so $Q = 0$ and every outlet vanishes. Fractions then have no physical effect. Conversely a feasible physical solution gives $\theta_j = v_j/T$ when $T > 0$ and any simplex point when $T = 0$. Thus core feasibility and original feasibility are equivalent, including inactive pools and zero-demand products.

For unit input costs c_i and revenues R_j , profit is

$$\sum_{j \notin J_P} R_j b_j - \sum_i c_i a_i + \sum_{j \in J_P} R_j \left(v_j + \sum_i z_{ij} \right).$$

It is affine in the core. Outlet costs and a cost proportional to T are affine there as well. Fixed-dimensional algebraic optimization gives an attained optimum, and stored path relations lift its sample in the same represented field. Recover feeds from Y_i afterward. $\square$

Arbitrary intake costs become a dense bypass objective after the substitution for Y_i , and arbitrary bypass costs are already dense; both are outside Theorem 5.3. A fixed number of additional cost-bearing arcs can be retained using Theorem 4.11; for a designated intake, retain its at most two adjacent bypass flows. The exact quality contract at nonreceiving products is needed. For example, let two unit supplies have qualities zero and two. Only the clean input bypasses to a product of demand one and upper quality one; the other product receives one unit from the pool and has upper quality $3/2$. The first product consumes all clean supply, which forces pool quality two and hence infeasibility. Treating its upper quality one as an equality would wrongly subtract one unit of quality mass in (79).

5.3 Fully contracted signed flows

Lemma 5.4 (Contract conservation). *With exact source supplies and exact product contracts, first check*

$$\sum_i a_i = \sum_j b_j, \quad \sum_i C_i a_i = \sum_j B_j b_j. \quad (81)$$

After eliminating $Y_i = a_i - \sum_j z_{ij}$ and $V_j = b_j - \sum_i z_{ij}$, retain their actual bounds and every equation

$$\sum_i (C_i - q) z_{ij} = b_j (B_j - q). \quad (82)$$

Together with bypass bounds, these conditions are equivalent to all physical constraints except a possibly restrictive common pool bound.

Proof. The mass equality in (81) implies $\sum_i Y_i = \sum_j V_j$, since both sides subtract the same total bypass flow. Summing (82) gives

$$q \sum_j V_j = \sum_j b_j B_j - \sum_{ij} C_i z_{ij} = \sum_i C_i Y_i.$$

Thus actual pool mixing holds coordinate by coordinate. At positive throughput this identifies q as the feed average. At zero throughput nonnegative incident flows all vanish, so their quality masses vanish. Every product equation is exactly its original equation after substitution, even when its outlet is absent. The converse follows by summing original mass and quality conservation. No common bound is derived. $\square$

Lemma 5.5 (Signed node intervals and connected cuts). *For a directed graph with finite signed arc bounds $\ell_e \leq u_e$ and node bounds $\alpha_v \leq \beta_v$, a flow satisfying*

$$\ell \leq w \leq u, \quad \alpha_v \leq \text{div}_w(v) \leq \beta_v$$

exists if and only if, for every vertex subset S ,

$$\begin{aligned} \alpha(S) &\leq u(\delta^+ S) - \ell(\delta^- S), \\ \beta(S) &\geq \ell(\delta^+ S) - u(\delta^- S). \end{aligned} \quad (83)$$

Here divergence is outgoing minus incoming, and set arguments indicate sums. It suffices to check connected induced subsets. A disjoint union of paths, cycles, and isolated vertices requires $O(n^2)$ such sets, with at most two cut edges each. Prescribing boundary arc flows gives the same description after shifting their signed contributions out of adjacent node intervals and retaining their arc bounds separately.

Proof. This is the signed interval transshipment criterion of Schrijver [40, Corollary 11.2i, p. 175], with the divergence sign reversed. To derive it from circulation feasibility, add a root and an arc from the root to each v , bounded by $[\alpha_v, \beta_v]$. Conservation at v makes that arc equal its original divergence; conservation at the root follows by summing original divergences. The circulation inequalities on subsets excluding the root and on complements of subsets containing it give the two inequalities in (83). Conversely, those two families are all the circulation cuts, so the circulation criterion supplies the flow. Signed bounds are handled by subtracting lower bounds before the usual nonnegative-capacity feasibility test.

If S is disconnected in its induced underlying graph, both node sums and both cut expressions add over its connected components, since no edge joins distinct components within S . Their separate inequalities therefore imply its inequalities. The connected subsets of a path are intervals; those of a cycle are cyclic intervals and the whole cycle. Whole components and isolated vertices must be retained. For prescribed boundary flow t_e with divergence sign σ_{ve} at v , the internal divergence lies between $\alpha_v - \sum_e \sigma_{ve} t_e$ and $\beta_v - \sum_e \sigma_{ve} t_e$. This is an exact substitution. $\square$

Theorem 5.6 (Quadratic-field feasibility). *Suppose every source and product is exactly contracted, the quality is scalar, and the bypass graph has maximum degree two. If the common pool lower bound is zero and the upper bound is redundant, feasibility is polynomial-time decidable. A feasible original flow can be constructed in one real field of degree at most two over $\mathbb{Q}$, with polynomial encoding length. The same result holds for arbitrarily many attributes of affine input rank at most one.*

Proof. Check (81). If the pool has no allowed feed or no allowed outlet, force its incident flows to zero, check their bounds, and solve the remaining rational LP with all external contracts. Otherwise restrict q to the closed interval spanned by allowed feed qualities. This loses no active flow by mixing; an inactive pool can be assigned any value in that interval. At every distinct source quality in this interval, including its endpoints, solve the original fixed- q physical LP with actual pool arcs and balances. These are rational LPs.

On each remaining open interval, every $\gamma_i = C_i - q$ is nonzero with fixed sign. Set

$$w_{ij} = \gamma_i z_{ij}, \quad z_{ij} = w_{ij} / \gamma_i. \quad (84)$$

Keep the orientation input to product even when w is negative. Feed bounds $\ell_i^P \leq Y_i \leq u_i^P$ become a bound on $\sum_j z_{ij}$ between $a_i - u_i^P$ and $a_i - \ell_i^P$. Multiplication by γ_i , with order determined by its sign, gives an affine interval on input divergence $\sum_j w_{ij}$. Every original bypass interval likewise gives affine signed bounds on its w_{ij} . Equation (82) becomes

$$\sum_i w_{ij} = R_j(q), \quad R_j(q) = b_j(B_j - q), \quad (85)$$

so product divergence is exactly $-R_j(q)$.

The outlet interval is a bound $L_j \leq s_j \leq U_j$ on total bypass s_j , where $L_j = b_j - u_j^P$ and $U_j = b_j - \ell_j^P$. With two inlets of distinct qualities, write $\gamma_h = C_h - q$. Solving $w_1 + w_2 = R_j$ and $s_j = w_1/\gamma_1 + w_2/\gamma_2$ gives

$$w_1 = \frac{\gamma_1(R_j - \gamma_2 s_j)}{C_1 - C_2}. \quad (86)$$

Its coefficient of s_j is $-\gamma_1\gamma_2/(C_1 - C_2)$, nonzero of fixed sign. Thus the outlet interval is exactly the interval on w_1 with endpoints

$$\frac{\gamma_1(R_j - \gamma_2 L_j)}{C_1 - C_2}, \quad \frac{\gamma_1(R_j - \gamma_2 U_j)}{C_1 - C_2}. \quad (87)$$

They are quadratic polynomials in q ; retain (85) and all original arc bounds too. If the qualities coincide, $\gamma s_j = R_j$ instead, so the outlet interval gives the pure parameter condition that R_j lie between γL_j and γU_j . With one inlet it gives that affine arc interval together with $w = R_j$. With zero inlets require $L_j \leq 0 \leq U_j$ and $R_j = 0$. These cases include zero demands and missing outlets without division by a vanishing quality difference.

Each arc now has constantly many lower and upper quadratic candidates. Its effective bounds are the maximum lower and the minimum upper candidate. Include every lower-versus-upper consistency

inequality. Node bounds are affine. Apply Lemma 5.5; each connected cut uses at most two arc bounds. The first right side of (83) is the minimum over all upper-outgoing/lower-incoming candidate choices, and the second is the maximum over lower-outgoing/upper-incoming choices. Thus each cut is equivalent to the conjunction over its constantly many choices. The result is polynomially many univariate quadratic inequalities together with the pure parameter conditions, with no capacity disjunction or exponential subset enumeration.

Exact root isolation and sign tests decide their conjunction on the open source-quality interval. Isolated roots of these new polynomials are included; only the original source-quality endpoints are excluded from the nonsingular chart. Coefficients have polynomial bit length, since they use constant-degree products, rational quality differences, and polynomially many sums. A feasible open interval contains a polynomial-bit rational sample, and a feasible isolated boundary point is a root of a quadratic polynomial with polynomial coefficient bit length. Thus q can be chosen of degree at most two.

At this q , evaluate all effective bounds and solve the signed circulation in $\mathbb{Q}(q)$. The standard reduction in Schrijver [40, Corollary 11.3a, p. 176], or the algebraic LP algorithm, constructs a feasible point. Its constant incidence matrix has vertex coordinates given by rational linear combinations of bounds in this same field, with polynomial coefficient length. Dividing by the nonzero γ_i and recovering Y_i, V_j preserves the field and polynomial encoding, and Lemma 5.4 proves physical feasibility. Singular LP witnesses are rational.

For affine rank one, compute $C_i = C_0 + d\xi_i$ with rational $d \neq 0$ and rational scalars ξ_i . Reject a positive-demand product outside that line, and otherwise write $B_j = C_0 + d\bar{\xi}_j$. At zero demand its quality vector may be replaced by any vector on the line. Local mass conservation cancels the C_0 terms, and any nonzero coordinate of d makes all vector equations equivalent to the scalar ones in $\xi_i, \bar{\xi}_j$. Rational affine algebra has polynomial encoding and changes no physical flow bound. Rank zero is a rational LP after checking positive-demand product compatibility. $\square$

The proof also gives a reusable boundary description : after retaining a fixed number of transformed boundary flows, Lemma 5.5 gives polynomially many rows linear in those coordinates and quadratic in q . Capacity bounds on a removed boundary arc remain explicit. A physical boundary coordinate z_{ij} can be linked by the bilinear equation in (84).

Example 5.7 (A common bound is an additional constraint). Take unit supplies of qualities zero and two. The clean input bypasses only to a product of demand one and exact quality $1/2$; the dirty input bypasses only to a product of demand one and exact quality $3/2$. Both inputs feed the pool and both products receive its flow; all individual upper capacities are one and lower bounds zero. The local contracts force

$$Y_0 = V_{1/2} = \frac{1}{2q}, \quad Y_2 = V_{3/2} = \frac{1}{2(2-q)}, \quad \frac{1}{2} \leq q \leq \frac{3}{2}.$$

Hence $T = 1/[q(2-q)] \geq 1$. Local feasibility holds at $q = 1$, but common upper capacity $3/4$ makes the physical model infeasible. Thus the global identities do not imply a restrictive pool bound. The example marks the scope of Theorem 5.6; it is not a hardness result.

5.4 A bounded number of contract exceptions

Designate exceptional external sets E_I, E_J and put $s = |E_I| + |E_J|$. Ordinary inputs have exact supplies; ordinary products have exact demand and quality contracts. Exceptional inputs may have supply intervals, and exceptional products may have demand and quality intervals. In this subsection the maximum bypass degree is two, and common pool bounds are redundant: their lower bound is

zero and their upper bound is at least a valid total throughput bound, such as the minimum of the sums of input upper supplies, feed capacities, and outlet capacities. Individual pool-arc intervals remain unrestricted. The objective is standard economics. A pool with no allowed feed or no allowed outlet is handled first by forcing all its incident flows to zero, checking their bounds, and solving the remaining rational LP with every external requirement. The nonlinear constructions may therefore assume that both kinds of pool arc exist.

Lemma 5.8 (A fixed-dimensional pool-quality space). *For fixed s, d , suppose pool quality is restricted to a supplied rational affine space $q = q_0 + D\eta$ of dimension d , where the columns of D are independent, and to the closed coordinate box spanned by the allowed feed vectors. For a pool with both an allowed feed and an allowed outlet, the preceding exception model with these quality restrictions admits polynomial-time feasibility and standard economic optimization, with a polynomial-size common real-algebraic original optimizer. Ambient quality dimension and input-quality rank may grow. The retained dimension is at most $d + 5s$.*

Proof. Use the supplied closed coordinate box. At an active pool, its balance implies this box; an inactive pool in this branch must have a representative quality in the supplied intersection. If the supplied affine space misses the box, the branch is empty. Select d independent coordinate rows of D . They express η as a rational affine function of bounded quality coordinates and hence give explicit finite polynomial-bit bounds on η . Retain all box inequalities.

Retain each exceptional node's incident pool flow and every bypass arc in H incident to an exceptional node. There are at most s such pool flows and $2s$ bypass flows. For exceptional nodes define the core expressions

$$A_i = y_i + \sum_j z_{ij}, \quad D_j = v_j + \sum_i z_{ij}, \quad M_j = qv_j + \sum_i C_i z_{ij}.$$

Keep their supply, demand, quality, and arc bounds. At ordinary nodes eliminate feeds and outlets by exact contracts as before. Preserve (82) at every ordinary product. Put in the core one mass equation and all ambient quality equations

$$\begin{aligned} \sum_{i \notin E_I} a_i + \sum_{i \in E_I} A_i &= \sum_{j \notin E_J} b_j + \sum_{j \in E_J} D_j, \\ \sum_{i \notin E_I} C_i a_i + \sum_{i \in E_I} C_i A_i &= \sum_{j \notin E_J} b_j B_j + \sum_{j \in E_J} M_j. \end{aligned} \tag{88}$$

These equations suffice. Subtracting total bypass mass from the first equality gives actual pool mass balance. In each quality coordinate, sum the ordinary product equations and the exceptional masses M_j ; subtracting bypass quality mass from the second equality gives $q \sum_j v_j = \sum_i C_i y_i$. Thus actual pool quality balance holds for every local lift. At zero throughput all pool arcs vanish.

For each full source vector C_i belonging to the supplied affine space and box, solve the original fixed-vector-quality physical LP, including its objective. These cover all $q = C_i$ singularities, even for inputs without a feed. At every other q some vector in the explicit list

$$\beta(k) = (1, k, \dots, k^{K-1}), \quad 1 \leq k \leq |I|(K-1) + 1 \tag{89}$$

has $\gamma_i = \beta(k) \cdot (C_i - q) \neq 0$ for every input. Indeed each nonzero $C_i - q$ defines a nonzero polynomial of degree at most $K-1$ in k , excluding at most $K-1$ list members. The union of all excluded sets cannot exhaust the list. Its size and coefficient bit lengths, $O(K \log(|I|K+1))$ per entry, are polynomial even when K grows. For $K=1$ one direction suffices, and $K=0$ is an LP.

Fix a direction. Define $w_{ij} = \gamma_i z_{ij}$ and, for retained arcs, keep both w_{ij} and its bilinear link to z_{ij} . This gives at most $d + s + 2s + 2s = d + 5s$ variables. Projecting the ordinary product equation onto β gives

$$\sum_i w_{ij} = R_j, \quad R_j = b_j \beta \cdot (B_j - q).$$

Original bypass and ordinary input bounds become affine signed arc and divergence bounds, after partitioning by the nonzero signs of γ_i . For a two-inlet ordinary product set $c_h = \beta \cdot C_h$. If $c_1 \neq c_2$, its outlet bounds become the quadratic arc interval of (87), with $c_1 - c_2$ in the denominator and the projected R_j . If $c_1 = c_2$, use the corresponding pure parameter condition instead.

The projection alone does not enforce all quality coordinates. For each ambient coordinate h , substitute $w_2 = R_j - w_1$ into its original quality equation and multiply by the nonzero $\gamma_1 \gamma_2$. The exact residual equation is

$$\begin{aligned} a_h(q)w_1 &= f_h(q), \\ a_h &= (C_{1h} - q_h)\gamma_2 - (C_{2h} - q_h)\gamma_1, \\ f_h &= \gamma_1 [b_j(B_{jh} - q_h)\gamma_2 - (C_{2h} - q_h)R_j]. \end{aligned} \tag{90}$$

The quadratic terms of a_h cancel, and the quadratic terms inside the brackets of f_h cancel; hence a_h is affine and f_h quadratic in η . Include every a_h in the affine sign partition. If $a_h = 0$, impose $f_h = 0$. Otherwise impose both bounds $w_1 = f_h/a_h$ with known denominator sign. Keep all such equations, including redundant ones, on the same chosen incident arc.

With one inlet, projected quality fixes $w = R_j$, and each additional coordinate gives

$$(C_{ih} - q_h)R_j = b_j(B_{jh} - q_h)\gamma_i.$$

Keep these parameter equalities and the outlet's affine interval on w . With no inlets, require 0 in the bypass-total interval and $b_j(B_j - q) = 0$ in every coordinate. Thus every ordinary quality condition is preserved, including equal projected qualities and zero coefficients. The argument does not assume that the ambient vectors lie in the supplied quality space.

The signs of the polynomially many affine functions γ_i, a_h have polynomially many realizations in fixed dimension d . Keep lower-dimensional sign conditions, and require $\gamma_i \neq 0$ strictly. On each such cell every internal arc has polynomially many rational bound candidates, of numerator degree at most two and denominator degree at most one. Delete exceptional nodes and remove retained boundary arcs from the internal graph, subtracting their signed w contributions from adjacent divergence intervals. A candidate bound assigned to a boundary arc remains a core row. An arc between two exceptional nodes requires no internal projection.

Residual components are paths, cycles, or isolated nodes. Apply Lemma 5.5 to connected sets. Each cut crosses at most two internal edges; conjunction over their candidate combinations is still polynomial, even with polynomial-size candidate lists. Include every lower-versus-upper candidate comparison on each arc. Clear only the at most two selected affine denominators in each inequality, using their known signs. Node sums are affine in η and linear in boundary w coordinates, so degree four is a safe bound for all resulting polynomial inequalities. Pure parameter conditions and retained candidate rows have the same bound. Multiplying all internal-arc denominators would be unnecessary and would lose this constant-degree property.

The resulting formula has fixed dimension, constant degree, polynomially many rows, and polynomial coefficient length. Together with (88) and the retained physical rows it is exactly the original model on the chart. Standard profit reduces to

$$\sum_{j \notin E_J} R_j^{\text{econ}} b_j - \sum_{i \notin E_I} c_i a_i + \sum_{j \in E_J} R_j^{\text{econ}} D_j - \sum_{i \in E_I} c_i A_i, \tag{91}$$

where R_j^{econ} denotes revenue, unrelated to the projected quality residual R_j . This profit is affine in the original retained flow coordinates.

Use quantifier elimination to describe the attainable profit values on every actual sign cell, and unite these descriptions with the fixed-source-vector LP value intervals. The union is precisely the attainable set of the original model with the closed affine quality restriction and box, a compact set. If nonempty it has an attained maximum. Select that value and sample a point on a cell attaining it. Do not close a chart at a vanishing denominator: its polynomial rows can lose information there, while the singular LPs retain it.

At the sample, all residual incidence bounds belong to its common represented real-algebraic field. Solve each residual signed flow LP in that field, then divide by nonzero γ_i and recover ordinary feeds and outlets from their contracts. A constant incidence matrix has vertex coordinates that are rational linear combinations of its bounds, and the sample has polynomial degree and encoding, so recovery remains polynomial in the same field. Equations (88) prove the actual pool balances for every recovered flow. $\square$

Corollary 5.9 (Scalar and fixed-rank exceptions). *For fixed s and fixed affine input-quality rank t , the exception model admits polynomial-time feasibility and standard economic optimization. For scalar quality the core dimension is at most $1 + 5s$ and every chart row may be chosen quadratic. For fixed rank t the bound is $t + 5s$. Neither assertion gives a quadratic-field witness bound when exceptions are present.*

Proof. For scalar quality the above chart argument uses only the signs of $C_i - q$, the scalar intervals (87), the boundary cut description, and the two equations (88); there are no residual vector equations or variable denominators in arc candidates. Thus all rows are quadratic and the dimension is $1 + 5s$.

For affine rank t , compute rational coordinates $C_i = C_0 + D\xi_i$ with t independent columns. A positive-demand ordinary product must belong to this affine hull; reject it otherwise. Zero-demand qualities are immaterial. Express every original exceptional quality inequality in the coordinate quality masses and throughput; keep all such inequalities. Local mass conservation cancels C_0 , so coordinate quality equations are equivalent to all ambient equations. Apply Lemma 5.8 in t coordinates, or take its supplied space to be this full affine hull. Rank zero is a fixed-quality LP. All coordinate and bound computations are polynomial-bit rational linear algebra. $\square$

Theorem 5.10 (Arbitrary-quality bounded exceptions). *For fixed s , the exception model admits polynomial-time exact feasibility and standard economic optimization with a polynomial-size common real-algebraic original optimizer. The number of quality coordinates and their affine input rank are unrestricted.*

Proof. Consider an ordinary product j with positive pool inflow $v_j > 0$. Its nonnegative flow and exact contract imply $b_j > 0$, and

$$q = \frac{b_j}{v_j} B_j - \sum_i \frac{z_{ij}}{v_j} C_i.$$

The coefficients sum to one by product mass conservation. Since at most two bypass inlets exist,

$$q \in \text{aff}(\{B_j\} \cup \{C_i : ij \text{ is a bypass}\}), \quad (92)$$

an affine space of dimension at most two. Enumerate this space for each positive-demand ordinary product and apply Lemma 5.8 with $d \leq 2$. Rational Gaussian elimination gives polynomial-bit bases and bounds. Dimension zero is a physical fixed-quality LP. There are polynomially many spaces. A branch need not require its designated outlet to be positive: every branch retains all original

constraints and is therefore sound, and every solution with an active ordinary outlet belongs to some branch by (92).

It remains to handle flows whose only active outlets are exceptional. Force all ordinary outlets to zero, checking their lower bounds. Let $J_E \subseteq E_J$ be the products with an allowed pool outlet and put $r = |J_E|$. For $j \in E_J \setminus J_E$, put $v_j = \theta_j = 0$ by convention. If $r = 0$, force all pool arcs to zero and solve the remaining rational LP. Otherwise retain exceptional input feeds, the r outlet flows indexed by J_E , every bypass arc incident to an exceptional node, and r outlet fractions θ_j , $j \in J_E$. There are at most $3|E_I| + 4|E_J|$ coordinates. Exceptional source withdrawals $A_i = y_i + \sum_j z_{ij}$ are core expressions. Ordinary inputs give the local affine feed bounds, while ordinary products now give bypass-only exact mass and vector-quality equations. Delete the exceptional nodes and apply Theorem 4.8 to all remaining rational path relations; solve detached components by LP.

Define core expressions

$$\begin{aligned} S &= \sum_{i \notin E_I} a_i + \sum_{i \in E_I} A_i, & G &= \sum_{i \notin E_I} C_i a_i + \sum_{i \in E_I} C_i A_i, \\ T &= S - \sum_{j \notin E_J} b_j - \sum_{ij: j \in E_J} z_{ij}, & Q &= G - \sum_{j \notin E_J} b_j B_j - \sum_{ij: j \in E_J} C_i z_{ij}. \end{aligned} \quad (93)$$

Every term uses only retained coordinates or constants. Summing ordinary product contracts and all source equations proves $T = \sum_i y_i$ and $Q = \sum_i C_i y_i$ for every local lift. Impose $\theta_j \geq 0$ and $v_j = \theta_j T$ for $j \in J_E$, and $\sum_{j \in J_E} \theta_j = 1$. Keep the demand and homogeneous quality bounds at every $j \in E_J$, including products without pool outlets, using quality mass $\theta_j Q + \sum_i C_i z_{ij}$ and throughput $v_j + \sum_i z_{ij}$, as well as all retained arc and source bounds. These rows are quadratic in a fixed number of variables independently of the ambient quality count. At positive T they give actual pool quality Q/T . At zero T , all nonnegative feeds vanish and $Q = 0$, so fraction choices are harmless. Conversely, every physical flow of this branch supplies its fractions, or any simplex point if inactive. Thus this branch is exact, closed, bounded, and polynomially described.

Standard profit is again (91) and depends only on the core. Optimize this branch and all active-space branches and select the largest feasible value. The branches cover every physical flow, and each feasible branch is compact before its internal chart partition, so an optimum is attained. The quality branches use incidence recovery; the exceptional-only branch uses rational endpoint lifting and local feed recovery. Both yield one common field of polynomial degree and encoding. $\square$

A fixed number of designated additional arc costs can be included by declaring their external endpoints exceptional, with their original contracts, so that their actual arc flows are retained. The increase in s is fixed, so the proof applies with these affine core costs. This does not give unrestricted dense arc-cost optimization. The argument also needs the redundant common bounds of this subsection, because total pool throughput need not be a core coordinate. Theorem 3.20 shows that bypass degree three at ordinary products already gives strong feasibility hardness with five exceptions.

5.5 Restrictive common pool bounds

The signed-flow feasibility cuts do not constrain the total unscaled bypass flow, which determines pool throughput. We compute its extreme values as a linear objective on node divergences. At a fixed quality, the classical box-truncated submodular rank formula gives this support value (Lemma 5.12). The additional step is to compute all prefix rank values uniformly as the quality changes; the binary path recurrence of Lemma 5.11 does this without enumerating bypass flow bases.

Lemma 5.11 (Symbolic binary path and cycle minimization). *Let η have fixed dimension, and consider binary path labels $x_1, \dots, x_n$ with energy*

$$\sum_i U_i(x_i; \eta) + \sum_{i=2}^n [a_i(\eta) \mathbf{1}_{(x_{i-1}, x_i)=(0,1)} + b_i(\eta) \mathbf{1}_{(x_{i-1}, x_i)=(1,0)}].$$

Suppose all terms are rational polynomials of fixed degree and $a_i, b_i \geq 0$ on the specified domain; unary terms may be signed. The minimum has an explicit polynomial formula on each of polynomially many realizable polynomial sign conditions, with polynomial coefficient length and the same fixed degree. The same conclusion holds for cycles.

Proof. Let $E_i(t)$ be the minimum prefix energy ending in label t , put $D_i = E_i(1) - E_i(0)$, and $u_i = U_i(1) - U_i(0)$. The two ordinary prefix recurrences give

$$\begin{aligned} E_i(0) &= E_{i-1}(0) + U_i(0) + \min(0, D_{i-1} + b_i), \\ D_i &= u_i + \min(D_{i-1}, a_i) - \min(0, D_{i-1} + b_i) \\ &= u_i + \text{clamp}(D_{i-1}, -b_i, a_i), \quad D_1 = u_1. \end{aligned}$$

The last identity follows by the three cases $D < -b_i$, $-b_i \leq D \leq a_i$, and $D > a_i$; nonnegative a_i, b_i put the endpoints in the stated order. Set $P_i = \sum_{h=1}^i u_h$ and $Z_i = D_i - P_i$. Then

$$Z_1 = 0, \quad Z_i = \text{clamp}(Z_{i-1}, -b_i - P_{i-1}, a_i - P_{i-1}). \quad (94)$$

Every Z_i is selected from zero and the $2(i-1)$ preceding clamp endpoints. Form the common list of those polynomials and all $-P_i$. Signs of its pairwise differences determine every clamp selection, every increment $\min(0, D_{i-1} + b_i)$, and the final comparison between $E_n(0)$ and $E_n(1)$. On each realizable sign condition, execute the recurrence symbolically; its value is a sum of polynomially many input terms of the original degree. Equal values can be resolved by either choice, including on lower-dimensional conditions. Coefficient bit lengths remain polynomial. A polynomial-size list of fixed-degree polynomials has polynomially many realizable sign conditions in fixed dimension, constructible by the cited real-algebraic algorithms.

For a cycle, fix x_1 to zero and then to one. The edge from the last vertex to the fixed first vertex is a unary cost at the last vertex; the first edge is handled by the fixed initial condition (or absorbed at the second vertex). Apply the path construction and compare the two value formulas on their common sign refinement. A product of two polynomial bounds is polynomial. This proof is specific to the binary submodular energy and its clamp recurrence. $\square$

Lemma 5.12 (Box-bounded divergence support). *Let $\ell \leq u$ be finite signed arc bounds on a directed graph with vertex set V , and let $\alpha \leq \beta$ be finite node bounds. Assume*

$$\mathcal{D} = \{\text{div}(w) : \ell \leq w \leq u, \alpha \leq \text{div}(w) \leq \beta\}$$

is nonempty. Define

$$f(S) = u(\delta^+ S) - \ell(\delta^- S), \quad g(S) = \min_{T \subseteq V} \{f(T) + \beta(S \setminus T) - \alpha(T \setminus S)\}. \quad (95)$$

Then g is submodular, $g(\emptyset) = g(V) = 0$, and

$$\mathcal{D} = B(g) := \{d : d(S) \leq g(S) \ (S \subseteq V), \ d(V) = 0\}.$$

If $c_1 \geq \dots \geq c_n$ is any ordering of a real cost vector and $S_k = \{1, \dots, k\}$, then

$$\max_{d \in \mathcal{D}} c^T d = \sum_{k=1}^{n-1} (c_k - c_{k+1}) g(S_k), \quad d_k^* = g(S_k) - g(S_{k-1}) \quad (96)$$

gives an attaining divergence vector. Minimum support is obtained by negating the maximum for $-c$.

The box rank formula and greedy rule are classical; Shioura et al. [41, Theorems 1–2, equation (11), p. 192] state them for the maximal vectors of a box-truncated submodular polyhedron. We prove the base-polyhedron specialization needed here, including its total-sum condition.

Proof. Before the node box is imposed, feasible divergences are exactly $B(f)$. Necessity follows by summing divergences. Conversely, prescribe a vector d with $d(V) = 0$ and $d(S) \leq f(S)$; its complementary inequality is $d(S) \geq \ell(\delta^+ S) - u(\delta^- S)$. Lemma 5.5 with $\alpha = \beta = d$ supplies the flow. Also

$$f(T) = \sum_{e \in \delta^+ T} (u_e - \ell_e) + \sum_{v \in T} \text{div}(\ell)_v.$$

A nonnegative directed cut function is submodular, as verified edge by edge on the four membership possibilities, and the second term is modular. Thus f is submodular even when ℓ, u are signed, and $f(\emptyset) = f(V) = 0$.

The expression minimized in (95) is jointly submodular in (S, T) . Its node term takes values $0, \beta_v, -\alpha_v, 0$ at membership pairs $(0, 0), (1, 0), (0, 1), (1, 1)$; the only nontrivial submodularity inequality is $\alpha_v \leq \beta_v$. Add the submodular term $f(T)$. For minimizing pairs $(S_1, T_1), (S_2, T_2)$, the product lattice inequality bounds their sum below by the values at union and intersection, which in turn bound $g(S_1 \cup S_2) + g(S_1 \cap S_2)$. Hence partial minimization gives submodular g .

For every $d \in \mathcal{D}$ and every S, T ,

$$d(S) = d(T) + d(S \setminus T) - d(T \setminus S) \leq f(T) + \beta(S \setminus T) - \alpha(T \setminus S).$$

Thus $d(S) \leq g(S)$. At $S = \emptyset, V$, this inequality and the choices $T = \emptyset, V$ respectively prove $g(\emptyset) = g(V) = 0$, using nonemptiness. This gives $\mathcal{D} \subseteq B(g)$. Conversely, choose $T = S$ to see $g(S) \leq f(S)$, so $B(g) \subseteq B(f)$. At $S = \{v\}$ choose $T = \emptyset$ to get $d_v \leq \beta_v$; at $S = V \setminus \{v\}$ choose $T = V$ to get $d(V \setminus \{v\}) \leq -\alpha_v$, hence $d_v \geq \alpha_v$. This proves equality of the two polyhedra.

The greedy vector is feasible by diminishing increments. For any subset A , and each $k \in A$, the marginal of adding k to S_{k-1} is at most its marginal when added to $A \cap S_{k-1}$, since the latter set is contained in the former. Summing these inequalities gives $\sum_{k \in A} d_k^* \leq g(A)$. The full sum telescopes to $g(V) = 0$. For any feasible d ,

$$c^T d = c_n d(V) + \sum_{k=1}^{n-1} (c_k - c_{k+1}) d(S_k) \leq \sum_{k=1}^{n-1} (c_k - c_{k+1}) g(S_k).$$

The greedy vector attains equality at each prefix. This handles signed costs and ties, and proves (96). $\square$

Theorem 5.13 (Common lower and upper throughput bounds). *The fully contracted scalar or affine-rank-one model of Theorem 5.6 admits polynomial-time feasibility with an arbitrary common throughput interval $[L_P, U_P]$. All individual arc bounds, including positive lower bounds, remain. A feasible physical flow is constructible in one real algebraic field of polynomial degree and encoding length; no quadratic-field bound is claimed.*

Proof. Intersect the common interval with nonnegativity and a valid finite throughput bound; reject an empty intersection. If the pool has no feed or no outlet, force all incident flows to zero and explicitly require $L_P = 0$, check incident bounds, and solve the remaining original LP. Check (81); treat every singular source quality by its original rational physical LP, now retaining $[L_P, U_P]$.

On a nonsingular quality interval perform the signed transformation of Theorem 5.6. Split at roots of all arc-candidate comparisons. There are polynomially many quadratic comparisons and hence polynomially many interval and singleton cells. On each cell the effective ℓ_e, u_e are explicit quadratic polynomials; node α, β are affine. Keep every pure parameter condition and connected cut. These enforce $\mathcal{D}(q) \neq \emptyset$ before Lemma 5.12 is applied.

Let $A = \sum_i a_i$ and assign costs

$$c_i(q) = \frac{1}{C_i - q} \quad \text{at inputs}, \quad c_j(q) = 0 \quad \text{at products.}$$

Since input divergence is the sum of its outgoing transformed bypasses,

$$Z := \sum_{ij} z_{ij} = c(q)^T \text{div}(w), \quad T = A - Z. \quad (97)$$

The cost order is fixed on each nonsingular source-quality interval: input cost differences have numerator $C_h - C_i$ and denominator $(C_i - q)(C_h - q)$, both of known sign; the comparison with zero is also fixed. Repeated input qualities merely give ties.

For every prefix in the orders for c and $-c$, evaluate $g(S)$ symbolically. To write its definition (95) as a path/cycle energy, fix $s_v = \mathbf{1}_{v \in S}$ and minimize over the binary labels $t_v = \mathbf{1}_{v \in T}$. With divergence defined as outgoing minus incoming, the expression minimized is

$$\sum_{e=(v,w)} (u_e - \ell_e) t_v (1 - t_w) + \sum_v [\beta_v s_v + (\text{div}(\ell)_v - \beta_v s_v - \alpha_v (1 - s_v)) t_v]. \quad (98)$$

Indeed the node box terms are $\beta_v s_v (1 - t_v) - \alpha_v t_v (1 - s_v)$, while the lower-bound shift of $f(T)$ adds $\text{div}(\ell)_v t_v$. The directed cut capacities $u_e - \ell_e$ are nonnegative, and the displayed node terms are unary. All these costs are quadratic polynomials on the cell. Lemma 5.11 gives an explicit quadratic minimum on polynomially many univariate sign cells. Different connected components, including isolated vertices, are minimized independently. Take the union of all univariate breakpoints over all prefixes, orders, and components; this union is polynomial, unlike a product enumeration of their partitions. Keep singleton and equality cells.

On the resulting cells, (96) supplies rational functions $Z_{\min}(q), Z_{\max}(q)$ with polynomial degrees and coefficient lengths. A common denominator is the product of the distinct nonzero factors $C_i - q$; its degree is at most the input count and its sign is fixed. Combining polynomially many quadratic numerators with these factors increases degree and coefficient bit length only polynomially. For a fixed feasible q , compactness and convexity of the signed flow polytope imply that its attainable bypass totals form exactly the closed interval $[Z_{\min}, Z_{\max}]$. It meets $[A - U_P, A - L_P]$ exactly when

$$Z_{\min}(q) \leq A - L_P, \quad Z_{\max}(q) \geq A - U_P. \quad (99)$$

Clear the known-sign denominator. These are univariate polynomial inequalities of polynomial degree and bit length. Root isolation and sign testing on each actual cell, together with retained cut and local conditions, decide feasibility in polynomial bit time.

Choose a feasible rational interval sample or an algebraic boundary point q of polynomial degree and encoding. For each support extremum, Lemma 5.12 constructs an attaining divergence by

differences of the computed prefix rank values. These values belong to $\mathbb{Q}(q)$ with polynomial encoding. Realize each divergence by a bounded-flow LP in the same field. Its incidence matrix has a polynomial-encoding vertex, which the cited algebraic LP algorithm constructs; the usual lower-bound shift and a polynomial-augmentation flow algorithm with exact comparisons in the represented field also suffice.

Choose Z^* in the nonempty intersection with $[A - U_P, A - L_P]$, for example $\max(Z_{\min}, A - U_P)$. If the extrema differ, interpolate the two realizing flows with coefficient $(Z^* - Z_{\min})/(Z_{\max} - Z_{\min})$. If equal, use either. This coefficient is one quotient in $\mathbb{Q}(q)$ of polynomial encoding. Convexity preserves all signed arc and node bounds and makes the total exactly Z^* . Divide by $C_i - q$, recover physical pool arcs, and use Lemma 5.4 to verify all balances. The resulting throughput is in the common interval. The rank-one compression of Theorem 5.6 changes no flow or capacity and therefore extends the conclusion to affine input rank one. $\square$

Corollary 5.14 (Exact pool-throughput optimization). *In Theorem 5.13, both the minimum and maximum pool throughput, and therefore standard economic cost plus a rational linear pool-throughput cost, can be optimized exactly in polynomial bit time with a polynomial-size common-field physical optimizer.*

Proof. On each nonsingular cell retain a second variable T and impose

$$L_P \leq T \leq U_P, \quad A - Z_{\max}(q) \leq T \leq A - Z_{\min}(q).$$

Clear the same known-sign denominators. These rows and the retained local feasibility conditions give a formula of polynomial size and degree in two variables. Every satisfying (q, T) is physically realizable by the extrema interpolation in the preceding proof. Conversely every physical flow in the cell satisfies them. At each singular quality, the original rational LP gives the full attainable throughput interval by minimizing and maximizing T .

The union of all these actual-cell descriptions and singular intervals is exactly the attainable (q, T) set of the original compact physical model with the closed feed-quality interval. Quantifier elimination projects this polynomial-size union onto T and supplies its attained endpoints. Sample an attaining (q, T) in its actual cell and use the same interpolation to recover an original optimizer, or use the singular LP. Fixed dimension and polynomial degrees give polynomial bit time and a polynomial-degree common field. Exact source and product throughputs make standard economics constant, so adding any rational multiple of T selects one of these extrema (or any feasible flow when its coefficient is zero). No other dense arc-cost objective is included. $\square$

5.6 Two full input-quality vectors

A different reduction permits arbitrary bypass topology and arbitrary source supply intervals. It uses the fact that all nonlinear outlet upper bounds share the same concave quadratic term. For interior pool quality, one auxiliary variable reduces the nonlinear feasibility test to one rational convex quadratic program, whose rational optimizer yields rational physical flows. This witness is stronger than the algebraic witnesses of the degree-two algorithms.

Theorem 5.15 (Two-vector feasibility with variable supplies). *Suppose the input qualities take at most two distinct full rational vectors. Products have exact demands and exact quality vectors, sources may have arbitrary finite rational supply intervals, and the bypass graph is unrestricted. Individual feed and bypass lower bounds may be positive. Every pool-outlet lower bound and the common pool lower bound must be zero; their finite upper bounds may be restrictive. Feasibility, including variable procurement, is polynomial-time decidable and, when feasible, has a constructible*

rational original-flow witness of polynomial encoding length. No economic or arbitrary arc-cost optimization is claimed.

Proof. If there are no inputs, all flows are zero and the original bounds and homogeneous contracts can be checked directly. With one distinct input vector, every active mixture has that vector, so checking compatible positive-demand product qualities leaves a rational LP. With two distinct vectors, choose a coordinate on their rational affine line that assigns the vectors the scalar values zero and one. Every positive-demand product vector must lie on the segment between them; otherwise it cannot be a convex mixture and the instance is infeasible. Its scalar coordinate is $B_j \in [0, 1]$. A zero-demand product has all incoming flows zero, so its stated quality is immaterial and may be assigned any coordinate on the line. Local mass equations cancel the affine offset, making scalar quality equivalent to every original vector equation. All this is polynomial-bit rational linear algebra. Write I_0, I_1 for the two input classes.

If the pool lacks a feed or an outlet, force all its arcs to zero, check their bounds, and solve the remaining original LP with source intervals and product contracts. Otherwise the only active scalar qualities are in $[0, 1]$. Solve the original physical LPs at $q = 0$ and $q = 1$, retaining every pool balance and capacity. An inactive pool can also be represented at either endpoint. Only the nonsingular case $0 < q < 1$ remains.

Let actual withdrawal at input i be A_i and feed flow y_i . Exact product contracts fix the total consumption of each class:

$$A_0^{\text{tot}} = \sum_j b_j(1 - B_j), \quad A_1^{\text{tot}} = \sum_j b_j B_j. \quad (100)$$

These identities follow by total mass and scalar-quality conservation; individual withdrawals need not be fixed. Put $\gamma_i = -q$ on I_0 and $\gamma_i = 1 - q$ on I_1 , and introduce

$$w_{ij} = \gamma_i z_{ij}, \quad t_i = \gamma_i A_i, \quad h_i = \gamma_i y_i. \quad (101)$$

A source interval $[L_i, U_i]$ becomes the interval with endpoints $\gamma_i L_i, \gamma_i U_i$ on t_i , in the sign order fixed by its class. Feed intervals give the corresponding bounds on h_i , with $h_i = 0$ for an absent feed. Bypass intervals give affine bounds on w_{ij} . This retains positive feed and bypass lower bounds. Impose

$$t_i = h_i + \sum_j w_{ij}, \quad \sum_{i \in I_0} t_i = -q A_0^{\text{tot}}, \quad \sum_{i \in I_1} t_i = (1 - q) A_1^{\text{tot}}. \quad (102)$$

Every row is affine in the transformed variables and q .

For each product define $W_{0j} = \sum_{i \in I_0} w_{ij}$ and $W_{1j} = \sum_{i \in I_1} w_{ij}$. Its exact quality equation after eliminating $v_j = b_j - \sum_i z_{ij}$ is

$$W_{0j} + W_{1j} = b_j(B_j - q). \quad (103)$$

Use $\sum_i z_{ij} = -W_{0j}/q + W_{1j}/(1 - q)$ in this identity to get

$$W_{0j} = q b_j(B_j - 1) + q(1 - q)v_j. \quad (104)$$

This uses no bound on the number of bypass inlets and remains valid when an input class is absent at a product. Thus the outlet bounds $0 \leq v_j \leq U_j^P$ are exactly

$$q b_j(B_j - 1) \leq W_{0j} \leq q b_j(B_j - 1) + U_j^P q(1 - q). \quad (105)$$

An absent outlet is $U_j^P = 0$. Summing (104) gives the common upper bound $T \leq U_P$ as

$$\sum_j W_{0j} \leq q \sum_j b_j(B_j - 1) + U_P q(1 - q). \quad (106)$$

All upper coefficients U_j^P, U_P are nonnegative.

Before solving these inequalities, we check the reverse physical correspondence. At interior q , divide (101) by its nonzero class factor to recover A_i, y_i, z_{ij} . Their original bounds and input conservation follow from the affine interval and local rows. The class equations (102) give $\sum_i A_i = A_0^{\text{tot}} + A_1^{\text{tot}} = \sum_j b_j$, hence recovered pool intakes and outlets have the same total. Also

$$\sum_i t_i = -qA_0^{\text{tot}} + (1-q)A_1^{\text{tot}} = \sum_j b_j(B_j - q) = \sum_{ij} w_{ij}.$$

Summing the input equations therefore gives $\sum_i h_i = 0$, namely

$$-q \sum_{i \in I_0} y_i + (1-q) \sum_{i \in I_1} y_i = 0.$$

This is precisely the pool quality balance. The product equations give actual product quality, and (105) gives actual nonnegative outlet flow and its upper bound by (104). Equation (106) gives the shared capacity. At zero throughput every nonnegative pool flow vanishes. Conversely every original interior-quality solution satisfies all transformed rows, by these same identities.

Introduce one auxiliary variable r and replace every $q(1-q)$ on the right sides of (105) and (106) by $q-r$. Let P be the resulting rational polytope of all affine rows together with $0 \leq q, r \leq 1$. It is bounded: source, feed, and bypass capacities bound t, h, w uniformly since $|\gamma_i| \leq 1$. Empty P is rejected. The original transformed system at an interior q is feasible if and only if some point in P satisfies

$$F(q, r, t, h, w) := q^2 - r \leq 0. \quad (107)$$

For the forward direction take $r = q^2$. Conversely, $r \geq q^2$ makes every replaced upper bound at least as restrictive as the original one, since every multiplying capacity is nonnegative, and the affine lower bounds are unchanged.

The rational objective F has positive semidefinite Hessian of rank one. Exact rational convex quadratic programming over a rational polytope is polynomial-time solvable, including an exact optimal value and a rational optimizer of polynomial encoding length, by the classical algorithm of Kozlov et al. [32, pp. 1320–1323; numbered paragraphs 3 and 6 for rational optimal solutions and recovery]. This is a convex-QP result, not generic exact conic feasibility. To see rationality directly, choose the active affine rows at a minimizer. The corresponding KKT stationarity, active equalities, primal inequalities, and nonnegative multipliers form a rational linear system, and every solution is optimal by convexity. A rational basic feasible solution of that system has polynomial encoding by determinant bounds. The algorithm constructs such an optimizer without enumerating active sets.

The closed polytope may contain spurious transformed endpoint points, so interior quality must still be enforced exactly. First maximize ϵ over P with $0 \leq \epsilon \leq q$ and $\epsilon \leq 1-q$. If infeasible or its optimum is zero, P contains no interior-quality point. Otherwise retain a polynomial-bit rational point $p \in P$ with $0 < q_p < 1$. Now minimize F over P , obtaining a rational optimizer x^* and value m . If $m > 0$, there is no solution. If $m = 0$, all minimizers have the same q : for two minimizers x, y ,

$$F((x+y)/2) = \frac{1}{2}F(x) + \frac{1}{2}F(y) - \frac{1}{4}(q_x - q_y)^2,$$

so different qualities would contradict optimality. Thus this branch is feasible precisely when $0 < q_{x^*} < 1$; then x^* is a witness. If $m = -\rho < 0$, use $F(p) \leq 1$ (from $0 \leq q, r \leq 1$) and choose

$$\lambda = \frac{\rho}{2(\rho+1)}, \quad x = (1-\lambda)x^* + \lambda p.$$

This rational point has interior quality, and convexity gives $F(x) \leq -\rho + \lambda(\rho + 1) = -\rho/2 < 0$. All numbers have polynomial encoding. This handles exact zero minima and isolated feasible interior qualities without a numerical margin or a closure assumption.

Finally, divide by the nonzero rational class factors to recover A_i, y_i, z_{ij} and set $v_j = b_j - \sum_i z_{ij}$. Every recovered flow is rational with polynomial bit length, and the reverse mapping above verifies all physical rows. Endpoint branches already use the original rational LPs. Returning to the original affine quality line restores all full-vector quality equations. $\square$

Corollary 5.16 (Exact supplies and additional outlet resources). *Theorem 5.15 includes exact supplies as a special case. For exact supplies standard input costs and product revenues are constant. In either supply model it also permits any polynomial number of additional outlet rows $\sum_j \alpha_j v_j \leq U$ with rational α_j and $U \geq 0$.*

Proof. Exact supplies fix $A_i = a_i$, so the class totals require $\sum_i a_i = \sum_j b_j$ and $\sum_{i \in I_1} a_i = \sum_j b_j B_j$. The variables t_i in (102) may then be substituted by $\gamma_i a_i$. Eliminating h_i gives the affine signed bypass interval at each source, preserving the exact-source construction and all its lower bounds. Economic constancy follows from fixed source and product throughputs.

For an additional resource row, multiply (104) by α_j and sum. Since $q(1 - q) > 0$, the row is exactly

$$\sum_j \alpha_j W_{0j} \leq q \sum_j \alpha_j b_j (B_j - 1) + Uq(1 - q).$$

Replace its last term by $U(q - r)$ in P . Nonnegative U gives the same two-way equivalence with (107); the signs of the α_j affect neither the identity nor this comparison. Endpoint LPs retain the original resource rows. The proof and rational recovery are otherwise unchanged. $\square$

Positive outlet lower bounds would give $W_{0j} \geq qb_j(B_j - 1) + \ell_j^P q(1 - q)$, whose quadratic term has the opposite inequality direction, and a positive common pool lower bound has the same obstruction. This convex reduction covers neither; the degree-two algorithm of Theorem 5.13 permits both. Variable source withdrawals also make input procurement costs nonconstant, and Theorem 5.15 does not cover their optimization. The restriction is to two distinct complete quality vectors, not to two quality coordinates or a fixed number of physical inputs.

The ingredients of this section are established: planar elimination, difference constraints, signed circulation, submodular base optimization, convex quadratic programming, and fixed-dimensional real algebra. The fixed-input no-bypass algorithm of Boland et al. [10, Theorem 1] and the uncapacitated fixed-demand model of Baltean-Lugoian and Misener [6, Assumption 2.2] have different hypotheses and do not give the unbounded-input, contracted-bypass classifications here. The additional contributions are the physical transformations and their uniform parameter analysis; this comparison is limited to the cited statements.

6 Consequences for exact algorithms and convexification

The $\exists\mathbb{R}$ reductions realize arbitrary bounded algebraic constraints within physical mixing equations. The one- and two-pool NP-hardness reductions retain only a few mixing variables but couple them to a large linear network. The structural and contract algorithms succeed when that network can be eliminated while retaining all balances and the objective. These statements classify decision or optimization; they do not predict the running time of a particular numerical solver. This section states the representation consequences of three supporting results, proved in Appendices A–C, and the main question left open.

6.1 Large response descriptions can coexist with exact optimization

A pool-free path is already enough for an exponential parametric response. Theorem A.3 constructs a rational physical family with $O(n)$ nodes, one upper-bounded attribute, only two distinct input qualities, zero flow lower bounds, and positive product revenues. Changing only the final product's price exposes 2^n distinct unique optimal flows and 2^n affine value regimes. The construction has polynomial encoding length; its exact penalties have large, polynomially encoded revenues. Every fixed-price problem is an LP. Corollary A.4 proves a matching lower bound for explicit piecewise-affine tables and for the total polynomial degree of formulas in price and value alone. It leaves compact extended descriptions and implicit evaluation available.

A single two-feed, two-outlet pool attached to the contracted path yields 2^n isolated global optima with ordinary nonnegative input costs and product revenues at most two (theorem A.6). A rational price perturbation yields 2^n strict local maxima with distinct values (corollary A.7); no tolerance or local-solver complexity bound follows. Nevertheless, the full feasible set of this family has a polynomial-size LP convex hull whose vertices are physically feasible (theorem A.9), so an optimal LP vertex is an exact physical optimizer for every linear objective. The same family requires exactly n integer coordinates in any convex lift of its finite optimal set (corollary A.8). The represented set is material: a convex hull preserves linear optimization but does not represent the original nonconvex feasible or optimal set exactly.

These constructions realize an established exponential Klee–Minty shadow [19] by physical blending constraints. Their role is to rule out enumeration arguments, not to establish hardness on paths. Indeed, the rank-one concave objective used in the construction can be minimized over the bare path by a linear program. The abstract hardness in proposition A.11 requires an additional dense weighted slab. No degree-two physical realization of that slab is proved here.

6.2 Cost restrictions and uniform hull restrictions are different

An isolated pool block records a nonnegative rank-one matrix with interval row and column margins. General matrix costs make its rational threshold problem strongly NP-complete even when every lower margin bound is zero and every upper bound is one (theorem B.4 and corollary B.6). Every rational yes threshold still has a polynomial-size rational matrix witness, and some exact optimizer always lies in one quadratic number field (theorem B.7). Thus threshold witnesses can be simpler than exact optimizers. These statements concern arbitrary source-to-terminal costs on an isolated block; with ordinary additive costs the quality-free margin problem is an LP, so the reduction is not a physical pooling hardness proof.

The useful cost parameter is its rank after subtracting arbitrary additive row and column costs. For each fixed value of this *interaction rank*, theorem B.1 gives polynomial exact optimization and an optimizer in one quadratic field. This yields a fixed-attribute Lagrangian oracle after terminal quality inequalities are dualized (corollary B.3). The theorem is an oracle for the stated isolated block; it implies neither zero duality gap nor an algorithm for coupled physical pools.

The corresponding convex hull can still require large uniform representations. Its unit-margin version contains an explicit face isomorphic to the correlation polytope (theorem C.1). Established correlation-polytope bounds then imply exponentially many PSD blocks of any fixed order, exponential total second-order-cone dimension, and superpolynomial unrestricted PSD order for an exact lift (corollary C.2). A quantitative face estimate extends these conclusions to uniform outer approximations: at the stated inverse-quadratic total entrywise ℓ_1 error, LP lifts need exponentially many inequalities and SDP lifts need superpolynomial total matrix order (corollary C.5 and theorem C.8). These bounds allow arbitrary real coefficients and do not assume that the lift is efficiently

constructed. They concern all bounded linear costs simultaneously, so they do not exclude a useful relaxation for one objective or a fixed-rank cost family.

Dey et al. [16, Section 3.1, after Theorem 4] conjecture hardness of linear optimization with simultaneous row and column bounds; the reduction in Appendix B proves that conjecture for this exact set. Jalilian and Kocuk [31, Theorem 2] prove second-order-cone representability with row, column and total bounds by a generally exponential construction; our size lower bounds are compatible with it and say nothing about the empirical performance of their compact relaxations. The correlation lower-bound machinery is due to Braun et al. [12], Fawzi and Parrilo [17], Lee et al. [33]; the proofs here supply the rank-one face and the quantitative transfer.

6.3 What the parameter boundaries leave open

The main remaining structural question is standard pooling with a fixed number of pools, fixed affine input-quality rank, degree-two bypass graphs, and unboundedly many pool attachments. General flow and quality intervals are allowed; feasibility and dense-profit optimization should be treated separately. Two issues remain: local endpoint relations depend on unknown pool-quality parameters, and elimination must retain pool mass and quality aggregates as well as any objective. For one scalar parameter and general planar path relations the bound is quasipolynomial (theorem 5.1); several parameters with such relations remain open. The endpoint/objective scope is explained after corollary 4.13, and proposition A.11 shows why a fixed aggregate count alone does not suffice in an abstract system.

A shared fixed-quality helper pool does not automatically convert the degree-three copy network into a degree-two bypass network. Each original midpoint source has its own private supply equation; a shared helper preserves only their total. For example, in two full midpoint-copy pairs with endpoint qualities zero and two, midpoint quality one, and demand four at each copy product, choose both endpoint-pair signals zero in the first pair and two in the second. The required four midpoint flows are 4, 4, 0, 0, totaling eight. A common helper with throughput eight can supply them, but the original private midpoint supplies would demand four in each pair. The local replacement therefore loses a required relation, regardless of whether additional constraints elsewhere exclude this particular assignment. The coordinate-port obstruction in proposition A.5 gives a separate reason arbitrary degree-three LP circuits cannot simply be replaced by degree-two pool-free gadgets.

Several restrictions supply the missing aggregate information: a fixed number of attachments (theorem 4.9), exact supplies with a fixed number of receiving products (theorem 5.3), full contracts or a fixed number of exceptions (theorems 5.6, 5.10 and 5.13), and two full input vectors (theorem 5.15). Table 3 gives their scopes. They do not combine automatically: arbitrary exceptions with a restrictive common capacity, or dense arc costs, require an additional argument.

The unrestricted one-pool, one-quality bypass threshold question is settled by theorems 3.13, 3.17 and 3.19. Its tractable predecessors retain material hypotheses: subdividing bypasses into artificial pools changes pool count, and dropping source capacities changes shared resource coupling. The remaining degree-two question imposes restrictions that the hardness constructions do not meet. Similarly, the two-quality response construction is an LP with interval product specifications after penalties; it does not contradict the exact-product-contract feasibility theorem.

A Physical response geometry and limits of elimination

A.1 A telescoping certificate on a path

For $0 < \epsilon < 1/2$, let

$$P_\epsilon = \{x \in \mathbb{R}^n : 0 \leq x_1 \leq 1, \quad \epsilon x_{j-1} \leq x_j \leq 1 - \epsilon x_{j-1} \ (2 \leq j \leq n)\}, \quad x_0 = 0. \quad (108)$$

Every coordinate lies in $[0, 1]$, and the two inequalities in each pair cannot both be tight. Thus a vertex has exactly one tight inequality in each pair; conversely, any such choice gives a nonsingular triangular system and a feasible vertex. Its coordinates are $x_j(u) = u_j + (1 - 2u_j)\epsilon x_{j-1}(u)$ for $u \in \{0, 1\}^n$. All 2^n terminal values are distinct: the last bit chooses the disjoint intervals $[0, \epsilon]$ and $[1 - \epsilon, 1]$, and within either interval its affine recursion can be inverted, successively recovering all bits.

Lemma A.1 (A vertex certificate). *Put*

$$d_j = (1 - \epsilon)\epsilon^{2(n-j)-1} \quad (j < n), \quad L(x) = x_n - \sum_{j < n} d_j x_j, \quad F(x) = L(x) - x_n^2.$$

Then

$$F(x) = \sum_{j=1}^n \epsilon^{2(n-j)} (x_j - \epsilon x_{j-1})(1 - \epsilon x_{j-1} - x_j) \geq 0 \quad (x \in P_\epsilon), \quad (109)$$

with equality precisely at the vertices. Every vertex v is uniquely exposed by $\sum_{j < n} d_j x_j + (2v_n - 1)x_n$.

Proof. The j th product expands to $x_j - x_j^2 - \epsilon x_{j-1} + \epsilon^2 x_{j-1}^2$. The weighted squares telescope, leaving $-x_n^2$, and the remaining coefficients give L . All factors and weights are nonnegative, so equality requires a tight endpoint in every pair, exactly the vertex condition. For $t = v_n$, throughout P_ϵ ,

$$2tx_n - L(x) \leq 2tx_n - x_n^2 = t^2 - (x_n - t)^2 \leq t^2.$$

Equality requires $F = 0$ and $x_n = t$, which select v uniquely. $\square$

Exponential two-dimensional shadows of this Klee–Minty cube were established by Gärtner, Helbling, Ota and Takahashi [19, Section 4]; the exponential-shadow phenomenon is not new. The lemma gives a self-contained certificate for the physical constructions below. At $\epsilon = 1/4$, their direction uses $c_j = \epsilon^{3(n-j)}$ for $j < n$, $c_n = 0$, and exposes every vertex by $c^T x + \lambda x_n$ with $|\lambda| < 1/15$; the direction d in the lemma uses the interval $[-1, 1]$ instead. Both directions have polynomial rational encoding length.

A.2 Physical paths and one varying product price

Fix $n \geq 2$, $\epsilon = 1/4$, and $s_j = 4^{j-n}$. Source j has quality $C_j = n - j$ and exact supply s_j . Its only arcs go to products $j - 1$ and j , with flows $s_j - t_j$ and t_j , each bounded above by s_j . Product 0 has capacity s_1 and redundant upper quality C_1 ; product n has capacity one and redundant upper quality $C_n = 0$. For $2 \leq j \leq n$, product $j - 1$ has capacity s_j and upper quality $(C_{j-1} + C_j)/2$.

Proposition A.2 (Exact path realization). *This pool-free physical network is affinely bijective to $P_{1/4}$ by $t_j = s_j x_j$. Its bipartite bypass graph is one path, and every source and product has degree at most two. All upper flow bounds are at most one; scaling all qualities by $1/(n - 1)$ puts them in $[0, 1]$.*

Proof. The internal product receives $t_{j-1} + s_j - t_j$. Its capacity is $t_j \geq t_{j-1}$. Subtracting its midpoint quality from the two incident source qualities gives opposite coefficients $1/2$ and $-1/2$, so its homogeneous quality row is $t_{j-1} \leq s_j - t_j$. Thus its two rows are exactly

$$t_{j-1} \leq t_j \leq s_j - t_{j-1}. \quad (110)$$

After dividing by s_j , the predecessor coefficient is $1/4$. The endpoint and arc bounds give the other bounds, and the reverse substitution satisfies every physical row. All other flows are uniquely $s_j - t_j$; hence the correspondence is bijective. Scaling quality does not change any homogeneous quality inequality. $\square$

Theorem A.3 (Exponential physical price response). *There is a family of rational blending paths with $O(n)$ nodes, one upper-bounded quality, all upper flow capacities at most one, zero production costs, and positive product revenues, for which varying just the final product's unit revenue exposes 2^n distinct unique optimal flows. The family can have only two distinct input qualities and no positive lower flow bounds. Its optimal-value function has 2^n open affine pieces. Every fixed-price problem is an LP.*

Proof. First keep the exact source contracts in proposition A.2. Use the direction d from lemma A.1; its coefficient in physical rightward flows is $w_j = d_j/s_j = 3s_j$ for $j < n$. Put $w_n = \lambda \in [-1, 1]$ and define product revenues

$$R_0 = 1, \quad R_j = 1 + \sum_{h=1}^j w_h \quad (1 \leq j \leq n).$$

Since $\sum_{j < n} 3s_j = 1 - s_1 < 1$, all revenues lie in $[0, 3]$ and only R_n depends on λ . With zero costs, total revenue is

$$\sum_j [R_{j-1}(s_j - t_j) + R_j t_j] = C_0 + \sum_{j < n} d_j x_j + \lambda x_n, \quad C_0 = \sum_j R_{j-1} s_j. \quad (111)$$

The constant C_0 is independent of λ . The lemma exposes every vertex, including $x_n = 0, 1$ at $\lambda = -1, 1$. Strict optimality among a finite vertex set persists on a neighborhood of each exposing parameter, so each vertex is optimal on a nonempty one-sided interval inside $[-1, 1]$, and on an open interval in its interior. Distinct terminal values give distinct slopes, so there are exactly 2^n affine regimes. Alternatively, the published direction c with $w_j = c_j/s_j = 16^{j-n}$ gives revenues in $[0, 2]$ and the smaller price interval $[-1/15, 1/15]$.

To use only two input qualities, replace the descending quality path by the following relay path. Start with a quality-one source of supply s_1 . For each $j = 2, \dots, n$, attach a quality-zero source of supply s_j through a product with capacity s_j and upper quality $1/2$. If $j < n$, append a quality-one source of the same supply s_j through a product with exact demand s_j and redundant upper quality one. A descending step gives (110). At a reset product,

$$t_j + (s_j - \tilde{t}_j) = s_j \quad \Longleftrightarrow \quad \tilde{t}_j = t_j.$$

Thus the quality-one duplicate copies the signal before the next quality-zero source. End products have redundant upper quality one and capacities $s_1, 1$. There are $m = 2n - 2$ sources with supply sequence

$$S_1, \dots, S_m = (s_1, s_2, s_2, s_3, s_3, \dots, s_{n-1}, s_{n-1}, s_n).$$

All arcs of source i have upper bound S_i . The original free rightward flows and their equal duplicates determine all physical flows, giving the same affine bijection to $P_{1/4}$. Assign coefficient $3s_j$ to the free-coordinate position $j < n$, zero to reset duplicates, and λ to the last position. With these coefficients in path order denoted g_i , take $\rho_0 = 1$, $\rho_i = 1 + \sum_{h \leq i} g_h$. The same telescoping revenue computation gives $\sum_i \rho_{i-1} S_i + \sum_{j < n} d_j x_j + \lambda x_n$. Only the final revenue varies and every base revenue is in $[0, 3]$.

It remains to remove all source and reset-product lower contracts. Let Q be the nonempty exact-contract polytope in its $N = 2m$ arc variables f . Write these contracts $Ef = e$. Retain their upper bounds and drop their lower bounds, so the deficit vector $\Delta = e - Ef$ is nonnegative; put $\delta = \mathbf{1}^T \Delta$. All coefficient rows, after clearing the midpoint factor two, have entries in $\{-1, 0, 1\}$. By lemma 3.16, every relaxed flow has an exact-contract repair $f' \in Q$ with $\|f' - f\|_1 \leq H\delta$, where $H = N^N$. The base revenue vector has infinity norm at most three, uniformly in λ . Choose $M = 3H + 1$, add M to every product revenue, and add another M to every reset-product revenue. By conservation, the first bonus rewards all source throughputs and the second rewards reset throughputs. Writing $E_0 = \mathbf{1}^T e$, the new objective $B(f) + M(E_0 - \delta)$ obeys

$$B(f) + M(E_0 - \delta) \leq B(f') + ME_0 - (M - 3H)\delta.$$

Thus every optimizer satisfies every removed contract. The value function changes by exactly ME_0 , and every previously exposed optimizer remains unique. Revenues are now positive, all production costs remain zero, and M has $O(n \log n)$ bits; no bound on revenue magnitude is claimed. For the published direction c , the same proof uses $M = 2H + 1$. The version with varying input qualities needs no reset contracts; it uses $N = 2n$ and the same source-throughput bonus argument. $\square$

Corollary A.4 (Explicit elimination needs exponential output). *For either response in theorem A.3, a piecewise-affine table requires 2^n pieces. Any quantifier-free Boolean formula in price and value alone describing its graph, or its epigraph on the price interval, uses polynomials whose total degree is at least 2^n . The fixed-objective state message $G(t) = \max\{\sum_{j < n} d_j x_j : x \in P_{1/4}, x_n = t\}$ has $2^n - 1$ affine intervals.*

Proof. Remove identically zero polynomials from a proposed formula. Where none of its remaining polynomials vanishes, all signs, and hence its truth value, are locally constant. Each open affine graph segment is a boundary segment, so the product of the defining polynomials vanishes on that segment. Restriction to its supporting line vanishes on an interval and therefore identically; the line equation divides the product. The 2^n distinct lines require that many linear factors. This applies also after any upper clipping of the epigraph that leaves all those segments below the clip. For G , all projected vertex points $(x_n, \sum d_j x_j)$ are uniquely exposed with coefficient $+1$ on the second coordinate. They lie on its upper boundary, have distinct first coordinates, and consecutive points cannot be collinear. Their first coordinates run from zero to one, giving exactly $2^n - 1$ intervals. $\square$

These are limits on explicit descriptions only. Backward elimination evaluates the support at a rational price with $O(n)$ arithmetic operations: set $a_n = \lambda$ and $a_j = d_j - \epsilon|a_{j+1}|$ for $j = n - 1, \dots, 1$; the value is $\sum_j \max(0, a_j)$. Eliminating a variable with positive current coefficient selects its upper endpoint, adds that coefficient to the constant, and subtracts ϵ times its absolute value from the predecessor coefficient; a negative coefficient selects its lower endpoint and makes the same predecessor update without the constant. All intermediate bit lengths are polynomial, and the same recurrence works with c instead of d . Compact circuits, extended formulations, and implicit algorithms are therefore compatible with the lower bound. Note also that theorem 4.8 projects onto endpoint coordinates, whereas G retains a dense eliminated objective.

Proposition A.5 (A limit on individual coordinate ports). *Without nonphysical global rows, a pool-free network with source and product degrees at most two cannot represent every bounded rational polytope by a coordinate projection onto selected individual arc flows, even with unrestricted attribute count and two-sided bounds.*

Proof. Every physical row involves at most two arc variables. Fourier–Motzkin elimination preserves this property: a pair of rows of opposite signs in the eliminated variable contains at most one other variable each. Thus every coordinate projection has a finite two-variable-per-inequality description; its size need not be polynomial. The simplex $\{x \in \mathbb{R}_+^3 : x_1 + x_2 + x_3 \leq 1\}$ has no such description. The point $(1/2, 1/2, 1/2)$ satisfies every valid inequality involving at most two coordinates, because each of its coordinate pairs extends to a simplex point by setting the third coordinate to zero. Yet that point is outside the simplex. This does not restrict arbitrary linear images. $\square$

A.3 One pool: many optima and an exact LP hull

Use the contracted path of proposition A.2, retaining its unscaled qualities $C_j = n - j$, and put $t = t_n = x_n$. Modify its terminal product and add the following interface:

$$A \rightarrow n, \quad A \rightarrow P, \quad B \rightarrow P, \quad B \rightarrow W, \quad P \rightarrow W, \quad P \rightarrow V, \quad Z \rightarrow V.$$

Sources A, B have qualities one and two and exact supplies one; source Z has quality zero and supply interval $[0, 1]$. Every new arc has capacity one. Pool P has exact throughput one. Products n, W have exact demand one and upper qualities one and two; product V has capacity two and upper quality one. There are no further positive lower bounds. The original terminal source has quality zero. Figure 4 shows the interface and its forced flows.

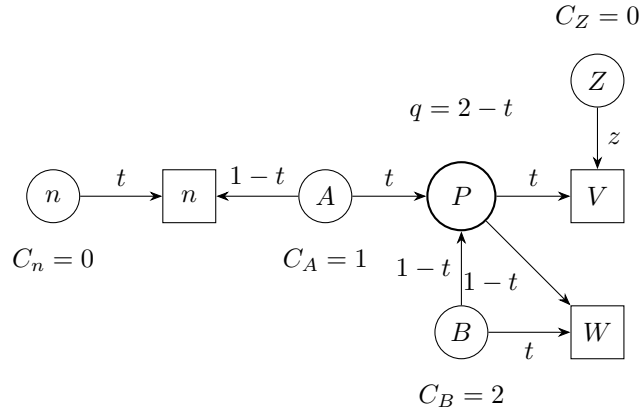


Figure 4: The vertex-forcing interface attached to the terminal source and product of the path. Circles are sources, the thick circle is the pool, and squares are products. Contracts force the displayed flows; the quality row at V becomes $z \geq t - t^2$. The triangle B, P, W is the only undirected cycle. The terminal source’s other path arc is omitted.

Theorem A.6 (A physical vertex-forcing family). *The resulting one-pool network has a feasible set affinely equivalent, in physical arc flows and active pool quality, to*

$$T = \{(x, z) : x \in P_{1/4}, \quad x_n - x_n^2 \leq z \leq 1\}. \quad (112)$$

There are ordinary nonnegative source costs and product revenues bounded by two for which the network has exactly 2^n isolated global maximizers. Every source and product has degree at most two,

the pool has two feeds and two outlets, and the full undirected network is connected with one cycle. All quality data can be scaled into $[0, 1]$.

Proof. Demand at n forces $f_{An} = 1 - t$; source A then gives $f_{AP} = t$. Pool throughput gives $f_{BP} = 1 - t$, whence source B gives $f_{BW} = t$. Demand at W forces $f_{PW} = 1 - t$, leaving $f_{PV} = t$. The pool is active with quality $q = 2 - t$. If $z = f_{ZV}$, the only nonredundant new quality row is

$$(2 - t)t \leq t + z \iff z \geq t - t^2.$$

Every $t \in [0, 1]$ has $0 \leq t - t^2 \leq 1/4$, so all other arc, source, throughput and quality bounds hold, and this describes every physical extension. In particular all flows and q are affine in (x, z) , despite the curved feasible boundary.

Give path source j cost s_j , sources A, B cost one, and Z cost two. Give path product $j < n$ revenue s_{j+1} and products n, W, V revenue one. Subtracting one from every source cost and every product revenue leaves profit unchanged by total mass conservation. The new interface prices are zero except for unit cost on Z . On the path, source j has cost $s_j - 1$, its left product has the same revenue, and its right product has revenue $s_{j+1} - 1$ when $j < n$ and zero when $j = n$. Thus profit is exactly

$$\sum_{j < n} (s_{j+1} - s_j)t_j - z = \sum_{j < n} 3s_j^2 x_j - z = \sum_{j < n} d_j x_j - z. \quad (113)$$

By lemma A.1, this is at most $-F(x) \leq 0$; equality forces a path vertex and the unique clean flow $z = t - t^2$. These are all 2^n optimizers, so all are isolated. The degree assertions follow from the displayed arcs. The original path and its attached trees introduce no cycle; the sole cycle is $B - P - W - B$. Dividing every quality and specification by $\max(n - 1, 2)$ preserves all flows and normalizes data. $\square$

Corollary A.7 (Strict local optima with distinct values). *A polynomially encoded nonnegative price perturbation of this network has at least 2^n strict local maxima with distinct values, exactly one of the constructed points being globally optimal. Its prices are at most three.*

Proof. Increase both the revenue of V and cost of Z by $\delta = 4^{-n}$. Their net contribution is $\delta[(t + z) - z] = \delta t$. At the least clean flow, the objective is $H(x) = -F(x) + \delta x_n$. At a vertex of terminal value t it is δt . Every terminal vertex value has denominator dividing 4^{n-1} , and these values are distinct. For neighboring vertices v, v' with terminal values t, t' , linearity of L and $F(v) = F(v') = 0$ give

$$F((1 - a)v + av') = a(1 - a)(t' - t)^2.$$

The outward derivative of H is $-(t' - t)^2 + \delta(t' - t) < 0$, since $|t' - t| \geq 4^{-(n-1)} > \delta$. The tangent cone at a polytope vertex is generated by its incident edge rays. Strict negativity on these finitely many rays bounds the linear term above by $-\eta\|x - v\|$ for some $\eta > 0$ on that cone; the quadratic remainder is $O(\|x - v\|^2)$. Hence the vertex is a strict local maximum. Extra clean flow decreases profit strictly, so the property transfers to (112) and then to the physical coordinates. Finally every feasible profit is at most $\delta t \leq \delta$, with equality only at the unique vertex with $t = 1$ and least clean flow. Additional local maxima are not excluded. The perturbation shrinks with dimension and gives no uniform tolerance or local-solver runtime bound. $\square$

Corollary A.8 (Exact integer dimension of the optimal set). *Let S be the complete optimal physical-flow set for the unperturbed objective in theorem A.6, optionally retaining the pool quality. The minimum number of integer coordinates in any exact convex lift of S is n , allowing arbitrary continuous auxiliaries and unbounded integer variables.*

Proof. For distinct optimal path vertices v, v' , $F((v + v')/2) = (v_n - v'_n)^2/4 > 0$, so their physical midpoint is not optimal. Suppose a convex set with $k < n$ integer coordinates projects exactly to S when those coordinates are integral. Choose a lift of each of its 2^n points. Two integer vectors have the same parity; their midpoint is integral and belongs to the convex lifting set. Its physical projection is the forbidden midpoint, a contradiction. For the upper bound, attach to the optimizer indexed by u the vector $u \in \{0, 1\}^n$ and take the convex hull of these finitely many lifted points. Any integral vector in its final n coordinates is binary; a convex combination yielding it must use only points with that same binary vector, so it selects exactly its original optimizer. This upper construction may have exponentially many vertices. The lower-bound argument applies the established midpoint/parity principle of Lubin, Vielma and Zadik [34, Section 4.2] to this physical optimal set. $\square$

Theorem A.9 (A compact exact hull and polynomial optimization). *The same physical feasible family admits an exact polynomial-size LP convex hull. Every rational linear objective in physical flows and active pool quality has a polynomial-time exact physical optimizer, with rational coordinates of polynomial bit length.*

Proof. Writing $d^T x = \sum_{j < n} d_j x_j$, the exact hull in (112) is

$$\text{conv}(T) = \{(x, z) : x \in P_{1/4}, \quad d^T x \leq z \leq 1\}. \quad (114)$$

The certificate gives $d^T x \leq x_n - x_n^2$, proving containment. Conversely write x as a convex combination of path vertices. At every vertex v , $v_n - v_n^2 = d^T v$, so $(x, d^T x)$ is the same convex combination of points in T . Also $(x, 1) \in T$. Their vertical segment contains every point on the right of (114). A vertex of that polytope must lie on $z = d^T x$ or $z = 1$, since otherwise it admits a vertical perturbation. On either affine boundary, a nonvertex x admits a nontrivial convex decomposition. Consequently every hull vertex has x a path vertex and is physically feasible. The two boundaries do not meet, since $d^T x \leq x_n - x_n^2 \leq 1/4$. Solve the rational LP, return an optimal vertex, and apply the affine physical map; a nonvertex point of the optimal face need not be physically feasible. Polynomial bit length follows from rational LP vertex bounds. The hull does not represent the original nonconvex set or its finite optimal set exactly, so it does not contradict corollary A.8. $\square$

Corollary A.10 (Three qualities and general supply sequences). *The conclusions of theorems A.6 and A.9 and corollaries A.7 and A.8 hold with only three distinct input qualities. The unperturbed geometry, integer dimension and LP hull also extend to any rational supplies with $s_n = 1$, $s_1 > 0$ and $s_{j+1} > 2s_j$ in the descending-quality path.*

Proof. For the first assertion attach the identical interface to the contracted two-quality relay path in the proof of theorem A.3. Its last source has quality zero, so every interface calculation is unchanged. Keep all its source, reset-product and interface contracts. At physical source position i give cost S_i , left endpoint product revenue S_1 , and product after position $i < m$ revenue S_{i+1} ; terminal and interface prices remain as before. The coefficient of its rightward flow is $S_{i+1} - S_i$. This coefficient vanishes when the next source is the same-supply reset duplicate. The coefficient $3s_j$ before the next descending step occurs on that duplicate, whose rightward flow equals t_j . At the first position it occurs directly on t_1 . Thus the sum is again (113). All conclusions transfer through the affine bijection, and the full quality alphabet is $\{0, 1, 2\}$.

For the second assertion keep (110), which remains the physical capacity/quality pair for these supplies. Direct expansion gives

$$t_n - t_n^2 - \sum_{j < n} (s_{j+1} - s_j) t_j = \sum_{j=1}^n (t_j - t_{j-1})(s_j - t_{j-1} - t_j), \quad t_0 = 0.$$

Every summand is nonnegative. The strict growth of supplies makes both endpoint branches distinct and their terminal ranges disjoint, so there are exactly 2^n endpoint vertices and distinct terminal values. The interface and economics now use coefficients $s_{j+1} - s_j$. The equality, midpoint and convex-hull proofs apply unchanged in these physical coordinates. Encoding and algorithmic bounds include the given supply bit lengths. The perturbation 4^{-n} is claimed only for the geometric sequence. $\square$

The linear exact-penalty argument of theorem A.3 does not remove the contracts in the nonlinear interface. Deleting only the lower pool-throughput bound already changes the optimum: set $x_1 = \dots = x_{n-2} = 0$, $x_{n-1} = 1$, $x_n = 1/4$, send $1/4$ from A through the pool to V , send the rest of A to n , send all of B to W , and set $z = f_{PW} = 0$. All other bounds and contracts hold, the pool quality is one, and profit is $d_{n-1} = 3/16 > 0$.

Multiple local optima in pooling have earlier geometric examples; Grothey and McKinnon [22, Section 3] describe a three-pool, two-quality example. The present family adds the explicit asymptotic count, degree restrictions and LP hull; we do not claim exhaustive priority for these refinements.

A.4 Why one unrestricted aggregate changes the abstract problem

The preceding geometry alone cannot prove degree-two pooling hardness. The following abstract reduction identifies an additional ingredient. Hardness of rank-one concave quadratic minimization in general was established by Pardalos and Vavasis [36]; the point here is the local path structure and the role of a dense slab.

Proposition A.11 (A path with one dense slab). *For rational $0 < \epsilon < 1/2$, deciding whether*

$$\min\{F(x) : x \in P_\epsilon, B - 1/4 \leq w^T x \leq B + 1/4\} \leq 0$$

is ordinarily NP-complete, even on nonempty compact instances. The Hessian of F has one negative eigenvalue and rank one. Local path coefficients can alternatively be fixed to $0, \pm 1, \pm 1/4$, with additional singleton upper bounds and polynomially many variables.

Proof. For a SUBSET SUM instance of positive integer weights w_i and integer $0 \leq B \leq W = \sum_i w_i$, set $\epsilon = 1/(8W)$. A vertex with endpoint bits u satisfies $|x_i - u_i| \leq \epsilon$. If $w^T u = B$, then $|w^T x - B| \leq 1/8$, so that vertex satisfies the slab and $F = 0$. Conversely $F \leq 0$ forces a vertex by lemma A.1; its bits satisfy $|w^T u - B| \leq 1/4 + 1/8 < 1$, forcing equality of the integers. The zero vertex has weighted sum zero. The all-one-bit vertex has weighted sum at least $W - 1/8$. If $B < W$, convex interpolation reaches B ; if $B = W$, the latter vertex already lies in the slab. Thus the domain is nonempty. The coefficients in F , including powers of ϵ , have polynomial bit length. The endpoint bits are an NP certificate: reconstruct the rational vertex and verify the slab. The Hessian is $-2e_n e_n^T$.

To fix $\epsilon = 1/4$, let r be the least nonnegative integer with $4^{-(r+1)} \leq 1/(8W)$. Place free coordinates at positions $1, r+2, 2r+3, \dots$ on a path of dimension $N = n + (n-1)r$. Impose $x_j \leq 1/2$ on each padding coordinate and use the full N -coordinate certificate F . At a zero of F , padding positions must take their lower branch: the upper branch is at least $3/4$. Successive free coordinates are separated by r multiplications by $1/4$, so each differs from its endpoint bit by at most $4^{-(r+1)}$. Every free-bit vector extends uniquely to a zero of F . Put the slab weights only on free positions; the same rounding, nonemptiness and certificate arguments apply. Since $r = O(\log W)$, the construction is polynomial. The dense weights and objective coefficients retain binary numerical data, so this gives no strong-hardness or inverse-polynomial gap claim. $\square$

Fixing the scalar nonlinear coordinate $q = x_n$ in this problem leaves a bounded LP. The local inequalities form a path, but the slab has two bounds on a dense aggregate, and L is another dense linear form. Thus one nonlinear parameter and a fixed number of aggregate rows do not justify replacing the independent blocks of theorem 4.1 by arbitrarily long coupled paths. We give no physical degree-two realization of the extra weighted slab. Without the slab, minimizing $a^T x - bx_n^2$ over the bare path, for $b \geq 0$, is polynomial: a concave function has a vertex minimizer, and $x_n^2 = L(x)$ at every vertex, so it suffices to minimize the linear function $a^T x - bL(x)$. The slab creates new vertices and invalidates this linearization.

A.5 Matrix parameters and nonlinear leaves

The interaction rank of a *cost*, treated in Appendix B, differs from the rank of a parameter perturbing a *constraint matrix*. Consider rational data and

$$\min c^T x \quad \text{subject to } Ax + \lambda Dx \leq b, \quad x \in X, \quad \underline{\lambda} \leq \lambda \leq \bar{\lambda},$$

where X is a rational polyhedron and the nonempty parameter interval is finite. If $D = uv^T$, introduce $z = v^T x$ and $w = \lambda z$. The exact projected feasible set is the union of the two polyhedra

$$\begin{aligned} Ax + uw \leq b, \quad x \in X, \quad z = v^T x, \quad z \geq 0, \quad \underline{\lambda} z \leq w \leq \bar{\lambda} z; \\ Ax + uw \leq b, \quad x \in X, \quad z = v^T x, \quad z \leq 0, \quad \bar{\lambda} z \leq w \leq \underline{\lambda} z. \end{aligned}$$

For $z \neq 0$, recover $\lambda = w/z$; at $z = 0$ both branches force $w = 0$ and any allowed parameter works. Thus two LPs solve joint feasibility and minimization. If the objective is finite and the set nonempty, a rational LP solution and that ratio give a rational optimum of polynomial bit length; the LPs also detect unbounded objectives. This auxiliary-variable extension is closely related to the one-nonzero-column case in the preprint of Boveroux, Carvalho, Lodi and Louveaux [11, Section 3.3].

Their positive-product construction already has perturbation rank two: a fixed-one variable s occurs in the opposing rows for $\lambda s = z_1$, and a second variable z_2 in $\lambda z_2 \leq t$. The two nonzero columns have independent supports. With the positive-product source of lemma 3.7 and its explicit bounds, these rows give ordinary hardness for joint minimization of t . This observation concerns dense base-polytope constraints, not physical path coefficients. A parameter-dependent right-hand side adds a fixed-one column before perturbation rank is measured; a direct parameter objective term and a true max-min objective are outside the two-LP argument.

Replacing the polyhedral blocks of theorem 4.1 by convex nonlinear leaves also meets an arithmetic obstacle. Given positive integers a_i and an integer B , let

$$P_i = \{y_i : 0 \leq y_i \leq \max(1, a_i), \quad y_i^2 = a_i\}.$$

Each leaf is a compact convex singleton in one scalar variable, described by rational quadratic data, and there is no nonlinear core. The single aggregate inequality $\sum_i y_i \leq B$ asks exactly whether $\sum_i \sqrt{a_i} \leq B$. Hence an unrestricted exact polynomial-bit extension to such leaves would solve Square-Root Sum in polynomial time. This is an arithmetic implication, not an NP-hardness theorem. Independent algebraic leaf values may generate a growing common field, whereas the polyhedral recovery in theorem 4.1 stays in the field of the fixed core.

B Optimization of an isolated rank-one flow block

B.1 Exact optimization of low-interaction-rank pool costs

For positive integers m, n , consider the isolated rank-one margin block

$$\mathcal{K} = \{W \in \mathbb{R}_+^{m \times n} : \text{rank } W \leq 1, \quad \ell \leq W\mathbf{1} \leq u, \quad \ell' \leq W^T\mathbf{1} \leq u'\}, \quad (115)$$

with finite nonnegative rational bounds in the stated order, and objective $\langle C, W \rangle$. For $W \neq 0$, its margins x, y and common total S give $W = xy^T/S$ by (8). The feasible-total interval is $I = [\max(\mathbf{1}^T \ell, \mathbf{1}^T \ell'), \min(\mathbf{1}^T u, \mathbf{1}^T u')]$; reject an empty interval. Define the *interaction rank*

$$k = \text{rank } D, \quad D_{ij} = C_{ij} - C_{i1} - C_{1j} + C_{11}.$$

This is the least rank after subtracting arbitrary additive row and column costs. Indeed, multiplying $C - a\mathbf{1}^T - \mathbf{1}b^T$ on the left by $I_m - \mathbf{1}e_1^T$ and on the right by $I_n - e_1\mathbf{1}^T$ gives D , so no subtraction has smaller rank, and $a_i = C_{i1}$, $b_j = C_{1j} - C_{11}$ attain it. Rational elimination therefore constructs $C = a\mathbf{1}^T + \mathbf{1}b^T + UV^T$ with k factor columns, and the objective is

$$a^T x + b^T y + \frac{(U^T x)^T (V^T y)}{S}. \quad (116)$$

Theorem B.1 (Fixed interaction rank). *For each fixed k , exact linear optimization over (115) is polynomial in the rational input bit length, with dimension dependence $(m+n)^{O(k+1)}$; this is not an FPT bound in k . An optimizer uses rational numbers and at most one square root, all with polynomial encoding length.*

Proof. A linear image of an h -coordinate box in dimension d is a translated sum of h segments. Its vertices correspond to full-dimensional sign regions of the central hyperplanes orthogonal to its nonzero generators. The region recurrence obtained by adding one hyperplane gives at most $2 \sum_{i=0}^{d-1} \binom{h-1}{i}$ vertices for positive effective dimension, and one in dimension zero. Degenerate or repeated generators only decrease this number. For fixed d , enumerate these vertices constructively by adding segments one at a time: union the existing vertices with their translates and retain only vertices of the resulting convex hull. Membership of one candidate in the convex hull of the others is a rational LP test. Store with every retained candidate a corner of the original box mapping to it. Intermediate lists have the same polynomial bound, and every stored coordinate is a rational generator sum of polynomial bit length.

Apply this construction to

$$Z_x = \{(\mathbf{1}^T x, U^T x, a^T x) : \ell \leq x \leq u\}, \quad Z_y = \{(\mathbf{1}^T y, V^T y, b^T y) : \ell' \leq y \leq u'\}.$$

Both projected boxes have dimension at most $k+2$ and polynomially many vertices for fixed k . At any fixed $S > 0$, the objective (116) is affine in either projected margin when the other is fixed. Starting from an optimum, minimize first on one slice at total S and choose a slice vertex, then do the same on the other. Both steps preserve optimality, so there is an optimum at a pair of slice vertices.

Every vertex of a hyperplane slice of a polytope lies on an original vertex or edge. Indeed, the point lies in the relative interior of its smallest containing face. If that face had dimension at least two, intersecting it with one hyperplane through the point would leave a nonzero two-sided feasible direction, contradicting extremality in the slice. Enumerate all pairs of projected-box vertices, including every edge. A segment with different endpoint totals parametrizes a projected margin

affine in S on the closed interval between those totals. Interpolating its two stored box corners gives a feasible original-margin witness. The additional nonedge segments are harmless, since they stay in the projected box. Include each original vertex as a singleton candidate at its fixed total. Equal-total segments can be omitted: an interior point of an edge parallel to the slicing hyperplane cannot be a slice vertex, and its endpoints are already included.

Pair row and column candidates and intersect their intervals with I . On every positive interval, their projected coordinates and witnesses are affine in S , and substitution gives

$$f(S) = \alpha S + \beta + \gamma/S \quad (117)$$

with rational coefficients of polynomial bit length. Evaluate positive endpoints and the stationary point $\sqrt{\gamma/\alpha}$ if $\alpha, \gamma > 0$ and it lies in the interval. Apart from a constant function, where any feasible total suffices, these are all possible interior minima: when both coefficients are negative a stationary point is a maximum, and the other sign cases are monotone. Singletons are evaluated directly.

If $0 \in I$, zero is feasible and is included separately; if $I = \{0\}$, it is the only feasible matrix. No infimum is missed at zero, since $|\langle C, W \rangle| \leq \|C\|_\infty S$ for every nonnegative matrix of total S . In particular an interval candidate reaching zero has the correct continuous zero extension. Every enumerated candidate is feasible, and every fixed-total optimum has representatives on the list, so minimizing the list gives a global optimum. There are polynomially many candidates for fixed k . Interior values are $\beta + 2\sqrt{\alpha\gamma}$; comparisons of two such values require sign checks and at most two squarings, or degree-four algebraic comparison. Interpolation at the chosen total and $W = xy^T/S$ stay in one quadratic number field. These operations and all rational LP tests have polynomial bit complexity, proving the stated bound and output representation. $\square$

Low-rank box optimization and projected-box methods are established. Punnen, Sripratak and Karapetyan [38, Sections 3.2–3.3] treat bipartite box objectives, and Hladík, Černý and Rada [28] use zonotope faces for continuous quadratic box optimization. The additional feature here is the shared variable total and the division by it in (116), which the proof handles by slice segments and one-variable rational optimization.

Proposition B.2 (Rank-one cost specialization). *If $C = pq^T$ is supplied in factored form, the optimum and its margins can be found in $O((m+n)\log(m+n))$ arithmetic operations and comparisons, with polynomial bit complexity. Factoring a supplied dense rank-one matrix or writing every entry of W adds $O(mn)$ arithmetic work.*

Proof. At each positive total define $A_-(S), A_+(S)$ as the minimum and maximum of $p^T x$ on $[\ell, u] \cap \{\mathbf{1}^T x = S\}$, and define B_-, B_+ similarly. The two scalar images are intervals, so their product is minimized at one of the four endpoint pairs, regardless of signs. The optimal value is $\min_{\sigma, \tau \in \{-, +\}} A_\sigma(S) B_\tau(S)/S$. Each endpoint function is continuous knapsack: start at the lower bounds and fill residual capacities in increasing coefficient order for a minimum, decreasing order for a maximum. Sorting and prefix sums give all affine pieces and $O(m+n)$ total breakpoints. Merge those breakpoints with I and sweep the four paired functions. Each is of form (117); process its endpoints and possible minimum stationary point as above. Recover the selected greedy margins in linear work. The same exact comparisons prove polynomial bit complexity. $\square$

A rank-one cost need not have a rational optimizer. For example, fix the first row and column margins to one, bound the other two margins by $[0, 1]$, and take $p = (3, 1)$, $q = (2, 1)$. The cost is $S + 3 + 2/S$ for $1 \leq S \leq 2$, minimized at $S = \sqrt{2}$.

Corollary B.3 (A fixed-attribute Lagrangian pool oracle). *Suppose an isolated pool block has additive operating costs, row and column margin bounds as in (115), and K attributes. For any supplied rational nonnegative multipliers on terminal quality inequalities, the Lagrangian subproblem obtained by dualizing those inequalities is exactly solvable in polynomial time for fixed K .*

Proof. Writing input attributes Q_{ik} and product upper bounds H_{jk} , the homogeneous quality rows are $\sum_i (Q_{ik} - H_{jk})W_{ij} \leq 0$. For minimization, multipliers $\lambda_{jk} \geq 0$ add matrix cost

$$Q\Lambda^T - \mathbf{1}h^T, \quad h_j = \sum_k \lambda_{jk} H_{jk}.$$

The additive part does not affect interaction rank, and $\text{rank}(Q\Lambda^T) \leq K$. Apply theorem B.1. Lower quality rows reverse the corresponding signs; both upper and lower multipliers can be combined into the same K input-attribute columns. An affine basis of the input qualities can further replace K by its rank. All remaining constraints must have exactly the margin/rank form; other pool couplings need separate treatment. The result is an exact subproblem oracle; it claims neither zero duality gap nor tractability of full pooling. $\square$

If a supplied cost approximation $\tilde{C}$ has fixed interaction rank and $\|C - \tilde{C}\|_\infty \leq \eta$, where this is the maximum absolute entry norm, its optimizer has true cost at most $2\eta S_{\max}$ above optimum, with $S_{\max} = \min(\mathbf{1}^T u, \mathbf{1}^T u')$. Indeed every feasible matrix has objective perturbation at most $\eta S_{\max}$. The bound assumes both the approximation and its error certificate are supplied.

B.2 General costs: hardness and short exact witnesses

The fixed-interaction-rank algorithm does not extend to arbitrary costs unless $P = NP$. We prove this for the same margin set (115), also with zero lower bounds. The reduction answers the conjecture stated after Theorem 4 of Dey et al. [16]. It uses the established NP-completeness of maximum edge biclique [37]; we claim no priority for its underlying Boolean bilinear encoding. Gillis and Glineur [20] already connect maximum edge biclique with nonnegative rank-one approximation of a possibly signed matrix under a squared Frobenius objective; here the objective is linear and the constraints are simultaneous margin bounds.

Theorem B.4 (General-cost margin hardness). *Deciding whether $\min_{W \in \mathcal{K}} \langle C, W \rangle \leq 0$ is strongly NP-hard, even with all margin bounds in $\{0, 1\}$ and integer costs of polynomially bounded magnitude. The only positive lower bounds needed are one row margin and one column margin, both one.*

Proof. Given a bipartite graph with sides of sizes p, q , let $h = \max(p, q)$ and let $1 \leq k \leq |E|$ be the required biclique edge count. Use one distinguished row and column, each of exact margin one; p graph rows and q graph columns; and h dummy rows and h dummy columns. Every remaining margin has interval $[0, 1]$. Set $C_{00} = k$, set graph-block entries to -1 on edges and h^2 on nonedges, and set every other entry to zero. Writing x, y for graph margins, the numerator in (8) is

$$r^T C c = k + \sum_{i,j} C_{ij} x_i y_j = k + Q(x, y).$$

Dummy margins enforce equal totals for every $(x, y) \in [0, 1]^p \times [0, 1]^q$: their required total difference lies in $[-h, h]$, and each dummy side realizes every total in $[0, h]$. The common total S is at least one, so the objective is nonpositive exactly when this numerator is nonpositive.

Minimize the bilinear function Q first over the row box with y fixed, choosing a binary minimizer, and then over the column box with that row choice fixed. Hence its minimum occurs at binary x, y .

For their supports A, B , a selected nonedge gives $Q \geq h^2 - (|A||B| - 1) \geq 1$. If there is no nonedge, then $k + Q = k - |A||B|$. Thus the constructed threshold holds exactly when there is a biclique with at least k edges. Dimensions are linear in h and all numbers are bounded by h^2 , proving strong hardness. In a no instance the numerator is at least one everywhere, because its minimum occurs at a binary pair. Since $S \leq 2h + 1$, the optimal value is then at least $1/(2h + 1)$. $\square$

Lemma B.5 (Repair of two unit margins). *Let $\mathcal{U}_{p,q}$ be the nonnegative rank-one matrices whose row and column margins are at most one, and let $F = \{W \in \mathcal{U}_{p,q} : r_0 = c_0 = 1\}$. Every $W \in \mathcal{U}_{p,q}$ has $W' \in F$ satisfying*

$$\|W - W'\|_1 \leq 2(2 - r_0 - c_0), \quad \|W\|_1 = \sum_{i,j} |W_{ij}|.$$

Proof. Put $\delta = 2 - r_0 - c_0$. When $S \geq 1$, raise r_0 to one and reduce the other row margins by $1 - r_0$, preserving total S ; their total $S - r_0$ suffices. Do the same to the column margins. The repaired vectors r', c' remain in their unit boxes and give $W' = r'(c')^T/S$. The outer-product triangle inequality gives

$$\|W' - W\|_1 \leq \|r' - r\|_1 + \|c' - c\|_1 = 2\delta.$$

When $0 < S < 1$, take $W' = E_{00}$. From $(S - r_0)(S - c_0) \geq 0$ follows $W_{00} \geq r_0 + c_0 - S$, and hence

$$\|W - E_{00}\|_1 = 1 + S - 2W_{00} \leq 1 + 3S - 2r_0 - 2c_0 \leq 2\delta.$$

At $S = 0$ the same choice satisfies the bound directly. $\square$

Corollary B.6 (Unit upper bounds and zero lower bounds). *Rational threshold optimization over $\mathcal{U}_{p,q}$ is strongly NP-hard with integer costs and a negative threshold, all of polynomially bounded magnitude.*

Proof. For the matrix C in theorem B.4, put $B = \max_{ij} |C_{ij}|$ and $L = 2B + 1$. Remove the two positive lower bounds and replace the objective by $\langle C, W \rangle - L(r_0 + c_0)$, a linear matrix cost. For a point outside F , the repair in lemma B.5 changes this objective by at most $(2B - L)\delta < 0$. Every optimizer therefore belongs to F , and the optimum is exactly its previous value minus $2L$. Use threshold $-2L$. New cost magnitudes are at most $5B + 2$; the construction is strong and retains the gap $1/(2h + 1)$. $\square$

Theorem B.7 (Margin certificates and algebraic optima). *For finite rational data in (115), every nonempty instance has an optimizer in one real quadratic number field with polynomial encoding length. Every yes instance for a rational cost threshold has a rational matrix witness of polynomial bit length. Consequently rational threshold decision is strongly NP-complete.*

Proof. Compactness gives an optimum. Treat the feasible zero matrix separately. At an optimal positive total S_* , minimize first over the row-margin slice with the column margins fixed, then over the column slice with those rows fixed. Choose a vertex at each step; both steps preserve global optimality. Every vertex of a box intersected with one total hyperplane has all but at most one coordinate at a bound, since two strictly interior coordinates would allow opposite perturbations. The same at-most-one-free-margin structure appears in the margin lemma and SOC construction of Jalilian and Kocuk [31, Lemma 1 and the proof of Theorem 2]; here the common-total parametrization gives exact arithmetic and certificate bounds. Fix these two margin patterns. They are affine functions

$$r(S) = e_a S + b, \quad c(S) = e_h S + d$$

on a closed rational interval J , with all coefficients of polynomial bit length. Here e_a, e_h denote unit vectors in their respective spaces. Their objective for $S > 0$ is $f(S) = \alpha S + \beta + \gamma/S$, where

$$\alpha = C_{ah}, \quad \beta = e_a^T C d + b^T C e_h, \quad \gamma = b^T C d.$$

These expressions have polynomial bit length. The endpoint and stationary-point argument of theorem B.1 shows that a global optimum in this family is rational or obtained at $S = \sqrt{\gamma/\alpha}$ with $\alpha, \gamma > 0$. Its margins and matrix belong to the same quadratic field. If J reaches zero, nonnegative margins with total zero vanish, and the continuous zero extension is handled separately.

Now let the rational threshold be t . The family contains a positive threshold witness exactly when, for some positive $S \in J$,

$$q(S) = \alpha S^2 + (\beta - t)S + \gamma \leq 0.$$

A quadratic achieves its minimum on a rational closed interval at a rational endpoint or, for positive leading coefficient, at the rational stationary point $(t - \beta)/(2\alpha)$. If the minimum is negative, its minimizer cannot be zero: if $0 \in J$, then $r(0) = c(0) = 0$, so $q(0) = 0$. If the minimum equals zero, any positive threshold witness is a minimizer. Such a minimizer is a positive rational endpoint or the rational stationary point, unless q is constant; in the constant case take any positive rational point of J . This gives a rational total, margins, and matrix of polynomial bit length. A verifier checks margin bounds, rank at most one, and the rational objective inequality in polynomial time. Together with theorem B.4 and corollary B.6, this proves the last claim. $\square$

Theorem B.8 (One fixed margin dimension). *For $d = \min(m, n)$ and total rational input length B , exact optimization over (115) takes $d2^d \text{poly}(m, n, B)$ bit operations. In particular it is polynomial when either matrix dimension is fixed.*

Proof. Transpose so the row dimension is d . For any fixed positive total, minimizing over column margins makes the objective numerator a concave function of row margins. A minimum occurs at a row-slice vertex. Enumerate its free coordinate and a lower/upper choice for every other coordinate, at most $d2^{d-1}$ patterns. Each pattern gives an affine row vector $r(S)$ and a rational validity interval. The column weights $(C^T r(S))_j$ are affine in S . Their $O(n^2)$ pairwise crossings split the interval into pieces with a fixed weight order; identical weights use a fixed index order. On each piece, the continuous-knapsack rule fills column margins from their lower bounds in that order. Its prefix capacities contribute at most n further breakpoints. Both margins are therefore affine on $O(n^3)$ intervals per row pattern.

At ties either adjacent greedy order is optimal, and singleton intervals are checked as well. On every positive interval apply the endpoint and square-root calculation for (117); include zero when feasible. Every candidate is feasible and the enumeration covers an optimum at every total. All rational arithmetic and algebraic comparisons have polynomial bit complexity as proved in theorem B.1. The factor outside the polynomial is precisely the row-pattern count, giving the claimed dependence. $\square$

Rational threshold certificates do not imply rational exact optimizers. For example, fix the first row and column margins to one, let the other margins vary in $[0, 1]$, and use $C = I_2$. Equal totals force the remaining margins to be $S - 1$, and the objective is $S - 2 + 2/S$ on $[1, 2]$, with unique minimum at $S = \sqrt{2}$.

C Exact and uniform approximate convexification of a pool block

This appendix proves the convexification results stated in Section 6. The set is an isolated matrix block without physical quality or interpool constraints. Throughout, N, m are positive integers and all error radii are nonnegative. Define

$$R_N = \{W \in \mathbb{R}_+^{N \times N} : \text{rank } W \leq 1, W\mathbf{1} \leq \mathbf{1}, W^T\mathbf{1} \leq \mathbf{1}\}, \quad K_N = \text{conv}(R_N).$$

Both sets are compact: all entries of R_N lie in $[0, 1]$, and its rank condition is closed. Every point of K_N is a finite convex combination of generators in R_N . The norm $\|W\|_1$ below is the entrywise sum $\sum_{ij} |W_{ij}|$, with unit ball U_1 ; the dual coefficient norm is $\|H\|_\infty = \max_{ij} |H_{ij}|$.

A lift over a cone $\mathcal{C}$ means an expression $K = \pi(\mathcal{C} \cap L)$, where L is affine and π is linear (or affine). Affine sections and images preserve the cone: intersect L with the preimage of the section equations and compose π with the desired map. Every scalar inequality counts as a nonnegative cone factor. Semidefinite size means total matrix order, including scalar blocks of order one. Arbitrary real formulation coefficients are allowed.

C.1 An explicit correlation-polytope face

Theorem C.1 (Correlation face). *The affine section*

$$F_m = \{W \in K_{2m} : \text{tr } W = 1, \sum_{i=1}^m W_{i,m+i} = 0, \mathbf{1}^T W \mathbf{1} = m\}$$

is a face affinely isomorphic to $\text{COR}(m) = \text{conv}\{xx^T : x \in \{0, 1\}^m\}$. The map is $X = mW_{[m],[m]}$; writing $x = \text{diag } X$, its inverse is

$$W = \frac{1}{m} \begin{pmatrix} X & x\mathbf{1}^T - X \\ \mathbf{1}x^T - X & \mathbf{1}\mathbf{1}^T - x\mathbf{1}^T - \mathbf{1}x^T + X \end{pmatrix}. \quad (118)$$

Proof. For a nonzero generator $W = rc^T/S$, with $r, c \in [0, 1]^{2m}$ of common total S , one has $\text{tr } W = r^T c/S \leq 1$. Equality forces $\sum_i r_i(1 - c_i) = 0$ and also $\sum_i c_i(1 - r_i) = 0$. Hence $r_i > 0$ implies $c_i = 1$ and $c_i > 0$ implies $r_i = 1$: both margins are the same nonempty binary vector $a = \mathbf{1}_A$. The trace-one face is therefore the convex hull of the matrices $aa^T/|A|$, because every positive-weight generator in a convex combination attaining trace one must itself attain trace one; in particular, the zero matrix cannot occur in that combination.

Nonnegativity makes the cross-pair equation another face equation. It retains exactly those generators whose support contains at most one element from each pair $\{i, m+i\}$. They satisfy $|A| \leq m$. Their total matrix mass is $|A|$, so the last equation selects the maximal possible value m , exactly one element of every pair. These are the atoms

$$\frac{1}{m}(x, \mathbf{1} - x)(x, \mathbf{1} - x)^T, \quad x \in \{0, 1\}^m.$$

Successive faces are faces of the original set. The principal-block map sends these atoms to xx^T , and block expansion gives (118). Its entries are affine in X , so the inverse identity extends to the two convex hulls. $\square$

Corollary C.2 (Exact conic size). *For each fixed $d \geq 1$, an exact lift of K_{2m} over a product of PSD cones of order at most d needs $2^{\Omega(m)}$ blocks, with constants depending on d . Every exact second-order-cone lift has $2^{\Omega(m)}$ total cone dimension. Every exact SDP lift has total matrix order $2^{\Omega(m^{2/13})}$. For $N \geq 2$, K_N has no finite LP lift.*

Proof. The face and affine map transfer any lift to $\text{COR}(m)$ without changing its cone. The fixed-order bound follows from Fawzi and Parrilo [17, Theorem 1]; smaller blocks may be padded with zeros to order d . A Lorentz cone of dimension $h \geq 3$ can be represented with $h - 2$ three-dimensional Lorentz cones: introduce successive partial Euclidean norms. A three-dimensional Lorentz cone is linearly isomorphic to the cone of 2×2 PSD matrices. Dimensions one and two use at most two nonnegative scalar factors. Thus total Lorentz dimension D gives at most D blocks of order two. The fixed-order lower bound proves the second assertion. For unrestricted SDP order apply Lee et al. [33, Theorem 1.1]; several blocks combine as a block-diagonal PSD matrix of their total order.

The nonpolyhedral 2×2 margin example is due to Dey et al. [16, Theorem 4 and Appendix A.3, Proposition 12]; we give its unit-upper-bound face form. Consider the face of K_2 where the first row and column margins equal one. Its generating matrices are

$$W(t) = \frac{1}{1+t} \begin{pmatrix} 1 & t \\ t & t^2 \end{pmatrix}, \quad 0 \leq t \leq 1.$$

If $q = W_{11} = 1/(1+t)$, then $W_{22} = q + 1/q - 2$ for $1/2 \leq q \leq 1$. This function is strictly convex, so every graph point is extreme in its convex hull: equality in the convexity inequality for a convex combination requires all first coordinates to agree. Therefore this face has infinitely many extreme points. For $N > 2$, fixing all entries outside a principal 2×2 block to zero gives the same face. A projection of a finite polyhedron is a polyhedron, so no finite LP lift exists. Consequently the fixed-order block-count bound is only formal for LP lifts; the positive-error LP result below is the substantive finite-size statement. $\square$

C.2 A quantitative transfer to the face

For $W \in K_{2m}$ write

$$T(W) = \mathbf{1}^T W \mathbf{1}, \quad D(W) = 1 - \text{tr } W, \quad B(W) = \sum_{i=1}^m W_{i,m+i}.$$

The letter D in this appendix denotes trace deficit, not the interaction-cost matrix of Appendix B.

Lemma C.3 (Exposure and rounding). *Every $W \in K_{2m}$ satisfies*

$$T(W) \leq m + 2mB(W) + 4mD(W). \quad (119)$$

There is $V \in F_m$ such that

$$\|W - V\|_1 \leq 28mB(W) + 156mD(W) + 5[m - T(W)]. \quad (120)$$

The right side is nonnegative; its last bracket need not be.

Proof. First take a nonzero generator $W = rc^T/S$. For $a, b \in [0, 1]$, $|a - b| \leq a(1 - b) + b(1 - a)$, so

$$\|r - c\|_1 \leq 2SD(W). \quad (121)$$

The pair inequality $r_i + r_{m+i} - r_i r_{m+i} \leq 1$ yields

$$S \leq m + \sum_i r_i r_{m+i} \leq m + \sum_i r_i c_{m+i} + \|r - c\|_1 \leq m + SB + 2SD.$$

As $S \leq 2m$, this proves (119) for generators, including zero directly, and hence for their convex hull by affinity.

Round each r_i to a nearest binary value a_i . Since $\min(t, 1-t) \leq 2t(1-t)$ and $\sum_i r_i(1-r_i) \leq SD + \|r - c\|_1 \leq 3SD$, we obtain

$$u := \|r - a\|_1 \leq 6SD, \quad v := \|c - a\|_1 \leq 8SD.$$

Let h, z be the numbers of pairs containing respectively two and zero ones in a , and put $k = \sum_i a_i = m + h - z$. The elementary bound $|ab - cd| \leq |a - c| + |b - d|$ in the unit box gives

$$h \leq \sum_i r_i r_{m+i} + u \leq SB + \|r - c\|_1 + u \leq SB + 8SD.$$

Change a into a binary vector b with exactly one one in each pair by deleting one entry from every full pair and adding one to every empty pair. Then

$$\|a - b\|_1 = h + z = 2h + m - k \leq 2SB + 22SD + |m - S|.$$

Here $|S - k| \leq u$ was used. The point $V = bb^T/m$ lies in F_m . To avoid division by a possibly small S , put $\tilde{b} = (S/m)b$. All of $r, c, \tilde{b}$ have total S . Outer-product differences therefore give

$$\begin{aligned} \|rc^T/S - \tilde{b}\tilde{b}^T/S\|_1 &\leq \|r - \tilde{b}\|_1 + \|c - \tilde{b}\|_1, \\ \|\tilde{b}\tilde{b}^T/S - bb^T/m\|_1 &= |S - m|. \end{aligned}$$

Consequently

$$\|W - V\|_1 \leq u + v + 2\|a - b\|_1 + 3|S - m| \leq 8mB + 116mD + 5|S - m|. \quad (122)$$

For the zero generator, any paired atom has distance m , and (122) still holds.

Now express $W = \sum_a \lambda_a W_a$ as a finite convex combination of generators, with totals S_a . By (119), $(S_a - m)_+ \leq 2mB(W_a) + 4mD(W_a)$. Using $|t| = 2t_+ - t$ gives

$$\sum_a \lambda_a |S_a - m| \leq 4mB(W) + 8mD(W) + m - T(W).$$

Combine their respective paired atoms in the same proportions and apply (122). The coefficients become $8m + 20m = 28m$ and $116m + 40m = 156m$, proving (120). $\square$

Theorem C.4 (Outer-approximation transfer). *Let $K_{2m} \subseteq Q \subseteq K_{2m} + \eta U_1$, and let H_m be the three-equation affine space defining F_m . For every $Y \in Q \cap H_m$ there is $V \in F_m$ with*

$$\|Y - V\|_1 \leq (184m + 6)\eta.$$

In particular, putting $A_m = m(184m + 6)$ and $P_Q = \{mY_{[m],[m]} : Y \in Q \cap H_m\}$ gives

$$\text{COR}(m) \subseteq P_Q \subseteq \text{COR}(m) + A_m \eta U_1.$$

Every conic lift of Q gives a lift of P_Q over the same cone.

Proof. For $Y \in Q \cap H_m$, choose $W \in K_{2m}$ within distance η . The three affine functions B, D, T have coefficient norms one. Thus $0 \leq B(W), D(W) \leq \eta$ and $m - T(W) \leq \eta$. Lemma C.3 supplies V within $(184m + 5)\eta$ of W , and the triangle inequality gives the first claim. Principal-block projection cannot increase the norm; scaling it by m gives the second inclusion. The first follows from the exact face atoms. Adding the affine equations and composing the output map proves the lift statement. $\square$

Corollary C.5 (Uniform approximate LP size). *If a polyhedron Q satisfies $K_{2m} \subseteq Q \subseteq K_{2m} + \eta U_1$ with $\eta \leq 1/A_m$, every LP lift of Q has $2^{\Omega(m)}$ inequalities.*

Proof. The transferred polyhedron P_Q lies between $\text{COR}(m)$ and its entrywise unit neighborhood. For each $a \in \{0, 1\}^m$ put $M_a = 2 \text{diag}(a) - aa^T$, whose coefficient norm is at most one. The inequality $\langle M_a, X \rangle \leq 1$ is valid for the correlation polytope: its slack at xx^T equals $(1 - a^T x)^2$. Every $X \in P_Q$ therefore satisfies $\langle M_a, X \rangle \leq 2$. The fixed-factor sandwich theorem of Braun et al. [12, Theorem 6(i), author manuscript], with dilation factor two, gives $2^{\Omega(m)}$ inequalities for every LP lift of any polyhedron in this sandwich. A lift of Q transfers with no added inequalities, proving the result. $\square$

C.3 Uniform approximate SDP size

The correlation inequalities above do not by themselves give an SDP lower bound. We use a different slack family and first state the published quantitative input. For a nonnegative matrix M , its PSD rank is the least q such that $M_{st} = \langle A_s, B_t \rangle$ for PSD matrices A_s, B_t of order q .

The following two facts are from Lee et al. [33, Theorem 5.3 and equation (3.11)], following Theorem 3.8]. For odd $k \geq 3$, put

$$f_k(x) = \frac{(\sum_i x_i - k/2)^2 - 1/4}{k^2}, \quad x \in \{0, 1\}^k.$$

There is a degree- k pseudo-density D with

$$\mathbb{E}D = 1, \quad \|D\|_\infty \leq k^{3/2}, \quad \mathbb{E}Df_k = -1/(4k^2), \quad \mathbb{E}Dp^2 \geq 0 \quad (\deg p \leq k/2).$$

Expectations are uniform on the Boolean cube; a pseudo-density may be negative. More generally, if $f : \{0, 1\}^k \rightarrow [0, 1]$ and a degree- d pseudo-density obeys $\mathbb{E}Df < -\delta$ for $0 < \delta \leq 1$, then, for $m \geq 2k$, the matrix $M(S, x) = f(x_S)$ indexed by k -subsets of $[m]$ and $x \in \{0, 1\}^m$ satisfies

$$\text{rank}_{\text{psd}} M \geq \left(\frac{c_0 \delta m}{dk^2 \|D\|_\infty \log m} \right)^{d/4} \left(\frac{\delta}{\|D\|_\infty} \right)^{3/2} \sqrt{\mathbb{E}f}, \quad (123)$$

where $c_0 > 0$ is universal. We use these theorems without reproving their pseudo-density machinery; all remaining transfer steps are proved below.

Lemma C.6 (Factorization of valid slacks). *If a nonempty convex set R has a PSD lift of order q , every finite matrix $M_{st} = \ell_s(X_t)$ formed from valid nonnegative affine functions ℓ_s on R and points $X_t \in R$ has PSD rank at most $q + 1$.*

Proof. Restrict the lifting PSD cone to its smallest face containing all feasible matrices. That face is a PSD cone on a subspace of order $r \leq q$. A finite convex combination whose ranges span all feasible ranges is positive definite on this subspace, giving strict feasibility there. For each s , minimizing ℓ_s over the restricted lift has finite nonnegative infimum. Strict primal feasibility gives dual attainment and no duality gap. Thus, on the affine lifting space,

$$\ell_s(\pi(Y)) = \langle B_s, Y \rangle + \tau_s, \quad B_s \succeq 0, \quad \tau_s \geq 0.$$

Choose a feasible lift Y_t of each X_t . The PSD matrices $\text{diag}(B_s, \tau_s)$ and $\text{diag}(Y_t, 1)$ factor M with order at most $q + 1$. Facial reduction is used only for existence, so no computational regularity assumption is needed. $\square$

Lemma C.7 (Robust sign-correlation lower bound). *Put $\text{SCOR}(m) = \text{conv}\{yy^T : y \in \{-1, 1\}^m\}$. If $\text{SCOR}(m) \subseteq R \subseteq \text{SCOR}(m) + \rho U_1$ for $\rho \leq 1/2$, every PSD lift of R has order $\exp(\Omega((m/\log m)^{2/13}))$ for sufficiently large m .*

Proof. Fix odd $3 \leq k \leq m/2$. For a k -subset S define

$$L_S(Y) = \frac{\sum_{i,j \in S} Y_{ij} - 1}{4k^2}.$$

At $Y = (2x - \mathbf{1})(2x - \mathbf{1})^T$, this is $f_k(x_S)$. Oddness of k makes it nonnegative on $\text{SCOR}(m)$. Its coefficient norm is $1/(4k^2)$, so $\ell_S = L_S + \theta$, with $\theta = \rho/(4k^2)$, is nonnegative on R . By lemma C.6, the matrix $M(S, x) = f_k(x_S) + \theta$ has PSD rank at most $q + 1$ for a lift of order q . Repeated sign-correlation points merely repeat columns.

Since $\theta \leq 1/(8k^2)$, the pseudo-density above gives $\mathbb{E}D(f_k + \theta) \leq -1/(8k^2)$. The shifted function takes values in $[0, 1]$, and its mean is at least $(k - 1)/(4k^2) \geq 1/(6k)$. Apply (123) with $d = k$ and $\delta = 1/(16k^2)$, which satisfies the strict hypothesis. There are universal positive constants c_1, c_2 such that

$$q + 1 \geq c_1 k^{-23/4} \left(\frac{c_2 m}{k^{13/2} \log m} \right)^{k/4}.$$

Choose odd k comparable to $a(m/\log m)^{2/13}$ for a sufficiently small constant $a > 0$. For all sufficiently large m , this is an admissible k and the parenthesized base is at least sixteen. The resulting lower bound is $\exp(\Omega(k))$, even after the polynomial prefactor and additive one are absorbed. $\square$

Theorem C.8 (Uniform approximate SDP size). *For sufficiently large m , if a convex set Q satisfies $K_{2m} \subseteq Q \subseteq K_{2m} + \eta U_1$ with $\eta \leq 1/(2A_m)$, every exact PSD lift of Q has total matrix order at least $\exp(\Omega((m/\log m)^{2/13}))$.*

Proof. Partition matrices into four $m \times m$ blocks and use

$$J(W) = m(W_{11} - W_{12} - W_{21} + W_{22}).$$

On a face atom indexed by x , this is $(2x - \mathbf{1})(2x - \mathbf{1})^T$, so $J(F_m) = \text{SCOR}(m)$. Each input entry contributes to one output entry with coefficient m or $-m$, giving $\|J(U) - J(V)\|_1 \leq m\|U - V\|_1$. The whole-matrix estimate in theorem C.4 consequently shows that $R = J(Q \cap H_m)$ lies between $\text{SCOR}(m)$ and its $A_m \eta \leq 1/2$ neighborhood. Section and projection preserve the PSD cone, and lemma C.7 proves the bound. $\square$

For a convex outer set, the inclusion $Q \subseteq K_{2m} + \eta U_1$ is equivalent to a support gap of at most η for every cost matrix with coefficient norm at most one. One direction follows from the dual norm inequality; the other follows by separating a point outside the compact convex set $K_{2m} + \eta U_1$. The approximate lower bounds therefore concern uniform objective guarantees at the displayed inverse-quadratic accuracy. They do not exclude objective-specific relaxations, restricted cost families, or larger approximation errors, and the SDP result does not give an exponential-in- m lower bound for approximate second-order-cone formulations.

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